Ben-Gurion University of the Negev

Faculty of Health Sciences

Lymphocyte dysregulation impacts cell senescence with aging

Submitted in partial fulfillment of the
requirements for the degree of Master
of Medical Science (M.Med.Sc.)

By

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2024

Beer-Sheva

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Lymphocyte dysregulation impacts cell senescence with aging, by Anna Dranko, Ben-Gurion University of the Negev, 2024

Abstract

According to the World Health Organization and the United Nations Department of Economic and Social Affairs Population Division, the global population is continuously aging. This trend necessitates the preservation of health and cognitive abilities for more extended periods. In response to this need, the concept of "successful aging" has gained prominence, aiming to reduce age-associated diseases. Successful aging represented by supercentenarians, individuals who have reached the remarkable age of 110 years or older, presents valuable information for researchers seeking to enhance the quality of life of older individuals and prevent age-related ailments.

Cellular senescence, irreversible cell growth arrest, is a pivotal feature of the aging process. Senescent cells express their metabolic activity by releasing the senescence-associated secretion phenotype (SASP), a blend of cytokines and chemokines. Interestingly, while the SASP attracts immune system cells, it also promotes local inflammation and facilitates the accumulation of senescent cells. Through SASP, senescent cells impact adjacent cells, thus causing them to undergo senescence.

Cellular senescence is accompanied by immune aging (also termed immune senescence), which refers to the aging of the immune system. This aging impacts immune responses and likely contributes to the buildup of senescent cells due to the inefficient clearance of these cells. Age-related alterations in the $CD4^+$ T cell repertoire, a vital part of the adaptive immune system, have been observed in both humans and mice, with older individuals displaying a substantial increase of T effector memory (TEM), activated T regulatory (aTregs), exhausted, and cytotoxic subsets compared to younger individuals.

In our research, we applied bioinformatic analysis tools on single-cell RNA sequencing (scRNS-seq) datasets to examine the co-accumulation of age-related lymphocytes and senescent cells in aging. Our primary aim was to identify age-related SASP and explore the clearance of senescent cells by cytotoxic $CD4^+$ T lymphocytes. Our investigation revealed differences in the SASP among young, old, and very old mice, and pointed to the involvement of specific chemokines in the clearance of senescent cells.

In conclusion, the experimental results support our hypothesis that alterations in lymphocyte composition, including an increase in activated regulatory T cells and a decrease in cytotoxic $CD4^+$ T cells and specific age-related SASP, contribute to the accumulation of senescent cells with age. These insights are pivotal in understanding aging mechanisms and pave the way for interventions that promote successful aging and reduce age-related disease incidence.

Acknowledgments

I want to thank everyone who guided, helped, and supported me throughout my master's degree.

First, I would like to thank my supervisor, Prof. Alon Monsonego, from the Shraga Segal Department of Microbiology, Immunology, and Genetics, for his guidance, patience, and support throughout this research.

In addition, I would like to thank Yehezquel Elyahu, an M.D./Ph.D. student in our lab whose groundbreaking work in this area served as the basis of the research performed in this thesis and with whom we collaborated. His enthusiasm and enlightening ideas and insights were much appreciated and contributed to the success of our work. I am also grateful to Anna Nemirovsky, our laboratory manager, who assisted with lab maintenance and technical issues.

I thank Dr. Ekaterina Ereminko, a researcher in our lab, for her valuable tips and suggestions and thank M.Sc. student Ilana Feigin for her contributions to my work. I thank M.Sc. student Livnat Lerrer for her help with technical issues and her friendship, and I thank M.Sc. students Amit Shiht, Ilana Kreinin, Shir Zaccai, Tal Damari, and the other members of our lab for their help, support, and friendship.

Finally, I would like to thank my husband and kids for their support and patience and for encouraging me not to give up when I was discouraged or frustrated.

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Abbreviations

ANOVA - Analysis of Variance, a statistical method.

APC - Antigen-Presenting Cell

ApoE - Apolipoprotein E

aTregs - activated Regulatory T cells

CAR-T - Chimeric Antigen Receptor T

CCL - CC motif chemokine Ligand

CCR - CC motif chemokine Receptor

CD - Cluster of Differentiation

CDK - Cyclin-Dependent Kinase Inhibitor

CDKN1A - Cyclin-Dependent Kinase Inhibitor 1A

CDKN2A - Cyclin-Dependent Kinase Inhibitor 2A

CIP - CDK Interacting Protein

CKI - Cyclin-Dependent Kinase Inhibitor

CTL - Cytotoxic T Lymphocytes

CTLA - Cytotoxic T - Lymphocyte - Associated Protein

DGE - Differential Gene Expression

DN - Double Negative

DNA - Deoxyribonucleic Acid

DP - Double Positive

DREAM - Dimerization partner, Retinoblastoma-like, E2F and MuvB

ECM - Extracellular Matrix

EOMES - Eomesodermin

FACS - Fluorescence-Activated Cell Sorting (Flow cytometry)

FOXO3 - Forkhead box O3

FOXP3 - Forkhead Box P3

GATA6 - GATA binding protein 6

GEM - Gel bead in Emulsion

GLB1 - Galactosidase Beta 1

GNLY - Granulysin

GO - Gene Ontology

GSEA - Gene Set Enrichment Analysis

GZM - Granzyme

H2A - H2A histone family

H2AX - H2A histone family member X

ICD-11 - International Classification of Diseases, 11th Revision

ID2 - Inhibitor of DNA Binding 2

IKZF2 - IKAROS Family Zinc Finger 2

IL - Interleukin

IL-xR - Interleukin x Receptor

KEGG - Kyoto Encyclopedia of Genes and Genomes

KLRB1 - Kallikrein B1

kNN - k-Nearest Neighbour

LAG3 - Lymphocyte - Activation Gene 3

MF - Molecular Function

MHC - Major Histocompatibility Complex

MIP - Macrophage Inflammatory Protein

MIP-1α - Macrophage Inflammatory Protein 1α

MIP-1β - Macrophage Inflammatory Protein 1β

mRNA - Messenger Ribonucleic Acid

mTOR - mammalian Target of Rapamycin

MVP - Mean-Variable Plot

NKG7 - Natural Killer Cell Granule Protein 7

PBMC - Peripheral Blood Mononuclear Cell

PC - Principal Component

PCA - Principal Components Analysis

PD-1 - Programmed Death-1

PD-L1 - Programmed Death-Ligand 1

pRB - Retinoblastoma protein

PRF1 - Porphyrin 1

QC - Quality Control

RANTES - Regulated Upon Activation, Normal T Cell Expressed and Secreted

RNA - Ribonucleic acid

rTregs - resting Regulatory T cells

RUNX3 - Runt-related transcription factor 3

SASP - Senescence-Associated Secretion Phenotype

SA-β-gal - Senescence-Associated beta-galactosidase

scRNA-seq - single-cell RNA sequencing

SELL - L-Selectin (CD62L)

TCM - Central Memory T cells.

TEM - Effector Memory T cells

Tfh - T follicular helper cells

Th - T helper

TIL - Tumor Infiltrating Lymphocytes

TM - Tabula-Muris

TMS - Tabula-Muris-Senis

TRAC - T Cell Receptor Alpha Constant

TRDC - T Cell Receptor Delta Constant

t-SNE - t-distributed Stochastic Neighbour Embedding

UMAP - Uniform Manifold Approximation and Projection

UMI - Unique Molecular Identifier

UN - United Nations

UNDESA - United Nations Department of Economic and Social Affairs

WHO - World Health Organization

1 Introduction

1.1 Current Concepts of Aging

There has been a significant rise in the lifespan of individuals worldwide, exceeding previous records, with an average estimated lifespan for 2019 of 72.6 years [1]. By 2030, about 17% of the world's population will be over 60 years of age, and in the next 30 years, the number of individuals aged 80 years and older is expected to triple, reaching 426 million [1,2]. As the global population continues to age, there is a crucial requirement to understand and promote healthier aging approaches.

As the average lifespan lengthens and the number of elderly people increases, there is growing demand for approaches that emphasize "successful aging" - aging in a healthy way [3–7]. Successful aging focuses on maintaining physical and cognitive health as individuals age [8–13]. To appreciate the importance of successful aging, it is essential to delve into the biological processes that underpin aging.

Aging is considered an inevitable and natural decline in bodily and cognitive functions, leading to decreased tissue quality, cognitive abilities, and brain activity. Human aging starts at birth and accelerates in adulthood (the stages of aging are presented in Fig. 1). With this understanding, we can examine one of the primary biological mechanisms involved in this decline.

One of the key hallmarks of aging is cellular senescence, which is characterized by an irreversible arrest of the cellular division cycle [5,14,15]. Cellular senescence is a defense mechanism against DNA damage triggered by various stressors [5,16]. However, senescent cells do not clear solitary, and the accumulation of senescent cells results from a decline in the immune system functioning and impaired clearance function, contributing to aging by maintaining chronic inflammation [5,16–21].

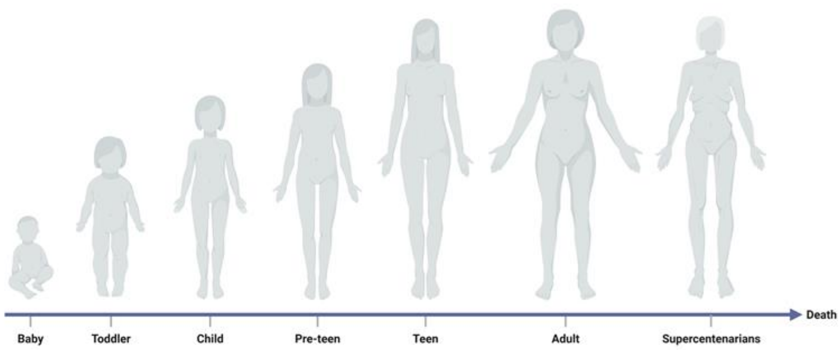


Figure 1: Key Milestones from Birth to End of Life. Comprehensive timeline displaying the stages of human aging, key milestones, and transition from birth through adulthood to the end of life.

1.1.1 The Evolutionary Dance of Aging

Aging can be viewed as a byproduct of natural selection, a concept supported by seminal works in the evolutionary biology domain [22–24]. While the lifespan of diverse organisms varies significantly, a universal truth remains that all organisms, including humans, are destined to face mortality [25,26].

Various theories have been proposed regarding the evolution of aging. Programmed theories, like Programmed Senescence, argue that aging is a predetermined consequence of genetic codes, where hereditary mutations, often expressed later in life, play a pivotal role (ref). This aligns with observations of decreasing hormone levels, diminished immune system capacity, and reduced proliferative potential [27–29]. Conversely, Damage or Error Theories, including the Mutation Accumulation and Free Radical theories, emphasize the accumulation of cellular damage or errors, mainly from free radicals [24,26,27,29–32]. Recent studies have also explored the molecular and genetic mechanisms underlying aging and longevity, offering a deepened understanding of how genetic factors contribute to lifespan [33,34].

The recurring motif of these theories is the notion that organisms allocate resources and prioritize reproduction rather than the maintenance of the body, mainly because organisms Strive to Fulfill their biological function as DNA “survival machines” [23,25,35]. This evolutionary perspective aligns with recent insights into the molecular mechanisms and interventions regarding aging and aging-related diseases [36].

This evolutionary perspective can be connected to immunosenescence: as the body prioritizes reproductive success during its prime, it might inadvertently allocate fewer resources to immune function as it ages. Consequently, a decline in the immune system's efficiency, or immunosenescence , leading to reduced capabilities in clearing out senescent cells, which further accelerates aging and age-related diseases [30,37].

Our study aligns with these evolutionary theories on aging, particularly focusing on Damage Accumulation [24,32] with Programmed Senescence [29], and the decline in immune system efficiency. Both are underlined by evolutionary trade-offs [35]. The latter perspective suggests that a dwindling immune system facilitates senescent cell accumulation, which further contributes to the aging process. The surveillance and clearance of senescent cells by the immune system are crucial, and impaired immune surveillance accelerates the accumulation of senescent cells and aging [38–41]. This finding confirms that our research is consistent with evolutionary theories of aging and particularly underscores the interplay between immune system efficiency and senescent cell accumulation [16,19].

1.1.2 Aging as a Disease

Aging is frequently associated with various diseases, often referred to as age-associated diseases. According to the International Classification of Diseases (ICD-11), aging is identified as a primary factor behind these diseases [42]. Currently, two major perspectives of aging prevail. One perspective posits that aging is a disease, aligning with the definition of disease as an abnormality of bodily functions possessing distinct causes and recognizable signs and symptoms and includes social problems [43]. This perspective opens up a discourse for the treatment or management of aging. The opposite, which is more widely accepted, perceives aging as a natural, normal process due to its universality among most organisms [44]. Although universal, it can vary by population and progress with different degrees of success at maintaining health [43].

With the underlying assumption that aging is a disease and based on the concept that aging is an ancestral trait that poses challenges today, our research focuses on the physiological decline associated with aging. Given their remarkable longevity, we refer to supercentenarian individuals and "very old" mice as examples of successful aging.

1.1.2.1 Successful Aging: Characteristics, Factors

John Rowe and Robert Kahn introduced the term "successful aging" in the 1980s [12], establishing a seminal notion in the discourse surrounding aging. It encompasses three aspects: avoidance of disease and disease-related disability, maintenance of high cognitive and physical function, and active lifespan extension. Drawing upon this concept, we propose the term “good aging” to emphasize longevity and the quality of life during aging. We delve into examining supercentenarian individuals and “very old” mice as examples of successful aging due to their remarkable age. These three components are essential for maintaining health and vitality in older adulthood [11,45–47]. There is also evidence that diet restriction, avoidance of harmful behaviors, moderate physical activity, and social activity are associated with healthy aging [12,48–52].

“Successful aging” is characterized by individuals who remain free from age-related diseases in senior age (60-80 years old), have delayed onset of diseases, and have the potential for longer lifespans, such as centenarians and supercentenarians, represent a minor fraction of the total population [53,54]. Those individuals longevity is mainly considered as hereditary based [55]. Previous genetic studies have identified several genes associated with longevity: a polymorphism in the apolipoprotein E (ApoE) and forkhead box O3 (FOXO3) were found to contribute to slowing the aging process and associated with longevity, and GATA binding protein 6 (GATA6) has been associated with the regulation of aging in mesenchymal stem cells—a specific variant of which is notably prevalent among supercentenarians [56–58].

The lifespan in multicellular organisms exhibits significant variation; mice live around 24-30 months [59], while humans live between 40 to 80 years [1,2,60,61]. Aging differs among individuals, whether or not they maintain physical and social activity [12,62,63]. Individuals preserving higher levels of physical and social activity are frequently regarded as experiencing healthier or more successful aging trajectories [9,64].

Technological development can shed light on such a complex process as aging. In contrast to traditional bulk methods, which yield average results for bulk populations of cells, scRNA-seq emerges as a powerful method furnishing high quantitative resolution [65–69]. With next-generation sequencing technologies, it has become feasible to sequence multiple genes in numerous single cells simultaneously, leading to a paradigm shift by generating parallel transcriptomes at single-cell resolution. Each cell undergoes processing in conjunction with gel beads within a unique emulsion system termed GEMs. These GEMs encapsulate primers inside aqueous nano-droplets set against an oil medium backdrop, as depicted in Fig. 2 [67,70–75].

The primers utilized in this process serve multi-fold purposes: a distinctive cell barcode that is singular to each cell, facilitating cell differentiation; a Unique Molecular Identifier (UMI) that allocates reads derived from individual transcripts; and a PolyT tail specifically designed to bind with the PolyA tail of mRNA. Post the encapsulation phase, cellular lysis occurs, releasing the mRNAs that are subsequently ensnared by the primers. This captured mRNA undergoes phases of reverse transcription followed by amplification [70,71,73,74,76–78].

Once amplified, the resultant DNA sequences for each cell are aggregated to form distinct gene libraries, which are then sequenced. Following sequencing, data undergoes rigorous processing, most commonly through a dedicated pipeline like Cell Ranger or other equivalent FASTQ file analysis tools, referenced in Fig. 3. The culmination of this analysis is a structured matrix that outlines the relationship between cells and genes [73,75,79–82].

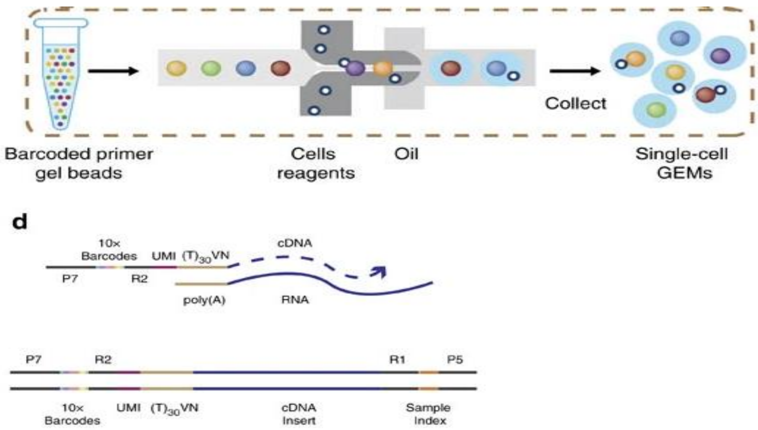


Figure 2: Diagram of the cell capture process by gel beads, and a barcode structure.

Gel beads loaded with primers and barcoded oligonucleotides are first mixed with cells and reagents and subsequently mixed with the oil-surfactant solution at a microfluidic junction. Single-cell GEMs are collected in the GEM outlet. Gel beads contain barcoded oligonucleotides of Illumina adapters, 10x barcodes, UMIs, oligo dTs, and prime RT of polyadenylated RNAs. Finished library molecules consist of Illumina adapters and sample indices, allowing the pooling and sequencing of multiple libraries on a next-generation short-read sequencer [83].

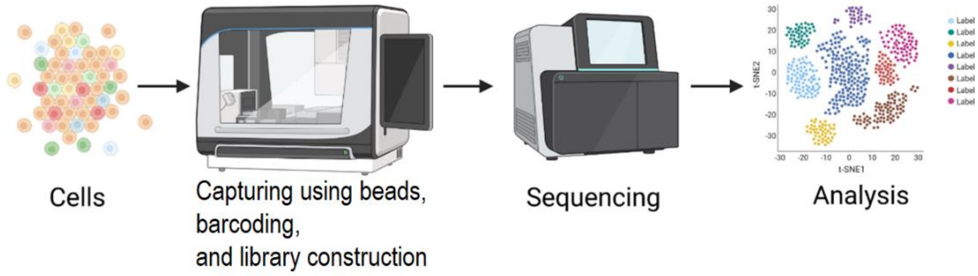


Figure 3: The 10x Genomics pipeline utilizes a unique droplet-based approach for single-cell sequencing. Each cell is paired within GEMs with primers that contain UMIs. These GEMs ensure each droplet contains only a single cell and a single gel bead coated with primers that contain not only UMIs but also a cell barcode and a polyT tail. This encapsulation takes place within aqueous nano-droplets suspended in an oil medium. After cell lysis and reverse transcription within these droplets, the resulting cDNA is pooled, amplified, and prepared for sequencing. Post-sequencing, the UMIs on each bead are used to attribute sequencing reads back to their original single cells, enabling high-resolution analysis of individual cell transcriptomes.

Our research emphasizes the physical aspect of aging displayed by our longevity. These aspects are central features of what we term successful aging. We analyze the lifespans of humans and mice, particularly those that exhibit longer lifespans since they present those aspects. Given that cellular changes underpin many physical manifestations, we delve into a bioinformatical analysis. We specifically examine senescent and $CD4^+$ T cells using the scRNA-seq method, as these cells are intricately tied to aging processes and can provide insights into the molecular mechanisms that contribute to prolonged lifespan and successful aging.

1.1.2.2 Hallmarks of Aging

Aging is a unique biological process characterized by specific molecular causes and common symptoms in all aging organisms. Numerous studies have identified these distinguishing features, often referred to as "aging hallmarks" [14,16,44,84–87]. As illustrated in Fig. 4, there are nine key hallmarks of aging. Prominent among these are the accumulation of senescent cells, stem cell exhaustion without proper clearance, and lower cell production capacity [5,14,16,20,88–90]. Our study focuses on understanding the mechanisms behind clearing these aging-related accumulations, which is particularly significant.

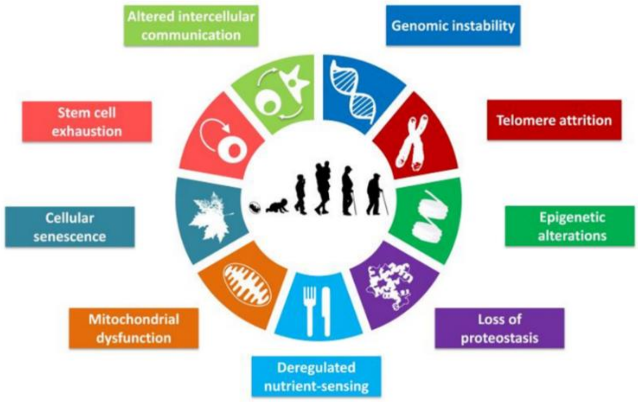


Figure 4: "The hallmarks of aging". Nine hallmarks characterized in the literature [16].

1.2 The Interplay of Immunosenescence and Inflammaging

Aging is characterized by a decline in the functions of various systems, including the immune system [18,19,91]. With age, the capacity of immune cells to proliferate decreases, leading to reduced regulation of inflammation [18,19,91]. Older individuals show a lower proportion of naïve cells and a higher proportion of memory and effector cells compared to young individuals [91–94]. Additionally, compared to young individuals, effector cells in older individuals exhibit significant functional decline [95–97]. These changes collectively contribute to a phenomenon observed in older people: low-grade chronic inflammation, often referred to as inflammaging [18,19,30,98,99].

Inflammation is a fundamental defense mechanism in the body, whether it is acute (short-term and regulated) or chronic (long-term and often low-grade) [100–103]. Inflammaging is the term used to describe the low-grade chronic inflammation observed in aging, characterized by increased levels of pro-inflammatory cytokines [18,19,30,104,105]. This process is linked to cellular senescence, where senescent cells release a mix of pro-inflammatory cytokines known as the senescence-associated secretory phenotype (SASP), further contributing to chronic inflammation [106–110].

Age-associated immune dysfunction, known as immunosenescence, is a progressive gradual decline in the immune system's functionality, resulting in increased susceptibility to infections and reduced ability of regulating inflammation [19,91,111–113]. $CD4^+$ T cells play an essential role in immunosenescence, and CD4+ T cells dysfunction advances with age [114–117]. As age increases, the amount of naïve lymphocytes decreases, while lymphocyte proliferation amplifies, contributing to immunosenescence [91,94,114,118]. Inflammaging and cellular senescence promote pro-inflammatory cytokine secretion, further contributing to immunosenescence [18,110,119]. Additionally, immune cells' cellular senescence can impair immune function [19,107,112].

As previously mentioned, "inflammaging" is a phenomenon marked by low-grade chronic inflammation seen with aging, characterized by elevated levels of pro-inflammatory cytokines.

This condition is linked to cellular senescence, where senescent cells release pro-inflammatory cytokines, forming the SASP and perpetuating chronic inflammation.

Building on this, cytokines, small proteins produced during the inflammatory response, play a vital role in maintaining inflammation [120–124]. Aged mice cytotoxic $CD4^+$ T cells have been found to produce several cytokines, which are also components of the SASP [125–128]. Two such cytokines, CC Motif chemokine ligand 3 and 4 (respectively CCL3 and CCL4), also known respectively as macrophage inflammatory protein 1α and 1β (respectively MIP-1α and MIP-1β), recruit cells expressing the Th1 phenotype [129–132]. Another cytokine, CC Motif chemokine ligand 5 (CCL5), also known as regulated upon activation, normal T cell expressed and secreted (RANTES), recruits both Th1 and Th2 cells [129,131,133–135]. Additionally, various cytokines linked to age-associated diseases, like interleukin-1 β (IL-1β), CCL2, CCL3, and CCL8, are known components of the SASP [107,109,125,136,137].

The immune landscape shifts during aging, leading to the appearance and accumulation of age-associated $CD4^+$ T lymphocyte subsets [91,116,138–140]. In contrast, the population of naïve subsets shrinks, resulting in decreased regulation of the immune response [91,94,118,141].

1.2.1 Immunosenescence and T Cell Dynamics

The aging process engenders immunosenescence, leading to intricate modifications in T cell dynamics. This natural yet complex transformation underscores the pivotal role of T cells in the unfolding narrative of age-related alterations within the immune system. Recent findings suggest a direct correlation between the age-induced alterations in T cells and the propensity towards certain age-related diseases [19,94,112,118,142]. These cells play a crucial role in the adaptive immune system's response to specific antigens and distinguishing between self and non-self [143]. As individuals age, their immune responses decline, affecting reactions to infections, vaccines, and tumor clearance [19,104]. This process also reduces the production of new $CD4^+$ T cells, leading to the accumulation of age-related subgroups [92]. The decline in immune competency with age not only predisposes the elderly to infections but also diminishes the efficacy of vaccinations, highlighting the necessity of advancing our understanding in this domain. Understanding these age-induced changes in T cell dynamics is fundamental to addressing the challenges posed by immunosenescence, setting the stage for a deeper exploration into the differentiation and maturation processes these cells undergo.

T cells undergo differentiation and maturation in the thymus, transitioning to a state known as single-positive and acquiring specialization in recognizing specific antigens. This prepares them for their roles in immunity. Subsequent paragraphs will elucidate T cell functions, categorizations in the context of aging, evolving populations with advancing age, and the broader immune response.

The decline in naïve T cells, coupled with dysregulation of memory T cells, leads to significant functional changes in the immune system, including an increase in memory T cells [91,94,112,139,144–146].

T cells are categorized into three major subtypes: cytotoxic or $CD8^+$ T cells, helper T cells ($CD4^+$), and regulatory T cells ($CD4^+$). Helper T cells coordinate immune reactions by responding to MHC class II molecules appearing on antigen-presenting cells (APCs) or regular cells, while cytotoxic T cells ($CD8^+$) recognize antigens presented by MHC class I molecules and are responsible for eliminating infected cells [143,147,148].

The memory T cells, when dysregulated, exhibit altered responsiveness. Part of the effector cells show changes in cytokine production and reduced cytotoxicity [94,116,149–153]. Both memory and naïve T cells exhibit comparable functional modifications as they age, suggesting that the effect of aging on T cell function is not limited only to memory T cells [91,94,118,140,154].

Dysregulated $CD4^+$ T cells , showing impaired mitochondrial activity, disrupt the effector function of Th17 cells, potentially exacerbating chronic inflammatory conditions and autoimmune disorders [155–159]. Th17 cells, as a subset of $CD4^+$ T cells, play a pivotal role in immune responses and inflammation. Dysfunctional Th17 cells may contribute to chronic inflammatory conditions and autoimmune disorders [156,160–164].

It is essential to understand the changes in T cell function and metabolic pathways with aging to develop strategies for enhancing immune responses and overall immune health in aging individuals [16,91,94,165–167]. This understanding builds upon a foundational framework while exploring the role of regulatory T cells in immune senescence.

As individuals age, their phenotype undergoes alterations in response to antigen stimulation, which results in modifications of their repertoire. This occurs as naïve cells' proliferation and clonal expansion take place while the thymus involutes [94,168]. Consequently, there is a decline in the number of immature cells, while subsets associated with the aging process tend to accumulate [91,94,144,168,169].

The function of recognizing cytokines and chemokines during inflammation is vital, as it results in their differentiation, killing, and regulation of immune responses [153,170–175]. This process encompasses both $CD4^+$ and $CD8^+$ T lymphocytes, with the naïve cells of these lymphocytes exhibiting typical markers such as the chemokine receptor CCR7, IL7R, and L-selectin CD62L (SELL/CD62L) markers [176,177].

Among the subsets that increase in aging individuals are the activated subsets, including TEM cells identified by the CD44 marker, responsible for immune cell memory [114,116,153,178]. However, these cells can become exhausted due to chronic inflammation and express genes such as

PD-1 and LAG3, which can reduce anti-inflammatory activity compared to younger individuals [95,97,179,180].

Additionally, there is a noted increase in the accumulation of regulatory T cells that express FOXP3, IKZF2, and CTLA4 [181–187]. This augmented presence of cells in older individuals contributes to immune senescence and chronic inflammation.

In summation, the age-induced metamorphosis of T cells plays a cardinal role in immune senescence, aligning with the broader spectrum of immunosenescence [19,91]. This investigation aspires to dissect the intricacies of T cell dynamics as individuals age, shedding light on potential therapeutic avenues to ameliorate age-associated immune deficiencies [138,139]. While numerous studies have delved into the complexities of immunosenescence and T cell dynamics [94,112,142], our research explores potential shifts in the $CD4^+$ T cell repertoire with aging. Specifically, we are interested in examining the equilibrium between cytotoxic $CD4^+$ T cells and aTregs, and its implications on the accumulation of senescent cells [19,140]. Through this probing exploration, we aim to glean insights into the variations in the SASP between successful and unsuccessful aging [110,125], offering a launchpad for future research on biomarkers and therapeutic strategies. The reverberations of this exploration extend beyond immunology, providing a valuable vantage point for gerontologists and clinical practitioners in understanding and managing the age-associated challenges on immune health and enriching the broader understanding of immune senescence [19,91].

1.3 Cellular Senescence: A Key Hallmark of Aging and Its Mechanisms

Cellular senescence characterized by the irreversible arrest of cell growth and division while maintaining metabolic activities [14,188–190]. It is important to recognize that this mechanism serves as a defensive measure against the growth of tumors [188,191–194]. During the secretion of SASP, senescent cells alternate their surrounding environment, influencing neighboring cells [20,108,110,110,195]. The interplay between cellular senescence and reduced lymphocyte function accentuates the accumulation of senescent cells, contributing to age-related diseases [5,14–16,196].

In 1961, Leonard Hayflick and Paul Moorhead identified a phenomenon of “senescence” upon discovering that human cells grown in culture exhibit a finite potential for division. The Hayflick limit which postulates that the longevity of human life is restricted by the finite number of cellular replications that can occur [188]. This condition of cellular senescence can be identified by distinctive alterations in the cell's morphology [15,190,194,197,198]. The phenomenon in question is thought to be caused by the activation of the mammalian target of rapamycin (mTOR) pathway, which triggers cell growth and reorganizes the cytoskeleton [199–204].

The senescent cell can go through apoptosis or autophagy, or if not cleared, may progress into acute or chronic senescence or even cancer, as presented in Fig. 5 [15]. The phenomenon of cellular senescence can be attributed to various factors, including telomere erosion, mitochondrial dysfunction, and non-repairable DNA damage (Figs. 3 and 5) [5,15,16]. Chronic inflammation and stem cell exhaustion may also contribute to this process [18].

Furthermore, one of the key indicators of cellular senescence involves the upregulation of the enzyme senescence-associated beta-galactosidase (SA-β-gal), marking a significant aspect of this intricate cellular process [197,205–209]. Additionally, senescent cells experience a loss of the nuclear protein lamin-B1 [15,210–214]. Furthermore, they demonstrate γ-H2A.X, a component of the histone H2A family, that serves as an indicator of the existence of double-stranded DNA breaks [15,215–217]. Some alterations to senescent cell hallmarks are presented in Fig. 6. Senescence mechanism goes through one or more of three pathways: $p53/p21^{WAF1/CIP1}$ pathway, $p16^{INK4\alpha}$/pRB pathway (Fig. 7), and a dimerization partner, retinoblastoma-like E2F and MuvB (DREAM) complex assembly (Fig. 7) [15,218–223].

The cellular senescence process has a profound impact on several fundamental physiological activities [194,224]. It is vital in wound healing and embryonic development [225,226]. Younger people primarily show acute senescence, whereas older people are more prone to chronic senescence. This is exacerbated since, with aging, the immune system's efficiency in clearing out these senescent cells diminishes [14,110,125,136,224]. This leads to a cell buildup, worsened by chronic inflammation and the influence of the SASP on surrounding cells [108,125,136].

Further complexities arise from the variability of the SASP among senescent cells, a factor that stems from their diverse causes. This variation can affect the immune system's capacity to clear senescent cells and the surrounding cellular environment [106,108–110,221,227–229].

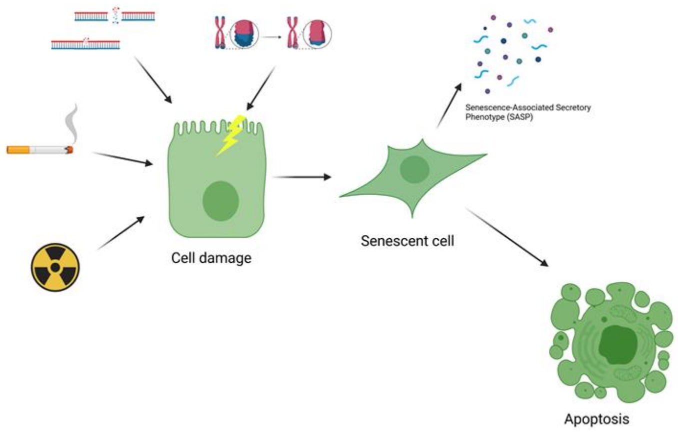


Figure 5: Overview of the Cellular Senescence Process. External and internal stressors can induce cellular damage, prompting cells to enter a state of senescence. While in this state, cells produce SASP. Ideally, these cells should undergo apoptosis (first scenario), but if not cleared, they can accumulate (second scenario) over time.

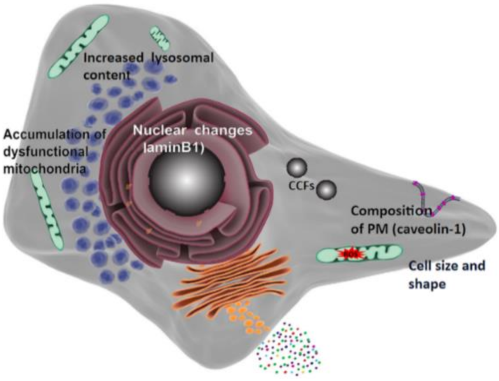


Figure 6: Alterations of senescent cell hallmarks seen in a cell. alteration of cell size and shape, SASP secretion, accumulation of dysfunctional mitochondria distressed by reactive oxygen species (ROS), nuclear changes, increased lysosomal content, cytoplasmic chromatin fragments (CCFs) and alterations in plasma membrane (PM) composition [15].

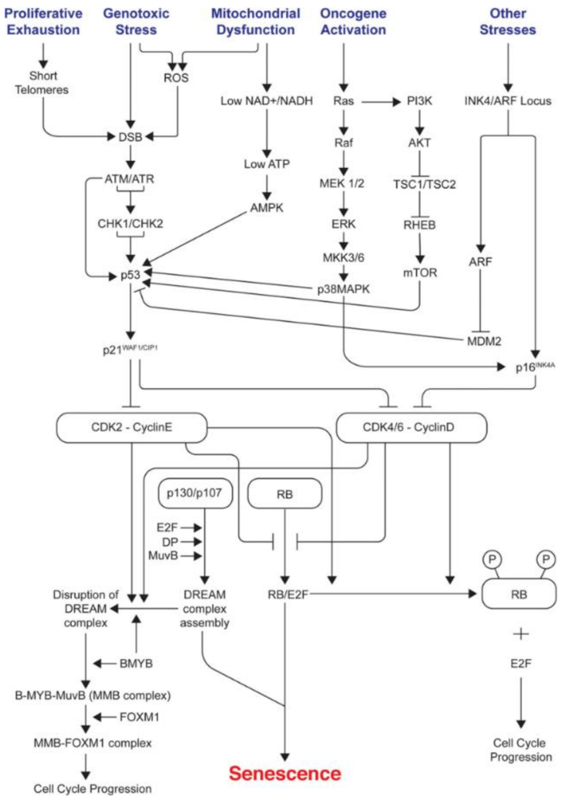


Figure 7: Scheme of signals and pathways involved in mediating senescence cell. The different causes that initiate cascade of the senescence, activating the three key proteins: p53, p21, and p16 [221].

2 Hypothesis

As individuals age, significant changes occur in their immune system, particularly in the composition of CD4$^+$ T lymphocytes [19,94,118,175]. Recent research in our laboratory suggests that aging gives rise to a novel subset of CD4$^+$ T lymphocytes exhibiting cytotoxic activity, more commonly found among supercentenarians who have a higher proportion of cytotoxic CD4$^+$ T lymphocytes than their younger peers [92,93,127,144,169,230]. The correlation between cytotoxic CD4$^+$ T lymphocytes and longevity suggests that they could serve as biomarkers for healthy aging and potentially open up novel therapeutic strategies.

An accumulation of senescent cells also characterizes aging and influences neighboring cells, driving their transition to a senescent state [5,189]. Furthermore, the SASP can vary based on tissue type and conditions and is likely to change with aging [106,189,206,227,228]. Cytotoxic CD4$^+$ T lymphocytes may help in the clearance of senescent cells, while activated regulatory T cells (aTregs) may promote their accumulation by suppressing cytotoxic activity [16,92,94,231,232].

In summary, we hypothesize that, as we age, our CD4$^+$ T cell repertoire undergoes significant shifts. There is an increase in activated regulatory T cells (aTregs) and a decrease in cytotoxic CD4$^+$ T cells, contributing to the accumulation of senescent cells, as reflected in the SASP variations between successful and unsuccessful aging.

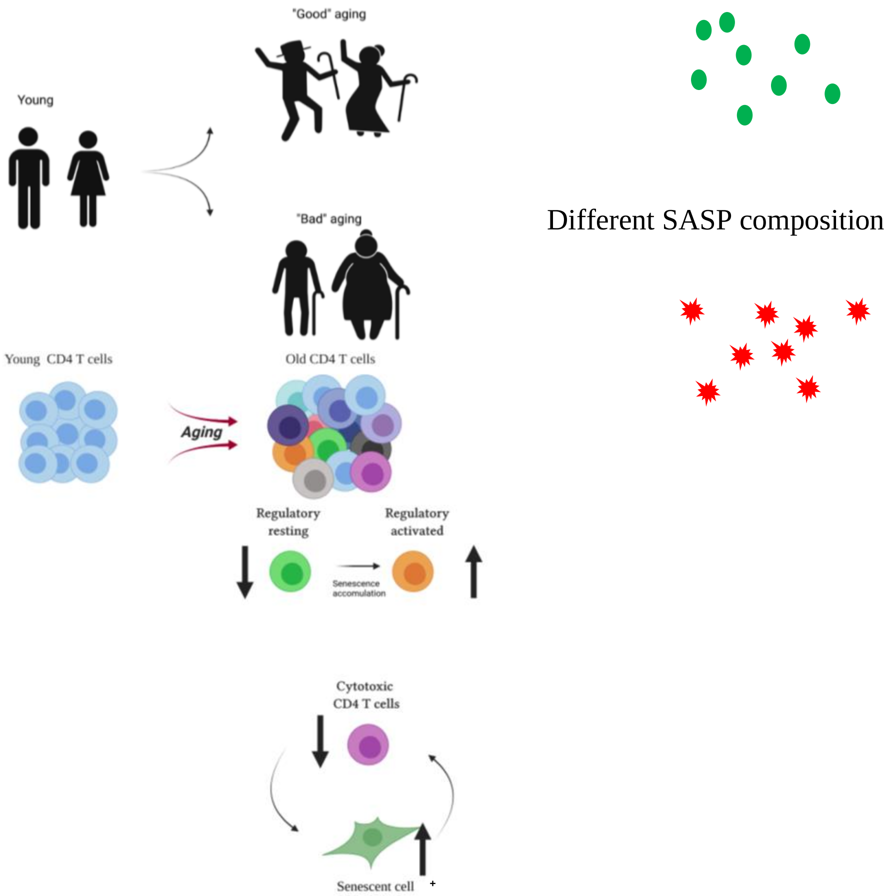


Figure 8: A visual representation of schematic overview of the hypothesis. Upper Panel - Illustrating the variations in the SASP between successful (good) aging and unsuccessful (bad)

aging. Lower Panel - As we age, the $CD4^+$ T cell repertoire changes. The number of activated regulatory T cells (aTregs) increases while the number of resting regulatory T cells (rTregs) decreases due to the accumulation of senescent cells. The decline in cytotoxic T cells may also contribute to this accumulation.

2.1 Aims

1. Determine whether $CD4^+$ lymphocyte subsets orchestrate senescent cell clearance.
1.1. Identify T cells in the Tabula-Muris-Senis droplet dataset (TMS) dataset.
1.2. Identify $CD4^+$ T cell subsets.
1.3. Define the relationship between the $CD4^+$ T cell subsets and senescent cells.
2. Describe cytotoxic $CD4^+$ T cells transcriptome within species.
3. Computationally characterize key cytokines involved in aging-related cellular senescence.
3.1. Identify senescent cells in the TMS dataset.
3.2. Describe senescent cell’s transcriptome through aging.
3.3. Identify age-related SASP.

Expected significance:

In this research, we aim to produce insightful knowledge about the significance of $CD4^+$ T cells, and particularly the cytotoxic subset in creating and maintaining senescent cells. Understanding aging can help develop strategies for healthy aging by identifying the causes and guiding effective approaches. This may contribute to increased longevity and improved quality of life.

3 Methods and Datastes

3.1 Analysis of Public Datasets

The datasets used in our research were strategically selected based on several critical attributes; for example, the inclusion of various age groups is crucial for aging research. The Tabula-Muris-Senis (TMS) dataset was selected for its comprehensive inclusion of different cell types and tissues from both males and females in six timepoints, encompassing lymphocytes and senescent cells, providing a robust platform for our investigations [233].

The supercentenarian dataset provides coverage of lymphocytes across different ages in humans and is invaluable in our work [93]. The use of the aging mice dataset of Elyahu et al. allowed us to perform a comparable investigation of $CD4^+$ T cells in a non-human model [92].

To further explore and compare cytotoxic $CD4^+$ T cells' characteristics in diverse conditions, we used a bladder cancer dataset [234]. This dataset enables us to examine cytotoxic $CD4^+$ T cells under pathological conditions, enriching our understanding of their behavior and potential roles across a broader spectrum of biological scenarios.

The datasets used in our research are publicly available open-source datasets in the format of tables, with additional metadata and matrices of cellXgene as input data, rows as genes symbols or IDs, and columns as cell barcodes. The datasets were analyzed using the R programming language for statistical analysis (version 4.0.3) and its packages [235].

3.1.1 General Workflow for the Seurat Package

To analyze the datasets, the Seurat package version 3.2.3 for R programming language version 4.0.3 was employed [236]. Its canonical workflow is described in next paragraphs and visualized in Fig. 9. Usually, its input data is the cell ranger pipeline output using the *Read10X()* function [73]. However, the input data was published in CSV and H5AD formats in TMS study and also published as FASTQ files [79,233].

The Seurat object, which is of the S4 file type, was created using the *CreateSeuratObject()* function. "Count" is the input data; in this case, it is the output of the *Read10X()* function, which is the cellXgene matrix. Data preprocessing is essential for preparing the data for analysis. The preprocessing, which consists of quality control (QC), and identifying highly variable genes, is aimed at creating a comparable, reliable, high-quality dataset [236,237].

All data from low-quality cells is filtered out, and the QC stage is crucial for reliable down-stream analysis. Commonly used criteria for detecting low-quality cells, doublets, or empty droplets include assessing the number of genes or UMIs per cell, with high counts suggesting doublets and extremely low counts indicating low-quality cells or empty droplets. Additionally, elevated expression of mitochondrial genes can signal compromised cells, usually present dying cells, while specialized tools can assist in detecting doublets and ambient RNA contamination.

Dying cells were identified by using the *PercentageFeatureSet()* function, which searches for genes based on the required pattern and calculates the proportion of needed gene set [73,79,81,82,237–239].

The filtered data was normalized, which means it was adjusted to a natural logarithmic scale (ln) with a virtual one count, as shown in the function (equation 1) below which aims to remove differences between gene expressions derived from technical differences in RNA capturing, amplification, and sequencing, enabling cell comparison [80,81,236,240–244].

$$Normalized = \ ln(\frac{reads\ per\ gene\ per\ cell}{total\ reads\ for\ the\ cell} \times scale.factor + 1)$$

Equation 1: Gene expression is normalized based on the number of reads per gene per cell. A "scale.factor" constant is used to ensure comparability across experiments or samples. The logarithmic function (Ln) is utilized with a "+" added to prevent taking the logarithm of zero.

FindVariableFeatures() function, which produces a vector of features, in this case genes, whose expression is highly varied between cells, is used to identify highly variable genes. These genes provide valuable information and were used to reduce the overall data size [79,80,237].

The data was then scaled using the *ScaleData()* function, and a linear transformation was applied to set the mean expression of each gene across all cells to zero with a variance of one. Regression against each gene in the data allowed us to remove batch effects introduced by highly expressed genes [80,237,245,246].

After preprocessing the data, the analysis could begin; before performing the clustering and cell identification steps, some other necessary tasks are completed, a linear dimensional reduction is performed using the *RunPCA()* function to identify principal components (PCs) for further analysis and determine dimensionality based on the selected PCs [80,237,247–250].

PCs enable dimensionality reduction with minimal information loss. The relationships between the variable reads are approximated, yielding corresponding vectors for the data. Selecting the most important PCs is a crucial step in the analysis. While various methods are available, two methods are presented in the Seurat package. One method is the Jackstraw procedure, which calculates the significance of each PC. Another widely used technique is creating an elbow plot, which ranks PCs by standard deviations. By examining this plot, one can choose the PCs located at the "elbow" as these represent the most informative dimensions [80,237,248,251]. Our analysis employed both methods but predominantly relied on the elbow plot due to its faster execution.

Then, the data was clustered graphically. This involved generating a k-nearest neighbor (kNN) graph with the *FindNeighbors()*. The package employed for the analysis defaults to using the Louvain unsupervised learning algorithm to create the kNN graph, likely due to the algorithm's recognized efficiency in community detection, especially in large-scale networks. Following this, the *FindClusters()* function was used to optimize modularity and group cell neighbors into communities. The choice of the Louvain algorithm ensures reliable clustering results, which aligns with the objectives of this research [237,252,253].

For visual representation of datasets, the Uniform Manifold Approximation and Projection (UMAP) technique was used; this method is rooted in the idea that data is uniformly distributed in a Riemannian space, and similar cells have local connections [80,237,254]. T-distributed stochastic neighbor embedding (t-SNE), another dimensional reduction method, was not employed, as UMAP offers superior visualization of distinct clusters [255,256].

Lastly, cell type identification requires that markers be defined for each cluster, either through the *FindMarkers()* or the *FindAllMarkers()* functions, applying *FindMarkers()* to compare certain clusters. With these markers in place, the cell type can be manually identified or determined using reference data and associated spatial packages [80,236,237].

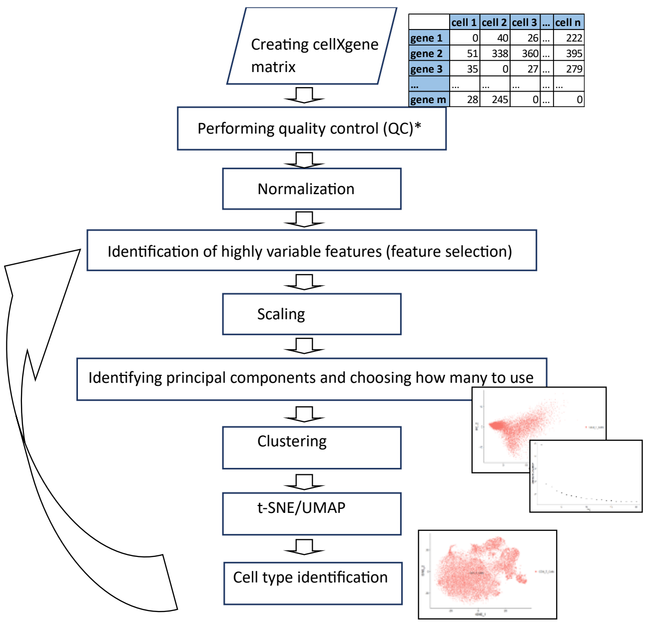


Figure 9: Seurat workflow utilized for scRNA-Seq analysis. The input in the process involves raw sequenced data encapsulated in a cellXgene matrix. The initial phase of the Seurat pipeline comprises data preprocessing, executed through quality control to filter out low-quality data, followed by normalization to render the data comparable across different cells. Subsequently, variable features (genes) were identified to spotlight genes exhibiting significant variation. The next phase involves Linear Dimensionality Reduction of PCs to simplify the data while retaining essential information. This is pivotal for the subsequent Clustering step, which aggregates cells with similar expression profiles. Then, Cell Type Identification leverages the clusters to determine the dataset's various cell types. The final segment of the workflow employs UMAP or t-SNE techniques to visually encapsulate the data in a two-dimensional representation, presenting the single-cell datasets in a comprehensible manner.

*The TMS dataset that was used is post-QC.

3.1.2 Cytotoxic CD4⁺ T Cell Subset Datasets

3.1.2.1 Aging Mice Dataset

scRNA-seq was employed to characterize the subsets of $CD4^+$ T cells, highlighting differences between young and old C57BL/6 mice. The aging mice dataset presents transcriptomic profiles from 24,007 $CD4^+$ T cells, sourced from the spleens of eight male mice, four mice in each age categories: young (2-4 months) and old (22-24 months) [92].

The schematic experimental design used is presented in Fig. 10. Purification of the $CD4^+$ lymphocytes from splenocytes was performed using fluorescence-activated cell sorting (FACS) based on recognized $CD4^+$ lymphocyte markers ($CD3^+CD8^-CD4^+$). This strategy aimed at excluding any possible interference from alternate cell types, thus optimizing accuracy, especially when characterizing rarer subsets.

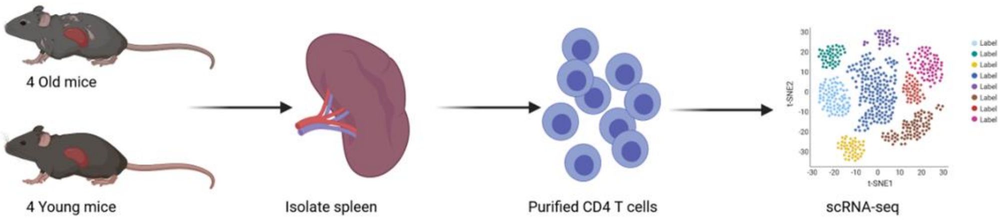


Figure 10: Schematic description of the $CD4^+$ T cells of the aging mice. Splenocytes from both old and young mice were purified using FACS to isolate $CD4^+$ T cells. The cells were then subjected to scRNA-seq and clustered into seven subsets [92].

3.1.2.2 Supercentenarian Dataset

The supercentenarian single-cell transcriptome dataset, comprised of 61,202 peripheral blood mononuclear cells (PBMCs), originates from a special cohort: seven supercentenarians (individuals over 110 years of age) and a control group of five adults between the ages of 50 and 80. As outlined in the work of Hashimoto et al. and depicted in Fig. 11, this dataset provides insights into the cellular compositions and distinctions related to age. Fig. 11 specifically offers a schematic description of the supercentenarians' cytotoxic $CD4^+$ T cells. Through the analysis of whole blood samples from people of varying ages, cell types, including T and B cells, were identified using scRNA-seq analysis of PBMCs [93].

They performed an in-depth bioinformatic analysis of this dataset, obtained from whole blood samples. The grouping into subsets was subsequently confirmed through FACS, targeting T cells via specific markers, including CD4 and GZMB. Our study also performed an in-depth bioinformatic analysis of this dataset. Multiple types of cells were distinguished by employing various computational approaches, and by focusing on pinpointing cytotoxic T cells, we were eventually able to identify the cytotoxic $CD4^+$ T cell subset precisely.

They showed a marked increase in the cytotoxic CD4$^+$ T cell subset in supercentenarians compared to the adult control group. This significant finding suggests a potential correlation between the presence of these specialized T cells and extended lifespan.

Following these discoveries, we delved deeper into the cytotoxic CD4$^+$ T cell cluster. Using the Seurat package for identification, we based our search on a series of cytotoxic genes, including GZMA, GZMB, PRF1, NKG7, and GZMH, paired with CD4 marker genes [236,257–259] . The main goal was to establish a comprehensive transcriptomic profile for this subset, pinpointing markers intrinsic to cytotoxic CD4$^+$ T cells and increasing our understanding of their broader biological implications.

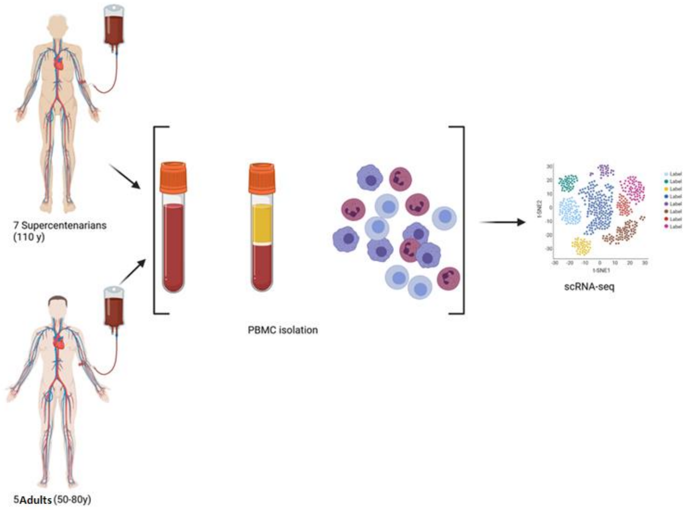


Figure 11: Schematic representation of cytotoxic CD4$^+$ T cells in supercentenarians. The figure showcases analyses of whole blood samples taken from two distinct age groups: seven supercentenarians over 110 years old and five adults between 50 and 80 years old. The scRNA-seq technique was applied to these samples to identify various cell types within the PBMCs, focusing on T and B cells. Subsequent bioinformatic analysis identified the cytotoxic CD4$^+$ T cells within the T cell subset [93].

3.1.2.3 Bladder Cancer Dataset

A comprehensive scRNA-seq dataset, the bladder cancer dataset contains 16,893 cells, focusing on CD4$^+$ tumor-infiltrating lymphocytes (TIL) from seven distinct human bladder tumors. The TILs were extracted following three treatment approaches: anti-PD-L1 treatment, chemotherapy, and no treatment (Fig. 12). The compilers of the dataset isolated both CD4$^+$ and CD8$^+$ T cells from the specified solid tumors utilizing FACS. Following this isolation, clustering was performed for each T cell type [234]. This collection contains 16,893 cells, with an average of around 2400 CD4-TIL cells per patient (Table S2).

Their analysis showed that the human bladder tumors host a variety of clonally expanded cytotoxic CD4$^+$ T cell states. Remarkably, these cytotoxic CD4$^+$ T cells can target and eliminate autologous tumors, relying on MHC class II mechanisms. The study also found that autologous regulatory T cells can potentially impede the functions of the cytotoxic $CD4^+$ T cells. A significant finding was that a specific cytotoxic $CD4^+$ gene signature could be a predictive marker for responsiveness to anti-PD-L1 treatment in bladder cancer patients [234].

Building on the foundation established in the earlier study, we used the Seurat (version 3.2.3) clustering algorithm [236] to distinguish cytotoxic $CD4^+$ cells within the CD4-TILs and zero in on the unique markers that define these cytotoxic $CD4^+$ cells.

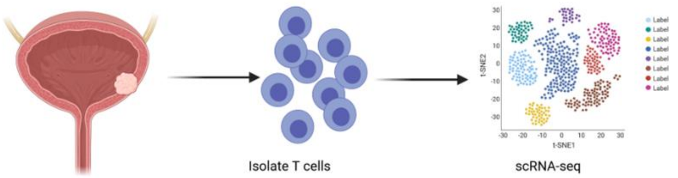


Figure 12: Schematic representation of cytotoxic intratumoral $CD4^+$ T cells in bladder cancer. The authors conducted scRNA-seq on both $CD4^+$ and $CD8^+$ T cells extracted from solid bladder tumors [234].

3.1.3 Tabula-Muris-Senis Droplet Dataset

The TMS data was described in an article titled ”A Single Cell Transcriptomic Atlas Characterizes Aging Tissues in the Mouse“ analyzed by the Tabula Muris Senis Consortium in 2019 provides a comprehensive understanding of cellular aging processes across multiple mouse tissues. One of our primary objectives was to examine how specific T cell subsets and senescent cells change with age and what implications these changes might have on the broader aging process. This atlas of aging mouse cells was designed to provide insight into the aging process across multiple tissues. Spanning 23 different mouse tissues, it enables comprehensive understanding of cellular changes as the mouse ages. The decision to span multiple age categories comes from an understanding of aging and its effects on organisms at a cellular level [16].

We exclusively used the droplet single-cell RNA sequencing dataset from the TMS data, containing 245,389 cells meticulously evaluated for quality by TMS researchers [233]. Sixteen tissues were sourced from 23 C57BL/6JN mice, incorporating both male and female specimens. The age range of these mice varied from 1 to 30 months, as detailed in Figure 13. scRNA-seq of two types was performed, droplet and FACS, from 30 mice (female and male) at six timepoints, we utilized only the droplet dataset. The cells were identified semiautomatically using the Tabula-Muris dataset as reference data [260].

Leveraging advanced bioinformatic tools and algorithms, the consortium’s researchers analyzed the scRNA-seq data, illuminating gene expression patterns, cell types, and other markers. By comparing cells from younger and older mice, they identified specific gene expression trends and pathways impacted by aging.

Their study found that the aging process manifests itself differently in different types of cells, with unique aging patterns found in different types of cells. Specific genes demonstrated variable expression patterns, with some being tissue-specific as mice matured. The research also pinpointed potential cellular markers indicative of aging. A notable decline in certain cellular functions, particularly mitochondrial activity, was evident in specific cell types. Furthermore, alterations in several signaling pathways became apparent as the mice aged, some of which had been previously associated with longevity and age-related diseases.

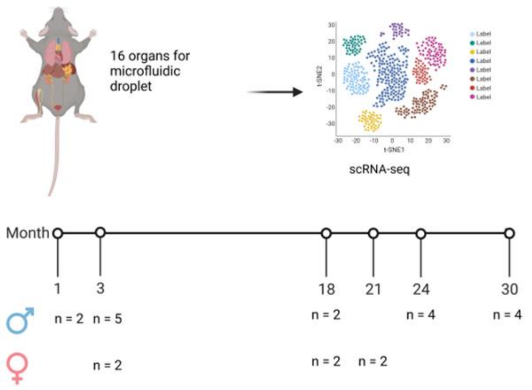


Figure 13: Schematic representation of the TMS microfluidic droplets dataset collection and analysis. The study analyzed tissue samples from 16 organs taken at six different time points from 18 males and six females. scRNA-seq was applied to each tissue and provided cumulative analysis [233].

Our analysis of the TMS dataset began by applying the *ReadH5AD()* function to the "tabula-muris-senis-droplet-official-raw-obj.h5ad" file. This raw input data contains the raw filtered data matrix and its metadata file "tabula-muris-senis-droplet-official-raw-obj-metadata.csv." As per best practices, this raw data had already undergone QC, and mitochondrial genes were excluded, consistent with previous studies that sometimes remove these genes due to their potential to introduce biases [233].

The *ReadH5AD()* function converts anndata h5ad files to Seurat objects , but in this case, the object generated was missing the "nCount_RNA" metadata. To rectify this, we reconstructed the Seurat object using the *CreateSeuratObject()* function. When using the "Counts" parameter it is essential to ensure the matrix is structured with cells as columns and genes as rows, as recommended by the Seurat team [236].

In alignment with various studies categorizing experimental animals by age, we regrouped the data into three age brackets. Subsequently, we merged these into two primary categories for enhanced statistical robustness.

Post-classification, normalization of the data was performed using the *NormalizeData()* function, an essential step in single-cell RNA-seq data preprocessing [72,237,241]. The dataset was then further subdivided by tissue, with each tissue undergoing a specific analysis. This was done in accordance with the standard Seurat workflow procedures, as depicted in Figure 9.

3.1.3.1 T Lymphocyte Subset Analysis

This section, we detail our methodology for analyzing the changes in $CD4^+$ and $CD8^+$ T lymphocytes using the TMS dataset. Firstly, genes with highly variable cell-to-cell expression were selected from every tissue in the dataset for further analysis. A mean-variable plot (mean.var.plot or MVP) was chosen to calculate the average expression of each gene and its dispersion. For each tissue, genes were meticulously selected based on their suitability to the plot. Consistent with findings by Satija et al., (2015)[237] and Stuart et al., (2019) [236], we observed that in datasets displaying high levels of homogeneity, the variability of gene expression was more pronounced than the average expression.

To refine our analysis and control for external factors, genes were then scaled and regressed out by age and mouse.id from the metadata to eliminate the batch effect of highly expressed genes resulting from aging, technical variety, and mice specificity and ensure that only the differences derived from cell types remained.

Next, we used the method described above in section 3.1.1. and for cell type annotation in the TMS dataset, we utilized the clustifyr package (version 1.0.0), referencing the Tabula-Muris data.

Once individual tissue datasets were annotated, they were combined for a cohesive analysis. From this combined dataset, a T cell subset was extracted. For these cells, the previous steps were reiterated, emphasizing regression by tissue, age, and mouse.id to identify $CD4^+$ and $CD8^+$ T lymphocytes. Following this, $CD4^+$ T lymphocyte subsets were identified, and Seurat's pipeline for these cells was once more applied [79,236,241,261]. Finally, they were annotated based on their differentially expressed genes as determined by the *FindMarkers()* function [79,236,261]. Through our methodology, we have accurately identified and annotated distinct subsets of T lymphocytes, including their age-related changes. These insightful findings will be expounded upon in the following section. Our analysis has yielded the $CD4^+$ T cell subsets, which we will compare with aging mice dataset for further investigation [92]. Furthermore, we will conduct a correlation analysis with the senescent subset to gain further insights.

3.1.3.2 Define Senescence Dataset

Our study utilized the TMS dataset to specifically outline senescent cells, facilitating subsequent correlation analysis with $CD4^+$ T cell subsets. These cells were determined through their concurrent expression of CDKN2A/p16INK4α and CDKN1A/p21 mRNA, a widely accepted criterion in the field [5,15,108,194,221,227,262–264]. This accurate approach enabled the identification of a total of 1,707 senescent cells, representing a pivotal step in unraveling the complications of cellular aging and its significance for our research. Additionally, we conducted a differential gene expression analysis comparing senescent and non-senescent cells, further illuminating the molecular distinctions between these cell populations. Finally, we explored the specific molecular pathways that characterize senescent cells within each age group, enhancing our understanding of the aging process.

3.2 Construction of pseudotime trajectory

To construct pseudotime trajectory for cell development, we purchase monocle3 version 0.2.3 for R on the Seurat object [265,266]. To convert the Seurat object into an appropriate momocle3 object, we applied *as.cell_data_set()* function on the Seurat object and adjusted clusters and UMAP projection by *the cluster_cells()* function. Next, we constructed a plot applying our subsets annotated with Seurat. This graph forms a base for the *learn_graph()* function to explore the cells' biological features and construct trajectory. We chose a subset as a root to build the pseudotime and projected it above the UMAP.

3.3 Statistical Analysis

For statistical analysis and data visualization in this study, we applied various software packages to enhance our research process. Specifically, we utilized the ggpubr package (version 0.4.0) to facilitate comparisons between different age groups [267]. Additionally, a significant portion of the graphical representations in our results section was created using ggplot2 (version 3.3.3) [268]. Furthermore, we utilized EnhancedVolcano (version 1.6.0) to generate volcano plots and Seurat (version 3.2.3) to produce UMAPs and heatmaps.

To compare two groups, we applied either the Wilcoxon signed-rank test for cell subsets or the student t-test for age groups, considering the normalized cell counts per mouse. This approach allowed us to accurately assess the differences and similarities between the two groups in a statistically sound manner. In addition, we apply ANOVA for the comparation of senescent cells in each mouse among the all six ages. Also, to explore gene sets involved in aging and their relation to biological pathways, clusterProfiler (version 3.16.1) was used along with the Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) databases. For molecular biology research, GO and KEGG are essential tools. GO focuses on analyzing gene, cell, and molecular functions, whereas KEGG examines metabolic, regulatory, and disease pathways [269–271].

Finally, for assessing correlations between the two groups, we opted for Pearson’s correlation, which was chosen over Spearman’s, because the data did not follow a normal distribution.

4 Results

4.1 Initial Analysis of Datasets

4.1.1 Aging Mice Dataset Analysis

We computed 24,007 CD4$^+$ T cells from eight male mice, from the young (2-3 months old) and old (22-24 months old) age groups. These cells, extracted from the spleen, are listed in Table S1.

The UMAP projection distinctly depicts the old cells in light blue and the young cells in brown (Fig. 14A). The young group consists predominantly of naïve CD4$^+$ T cells (Fig. 14G). Marker genes are instrumental in distinguishing these subsets. For instance, Cd44 is a recognized marker for the memory, effector, exhausted (gold), and TEM (purple) subsets (Fig. 14B). The identification or utilization of Cd44 as a marker in the context of these T cell subsets explored in several studies, including those by Gattinoni et al. (2011)[272], Henning et al. (2018)[273], Sen et al. (2016)[274], and Wherry & Kurachi (2015)[116]; Gzmk for the cytotoxic (green) subset (Fig. 14C) [257,275]; Foxp3 for the regulatory aTreg (red) and rTreg (pink) subsets (Fig. 14D) [276,277]; Pdcd1 for the activated and exhausted cells exhausted (gold) and aTregs (red) subsets (Fig. 14F) [278,279]; and Sell for the naïve rTreg (pink), naïve (turquoise), and naïve_Isg15 (blue) subsets (Fig. 14E) [177]. The comprehensive portrayal of all seven CD4$^+$ T cell subsets is provided in Fig. 14G.

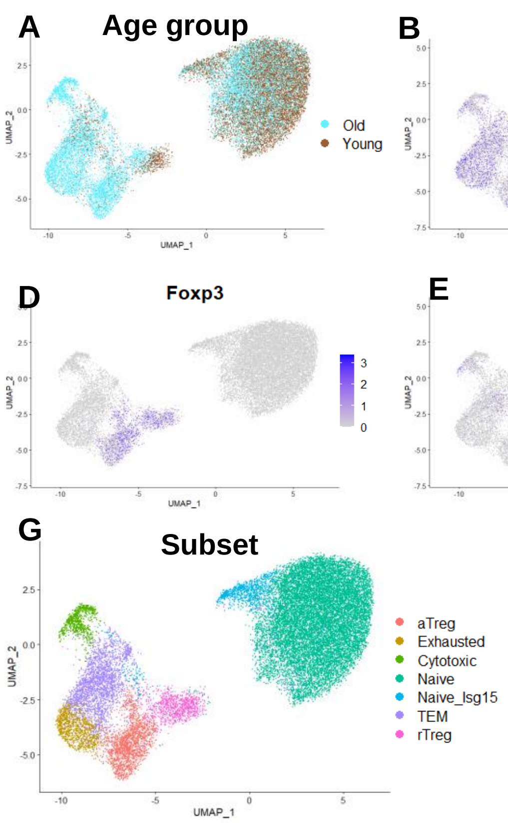


Figure 14: UMAP projection of the CD4$^+$ T cells of mice in the aging mice dataset grouped by age group and CD4$^+$ subsets. A) UMAP projection of CD4$^+$ T cell distribution between old and

young age groups. B-F) UMAP projection showcasing the expression of five marker genes in the aging mice dataset, with the purple scale on the right indicating the level of gene expression: B) Cd44, a marker of memory T cells; C) Gzmk, a cytotoxic marker; D) Foxp3, a marker of regulatory cells; E) Cd62l/Sell, a marker of naïve cells; F) Pdcd1, a marker of exhaustion; and G) UMAP projection of CD4^{+} T cell distribution of CD4^{+} T cell subsets.

Our analysis showcased the diversity of the CD4^{+} T cell subsets and the differences between the subsets of the young and old mice. Notably, the old mice manifested an evident accumulation of activated regulatory, exhausted, and cytotoxic CD4^{+} T cell subsets. Conversely, their younger counterparts predominantly exhibited naïve CD4^{+} T cells [92].

The utilization of this dataset in our study bolstered the verification process for identifying CD4^{+} T cell subsets in the TMS dataset. Furthermore, our specific focus on the cytotoxic CD4^{+} T cell subset contributed to a more comprehensive understanding of the distinct features of this subset. This comparative analysis between datasets allowed us to effectively validate our findings, thereby enhancing our results' credibility.

4.1.2 Supercentenarian Dataset Analysis

The 2019 study of Hashimoto et al. investigated the cellular mechanisms prevalent in supercentenarians who boast impressive immune features and longevity [93]. Their dataset consists of 61,202 PBMCs derived from seven supercentenarians (individuals 110 years old or older) and five controls (individuals 50-80 years old), which were clustered and annotated according to marker genes into 9 clusters (Fig. 15).

Unlike Hashimoto's team, we were particularly interested in the specific subset of cytotoxic CD4^{+} T cells [93]. When T cell clusters (presented in purple in Fig. 15A) with expression of both CD3D, one of the CD3 subunits (Fig. 15B) and T Cell Receptor Alpha Constant (TRAC) (Fig. 15C) were isolated, 25,138 cells were identified (Fig. 15D). There was an average of 2,094.83 cells per person. This average cell count per person is notable as it reflects the substantial representation of the cytotoxic CD4^{+} T cell subset in the analyzed supercentenarian dataset, underscoring the robustness and significance of our study's focus on this specific immune cell population.

To refine our analysis and ensure the specificity of our investigation into cytotoxic CD4^{+} T cells, we further filtered the dataset to exclude cells expressing CD8A/B and TRDC. It is important to note that CD8A/B and TRDC are markers associated with CD8^{+} T cells and other T cell subsets, and their presence in the dataset indicates the potential presence of these non-CD4^{+} T cell populations [280]. This decision was driven by the need to isolate and examine the cytotoxic CD4^{+} T cell subset with precision, eliminating potential confounding factors and ensuring a focused exploration of this immune cell population (Fig. 15E and Fig. 16) [281–283].

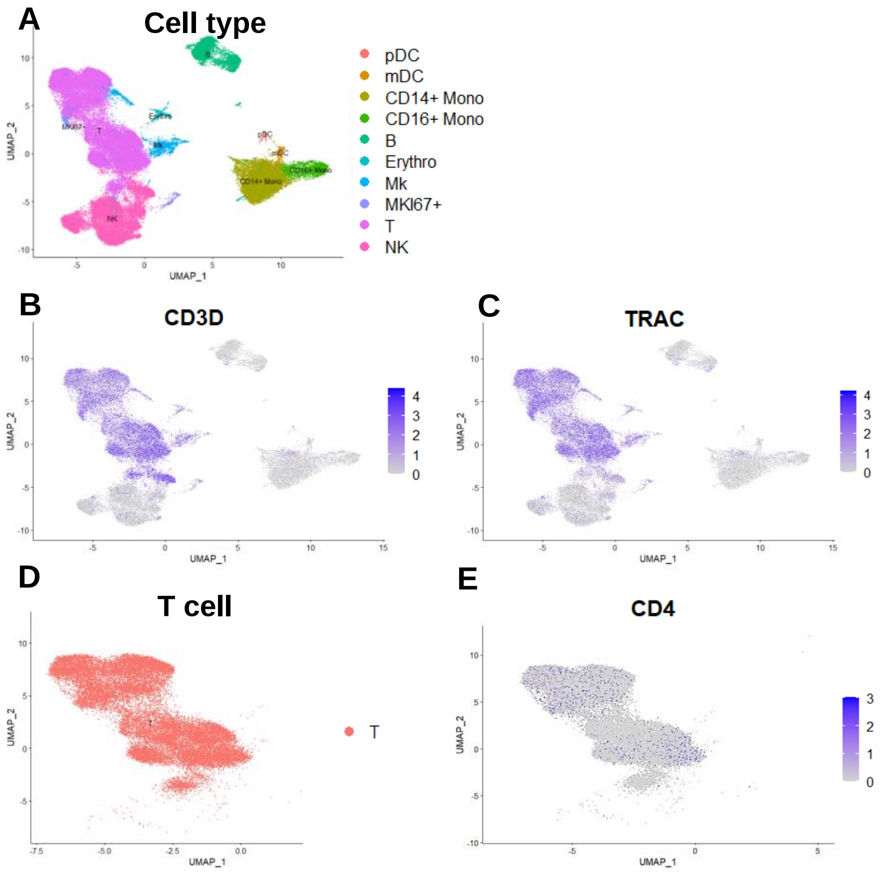


Figure 15: UMAP projections of PBMCs from Hashimoto et al.'s supercentenarian dataset (2019). A) UMAP projection by cell type; B) UMAP projection of CD3D gene expression; C) UMAP projection of TRAC gene expression; D) UMAP projection of T cell subset within PBMC; E) UMAP projection of CD4 gene expression in T cells [93].

Those $CD4^+$ T cells were clustered into four distinct subsets based on marker genes. Each subset was then annotated according to specific marker gene expressions that were identified as: naïve, presenting high CCR7 (Fig. 16F) and SELL (Fig. 16E) [177]; regulatory T cells expressing FOXP3 (Fig. 16B) [276,277,284]; TEM expressing low CCR7 (Fig. 16F) and SELL (Fig. 16E) [178,272]; and cytotoxic expressing GZMK (Fig. 16C) and GZMB (Fig. 16D) [275,285,286].

Notably, cytotoxic $CD4^+$ T cells are heterogenic, expressing either GZMK or GZMB (Fig. 16A, 16D-16E), a feature that may affect their functional diversity [257,275,285]. We observed that there is a distinct separation between the cytotoxic $CD4^+$ subset and the rest of the $CD4^+$ subsets, a differentiation that reflects the complexity of the T cell immune response (Fig. 16). Furthermore, the distinct cauterization between older adults and supercentenarians presented in Fig.

16B suggests age-related alterations in CD4^{+} T cell subsets, a phenomenon previously observed in aging studies [92–94,175].

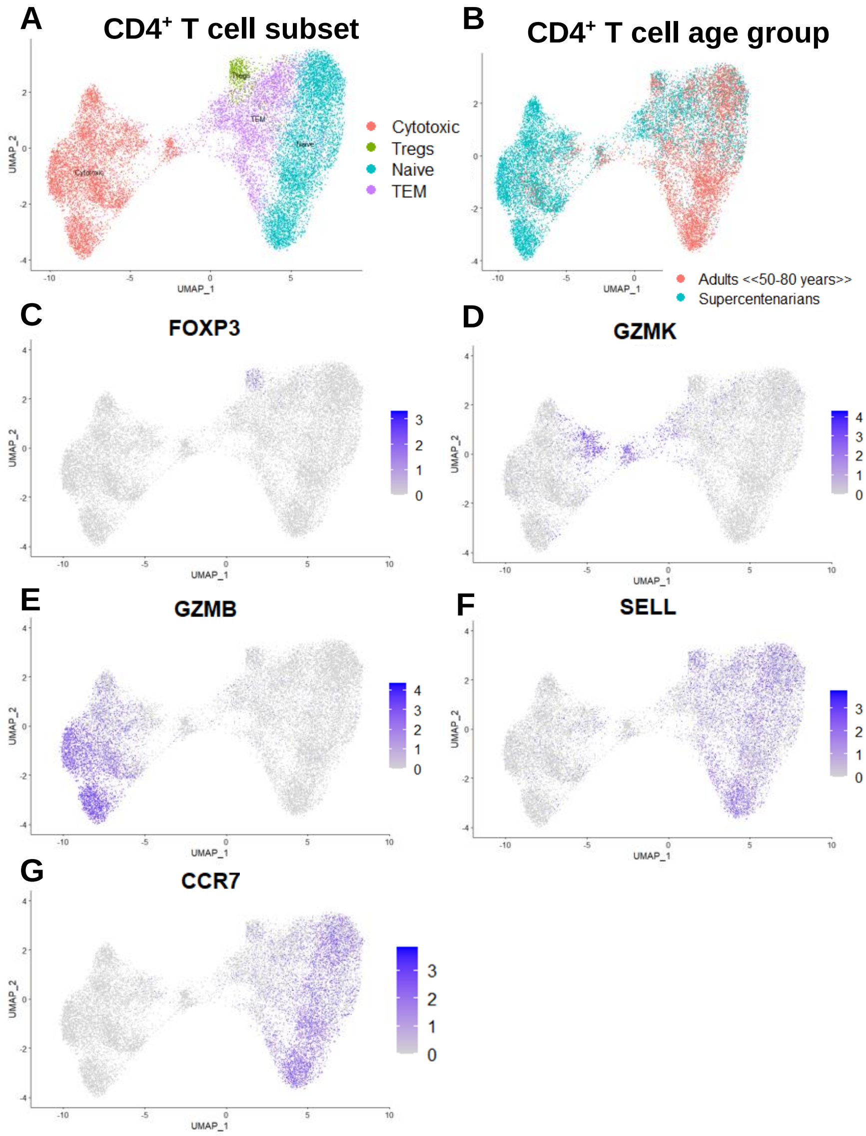


Figure 16: UMAP projection of CD4^{+} T cells from the Hashimoto's Supercentenarian dataset. A) UMAP projection of CD4^{+} T cells by subset, B) UMAP projection of CD4^{+} T cells by age groups, C) UMAP projection of FOXP3 gene expression; D) UMAP projection of GZMK gene expression; E) UMAP projection of GZMB gene expression; F) UMAP projection of SELL gene expression; and G) UMAP projection of CCR7 gene expression. The purple scale represents gene expression levels.

4.1.3 Bladder Cancer Dataset Analysis

We performed in-depth analysis using Seurat software, specifically version 3.0.2, to gain insights into the data. On average, each person in the dataset exhibited average of 505.57 cytotoxic CD4^{+} T cells, as detailed in Table S2. Through this analysis, we were able to identify and categorize the cells into 16 distinct clusters [236]. Our subsetting approach differed from the original research, which categorized subsets into $CD4_{GZMK}$, $CD4_{IL2RAHIGH}$, $CD4_{IL2RALOW}$, $CD4_{ACTIVATED}$, and others based on specific marker genes. In our study, we created distinct subsets, including naïve, TEM, Central Memory T cells (TCM), regulatory, and cytotoxic CD4 T cell subsets, based on different marker gene expressions, as described in the following paragraph. Then we grouped the clusters into distinct subsets based on their marker gene expressions. This categorization included activated and resting regulatory subsets expressing FOXP3 (Fig. 17C) [187,277,284]; TCM cells expressing CCR7 (Fig. 17E) and SELL (Fig. 17D) [287]; TEM cells uniquely marked by CCR7 expression (Fig. 17E) and lack of SELL (Fig. 17D) [178,287]; and a cytotoxic subset which exhibited cytotoxic genes such as GZMK (Fig. 17A), EOMES, PRF1, and GNLY [288,289].

Further examination of the cytotoxic $CD4^+$ T cell subset showed that it was categorized into five distinct clusters, mainly split between GZMK cytotoxic $CD4^+$ T cells (Fig. 17A) and GZMB cytotoxic $CD4^+$ T cells (Fig. 17B) categories. Like Hashimoto's research on supercentenarians [79], our findings revealed heterogeneity in the cytotoxic $CD4^+$ T cells present in the population. Pseudotime projection on UMAP projection suggests the differentiation of the Cytotoxic subset from the TEM subset (Fig. 18A-B). Also, the volcano plot comparing those two subsets presents the DGE for the Cytotoxic subset, mainly characterizes cytotoxic function CCL4, CCL5, GZMK, GZMB, NKG7, GNLY, PRF1 and for TEM are mostly characterize $CD4^+$ T cells CCR7, IL7R, SELL (Fig. 18C). The total percentage of all detected Cytotoxic $CD4^+$ T cells is 20.95% when the average percent is 21.43% (Table S2).

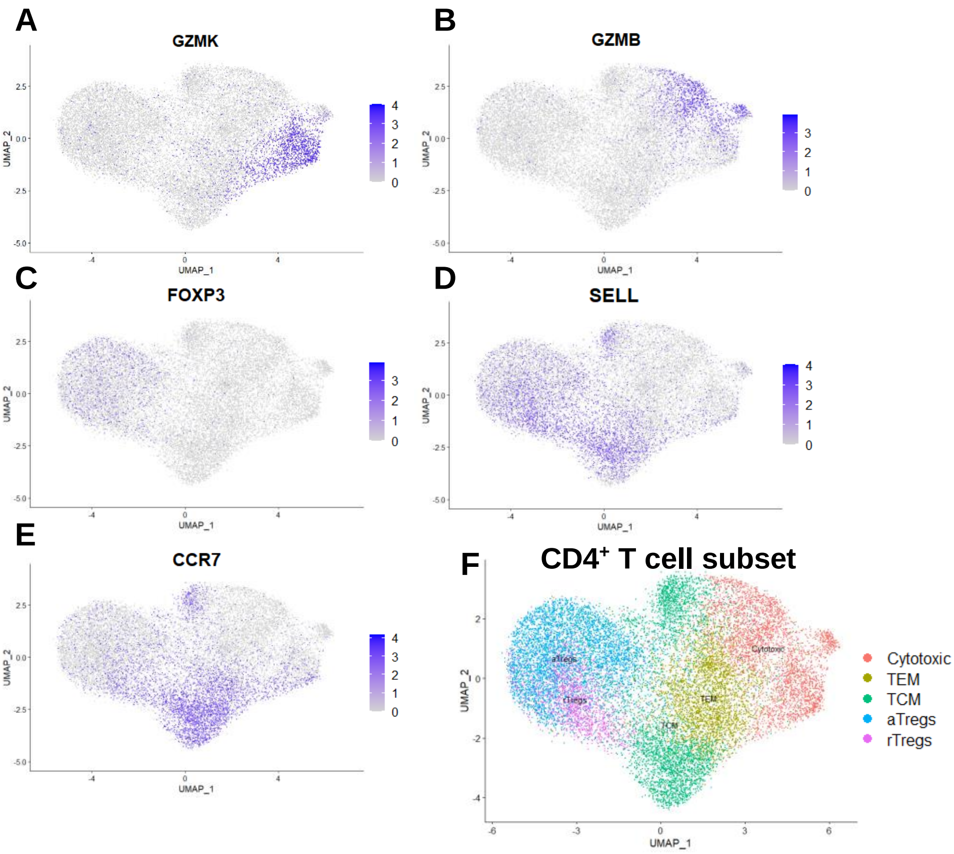


Figure 17: UMAP projection of the CD4 T cells in the bladder cancer dataset. A) UMAP projection of GZMK gene expression; B) UMAP projection of GZMB gene expression; C) UMAP

projection of FOXP3 gene expression; D) UMAP projection of SELL gene expression; E) UMAP projection of CCR7 gene expression; and F) UMAP projection of CD4^{+} T cells by subset. The purple scale represents gene ex-pression levels.

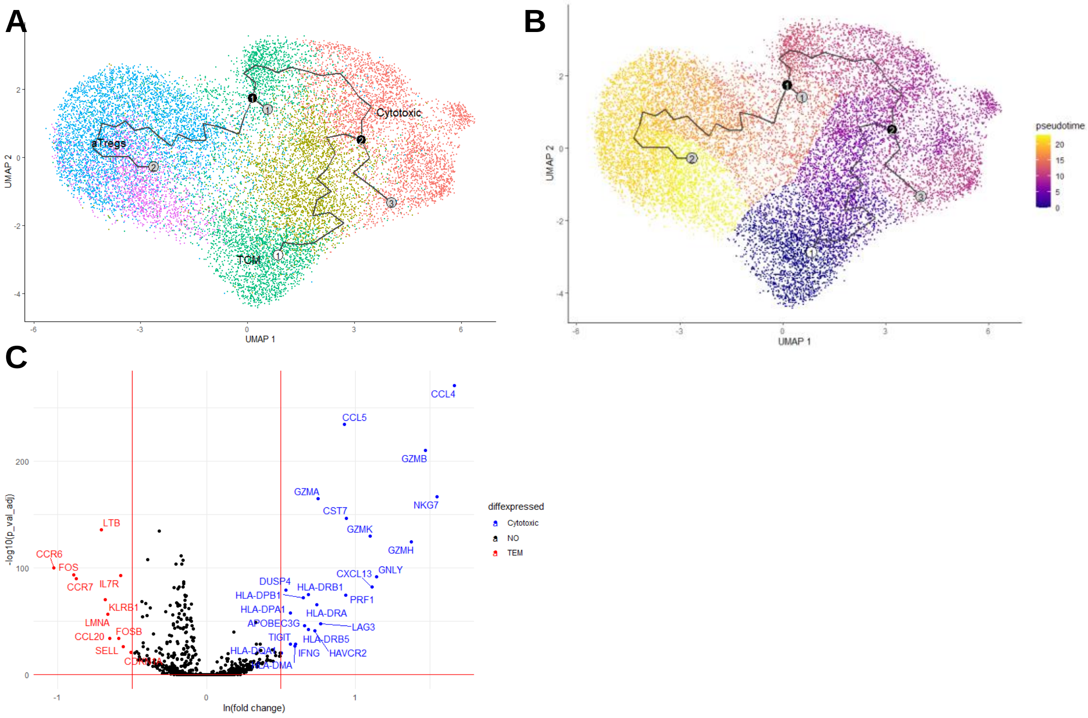


Figure 18: Pseudotime trajectory projected on UMAP projection and volcano plot visualization of DGE analysis between Cytotoxic and TEM CD4^{+} T cells. A) Pseudotime trajectory projected upon UMAP projection grouped by subsets as in Fig. 19F, B) Pseudotime trajectory stained by an abstract time units that present changes in cells as a function of its differentiation, C) Volcano plot of DGE analysis of Cytotoxic (blue) versus TEM (red) subsets. "NO" – not differentially expressed (black).

4.1.4 Tabula-Muris-Senis Droplet Dataset Analysis

In our analysis of the comprehensive TMS dataset, the 10x Genomics platform was employed for sequencing, a method known for its reliability in scRNA-seq [73,233]. This dataset consists of 245,389 cells sequenced using the droplet-seq method. The TMS dataset, derived from 23 mice, spans six distinct age categories and encompasses both genders [233]. The tissues sampled from the 16 organs were subsequently subdivided into two primary age groups: Young (comprising tissues from mice aged 1-3 months) and Old (comprising tissues from mice aged 18-30 months) (Table S5) [233].

To refine our analysis, we initially segregated the mice into three age groups, separating the old mice (18-24 months) from the supercentenarian mice (30 months), acknowledging the potential difference in their senescent cell accumulation, a significant factor in the aging process [16]. We refer to this division as the first reorganization. It was employed for the senescent cells analysis provided in subsection 4.4., titled “Computational Characterization of Key Cytokines Involved in Cellular Senescence”.

Subsequently, during the second reorganization, we conducted further refinements to our dataset. This involved a closer examination of the age groups, with a particular focus on comparing the young (1-3 months) and old (18-30 months) groups. This allowed us to investigate any nuanced differences between these two primary age categories, providing valuable insights into the aging process. The results of this second reorganization revealed a modest increase in the average cell count per mouse within the old group, as compared to the young group. Notably, a t-test yielded a p-value of 0.096, suggesting a trend toward a higher number of sequenced cells in the old mice category. This trend is visualized in Fig. 19A, where the old group is depicted in brown, and the young group in light blue. However, it is important to note that the data also indicated a broader dispersion in the old group. This analysis was conducted using data sourced from 16 distinct tissues (Fig. 19B and Table S4), with six of these tissues sequenced across all six age timepoints [233].

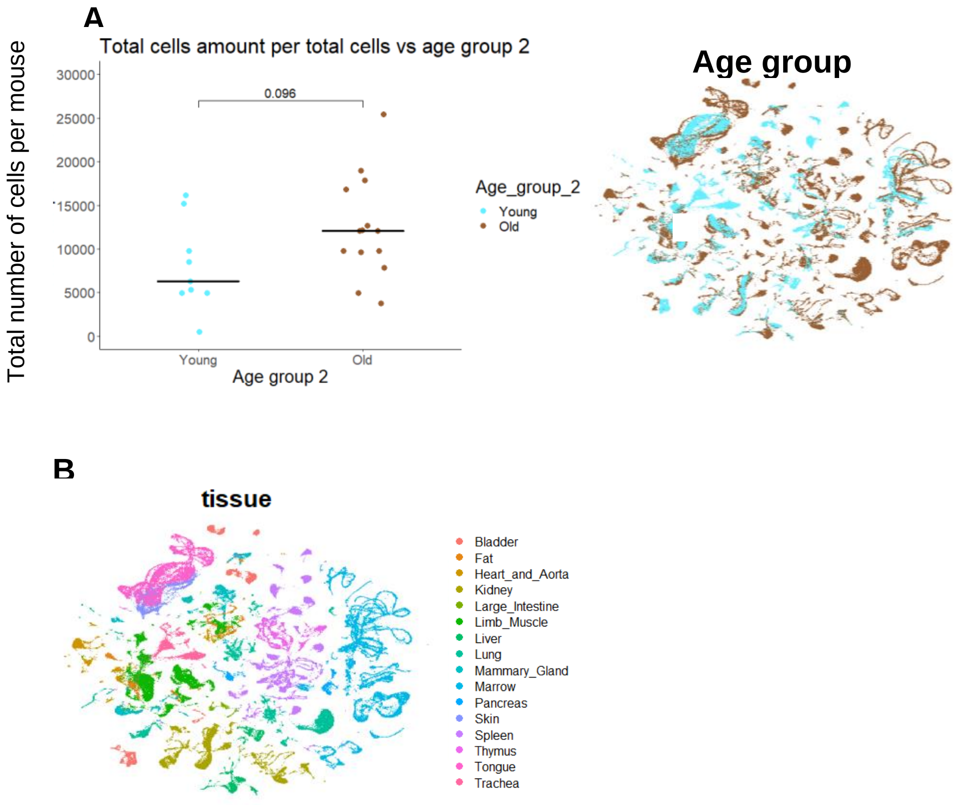
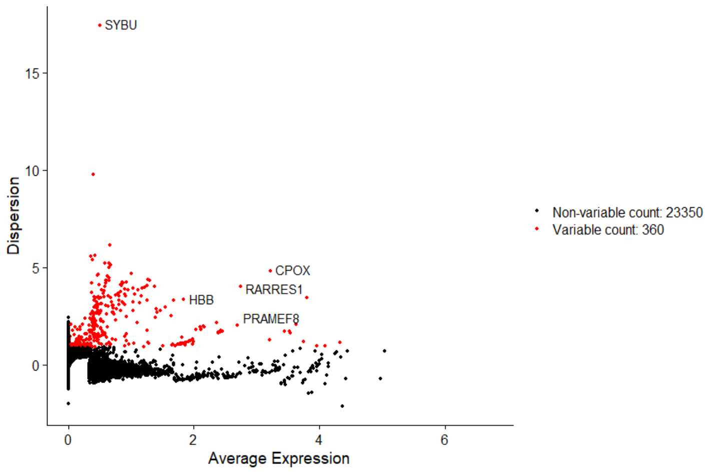


Figure 19: Presentation of TMS Dataset Across Age Groups and Tissue Types. A) Age Group Analysis: Left - Comparative total cell count in young (1-3 months) and old (18-30 months) mice. Right - UMAP projection of total cells by age groups: young (light blue), old (brown). B) Tissue-Specific Distribution: UMAP projection of cells from A Right, grouped by tissues of origin.

4.2 Observed Similarities Between the Datasets' Cytotoxic Subsets

In this section, we delve into an in-depth analysis to compare the cytotoxic CD4$^+$ T cell subsets identified across different species and ages groups. Our aim is to uncover key insights into the similarities and differences within these subsets, shedding light on the impact of aging and species-specific variations. To achieve this, we divided our data into two distinct age groups: an old group (Supercentenarians in human and Old in mice) and a young group (Adult in human and Young in mice), allowing us to perform a detailed comparative analysis. We combined data from two primary sources: Hashimoto's Supercentenarian dataset representing human aging data [93] and the aging mice dataset of Elyahu et al. [92], providing a comparative look at age-associated changes in T cell subsets. To shed light on disease-specific differences, we incorporated data from the bladder cancer dataset related to the cytotoxic subsets [234]; the results of the merged Cytotoxic CD4$^+$ T cells dataset from this aspect of our analysis are provided in the Appendix section (Fig. S1), sowing largely clusterization of bladder cancer Cytotoxic CD4$^+$ T cells with human aging Cytotoxic CD4$^+$ T cells in UMAP (Fig. S1B) and a distribution in gene sets presented in PCA (Fig. S1A). Our analysis leverages the Seurat package to dissect the merged supercentenarian and aging mice datasets, revealing a limited set of 360 variable genes out of 23,710 genes when analyzed using the MPV method (Fig. 20). MVP is known for its efficacy in assessing scRNA-seq data across different conditions [236,237].



	Low cutoff	High cutoff
dispersion	0.9	infinity
mean	0.0125	infinity

Figure 20: Plot of Mean Variability across Combined Dataset (Supercentenarians and Aging Mice). Highlighting 360 Highly Variable Genes out of 23710 total genes sequenced per cell, with a Dispersion Cutoff of 0.9 to Infinity and a Mean Cutoff of 0.0125 to Infinity.

As we progress, we will discuss the observed gene sets in both human and mouse datasets and highlight distinct expression profiles. Specifically, we will explore variations in the expression of genes such as GZMH, GZMA, GZMK, GZMB, PRF1, GNLY, KLRB1, CCL5, NKG7, and

EOMES between these species (Fig. 21). These genes play crucial roles in T cell cytotoxic function and may hold evolutionarily conserved significance. Interestingly, the gene sets identified in the human dataset had strong parallels with known human cytotoxic markers (Fig. 22). In contrast, the mouse dataset manifested distinct gene sets, possibly indicating species-specific variations in immune functions (Fig. 21A). This distinct separation between the human and mouse datasets is further emphasized by the UMAP projection (Fig. 21B), providing additional support for the conclusion regarding species-specific immune function variations. Similar observations are reflected in the results presented in Fig. S1.

Differential gene expression (DGE) analysis comparing the human and mouse datasets highlighted their distinct expression profiles. Human cytotoxic $CD4^+$ T cells exhibited elevated expression levels of GZMH and GZMA. In contrast, GZMK, GZMB, and EOMES exhibited consistent expression across both species, with EOMES having an ln(Fold change) of 0.46 (Fig. 22). Additionally, the PRF1 gene had marginally higher expression in humans, whereas GNLY, KLRB1, CCL5, and NKG7 were notably upregulated. This consistency in expression patterns across species underscores the universality of certain age-associated alterations in T cells. Specifically, slightly higher expression of GZMH (ln(Fold change) = 0.4), GZMB (ln(Fold change) = 0.48), and ID2 (ln(Fold change) = 0.35) in the older groups for both species. The increased expression of these genes has previously been linked with increased cytotoxic potential and T cell differentiation in the context of aging [286,290,291]. Notably, GZMK and GZMB appear to be age-specific, while the old group showed significant upregulation of GZMB, younger individuals' T cells predominantly expressed GZMK, as reflected by an ln(Fold change) of 0.9 (Fig. 23). Our study underscores the evolutionarily conserved roles of GZMK, GZMB, and EOMES in T cell cytotoxic function based on their cross-species expression.

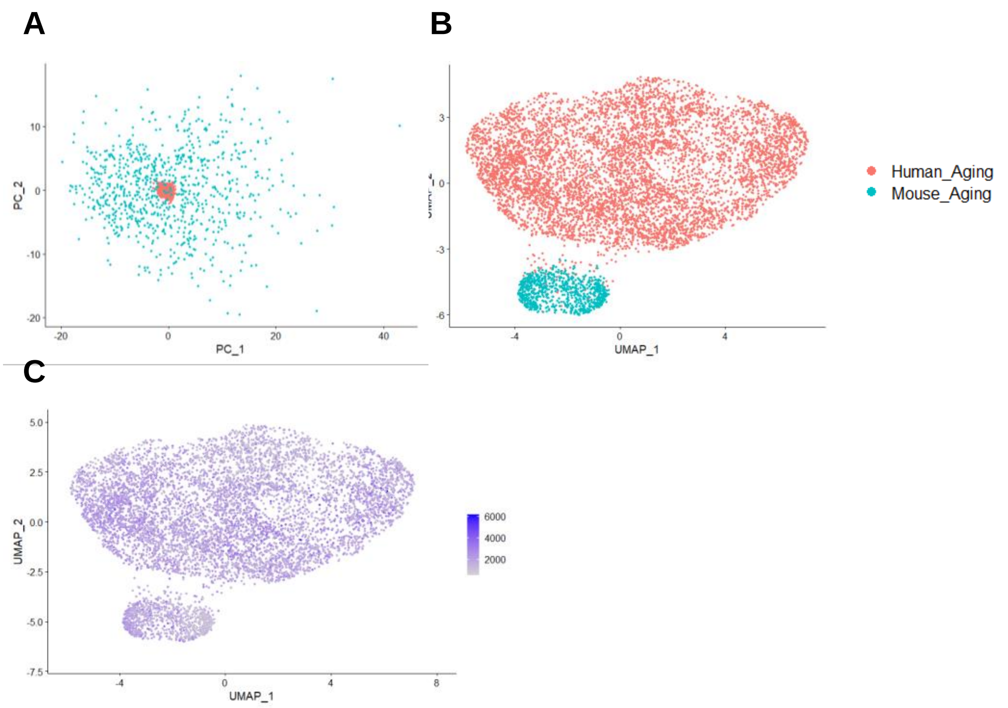


Figure 21: Comparative Analysis of Human and Mouse Cells Using Two-dimensional Projections. A) PCA illustrates a dense cluster of human (light red) cells distinct from the more dispersed clusters of mouse (light blue) cells; B) UMAP reveals a clear differentiation between the human (light red) and mouse (light blue) cell clusters; C) UMAP visualization of UMI levels in each cell, facilitating a quantitative comparison between human and mouse cells.

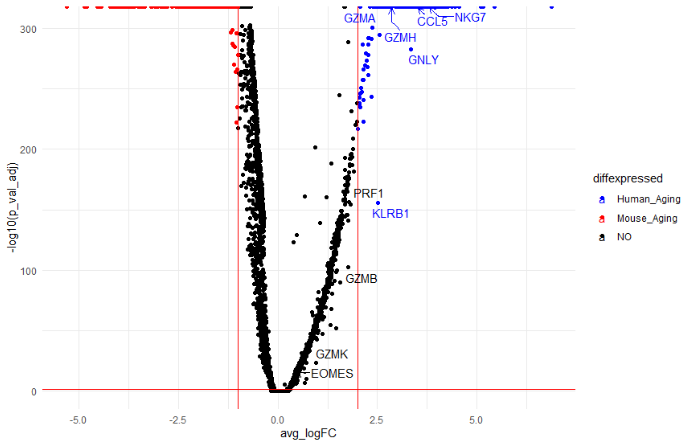


Figure 22: Volcano Plot of DGE Analysis in Cytotoxic CD4$^+$ T Cells: Comparing Species Variations in Gene Expression. The volcano plot illustrates the DGE analysis of the combined cytotoxic CD4$^+$ T cells dataset from both human (blue) and mouse (red). The plot highlights the differentially expressed genes, aiding in distinguishing between human and mouse gene expressions in cytotoxic CD4$^+$ T cell types. "NO" refers to non-differentially expressed genes (black).

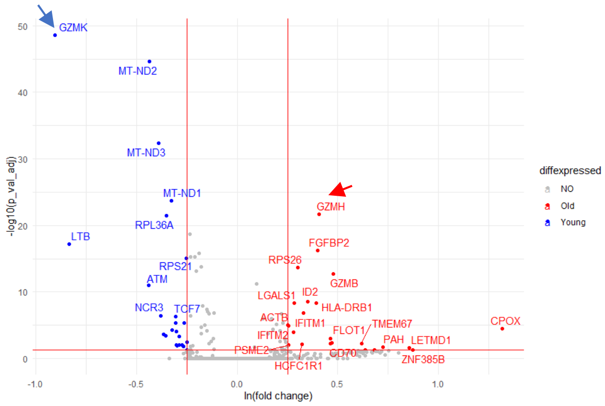


Figure 23: Volcano Plot of DGE Analysis in Cytotoxic CD4$^+$ T Cells: Comparing Age throughout Species Variations in Gene Expression. This volcano plot represents a Differential Gene Expression (DGE) analysis conducted on a combined dataset of cytotoxic CD4$^+$ T cells derived from both human and mouse samples. The plot distinctly color-codes the gene expression variations between older (red) and younger (blue) individuals within each species. Additionally, it highlights the differentially expressed genes that aid in distinguishing the age-related gene expression variations in cytotoxic CD4$^+$ T cell types across both species. The term 'NO' refers to the non-differentially expressed genes, depicted in gray, providing a baseline for comparative analysis.

Our comprehensive analysis, which incorporated data from the datasets mentioned earlier, identified 1,230 shared genes in the cytotoxic CD4$^+$ T cell subsets (Fig. S2). It shows some zero fills in the Venn diagram which could mean lack of intersection between the gene sets or a complete overlap. These findings resonate with previous research that emphasized the conservation of core genes in cytotoxic responses [292,293]. It is crucial to understand the implications of this shared gene set. For instance, while EOMES and RUNX3 were notably absent, their absence hints at their potential association specifically with aging, as underscored in prior research [294,295].

To provide clarity on the genes we observed, some commonly identified ones include NKG7, GNLY, GZMH, CCL5, CD3D, and GZMK, which can be further explored in Fig. S1-S2. To ensure a harmonized cross-species analysis, we employed human orthologs as substitutes for mouse genes. However, it is crucial to note that our consolidated dataset, derived from three distinct datasets, might present an appearance of specific genes being upregulated due to varying query gene sets, a complexity also hinted at by Wagner et al., 2018 [296].

4.3 Analysis of CD4$^+$ T Cell Identification and Clustering in the TMS Dataset and its Subsequent Validation

4.3.1 Detection of T Cells

In the TMS dataset, T cells in each tissue were identified and annotated using the clustifyr package (version 1.0.0), with the Tabula-Muris dataset serving as the reference standard [260,297].

In addition to reference data we based the identification on canonical T cell markers, CD3, Cd3d, Cd3e, Cd3g, and Cd247, well established in prior research as integral components of the T-cell receptor complex [298–300]. Cells that have shown expression of the markers are designated as 'T' and can be viewed in the UMAP representation of the TMS dataset, where they are highlighted in red (Fig. 24A).

In our analysis of the TMS dataset, we identified 26,326 T cells out of 245,398 cells. However, in extensive single-cell sequencings, misclassification due to overlapping gene expression patterns is not uncommon [301–303]. Consequently, some cells initially suspected to be T cells were later reclassified as misclustered based on the Tabula-Muris reference data [260], as shown in Fig. 24E and Table S6. To differentiate between the main T cell subsets, we employed canonical markers for $CD4^+$ and $CD8^+$ T cells, specifically CD4 (Fig. 24B), CD8a (Fig. 24C), and CD8b1 (Fig. 24D). These markers have been recognized in prior research to enable precise clustering of T cells into categories: $CD4^+$, $CD8^+$, double positive T cells (DP) and double negative T cells (DN), consistent with the findings of Manh et al. [304].

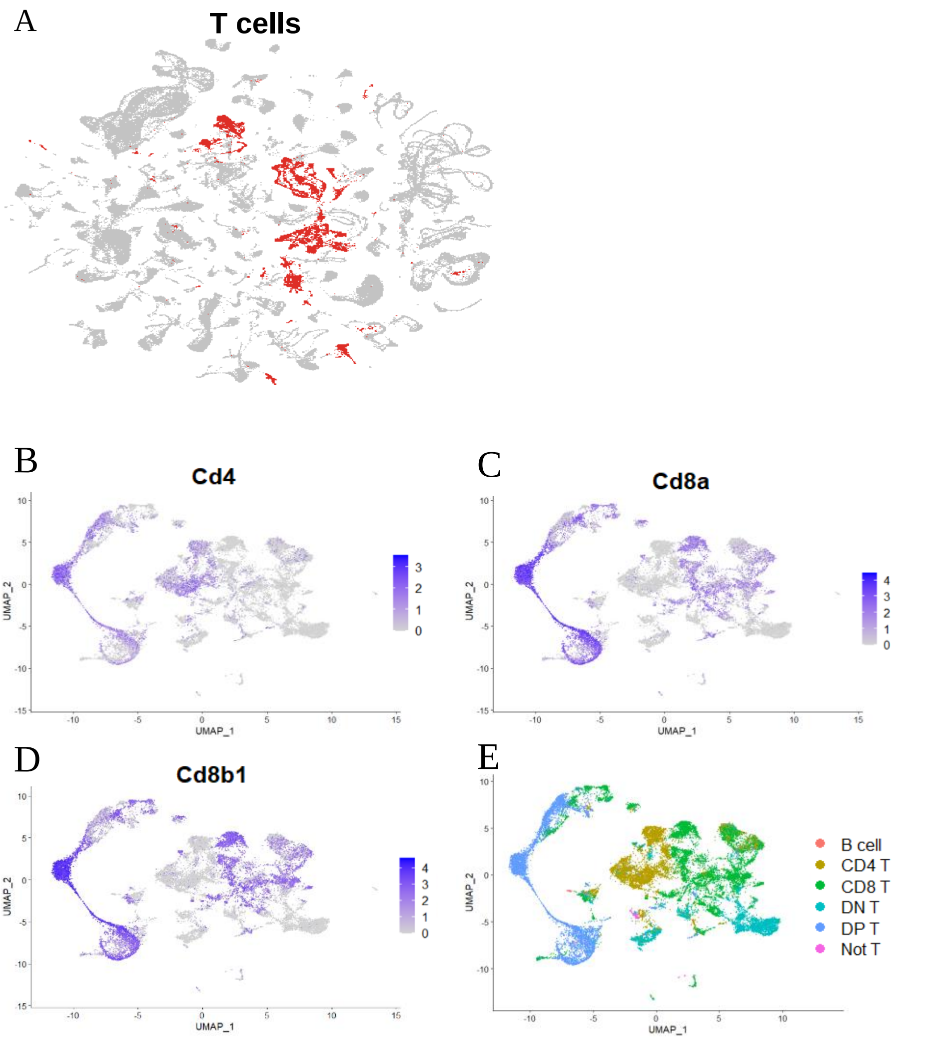


Figure 24: Visualization of T cells in the TMS dataset using UMAP projections. A) Overall distribution of T cells (red) within the total cell population of the TMS dataset; B) UMAP representation highlighting the expression of the CD4 marker in T cells, visualized using a purple

scale for gene expression intensity; C-D) UMAP projections showcasing the expression of CD8 markers in T cells, with gene expression intensity represented by a purple scale; E) Specific UMAP projection of T cell types. Pink represents "Not T" cells, blue indicates DP T cells, light blue showcases DN T cells, green signifies CD8 T cells, olive green denotes CD4 T cells, and light red is for B cells.

The clustering of T cells was executed using a two-step process, similar to methodologies used in other scRNA-seq studies [236,305]. Initially, the cells were grouped using standard clustering algorithms. This was followed by the re-clustering of heterogeneous clusters separately, specifically to differentiate between $CD8^+$ and $CD4^+$ T cells and to establish the CD4/CD8 ratio, a well-documented indicator of immune system aging (Fig. 25 and Table S7).

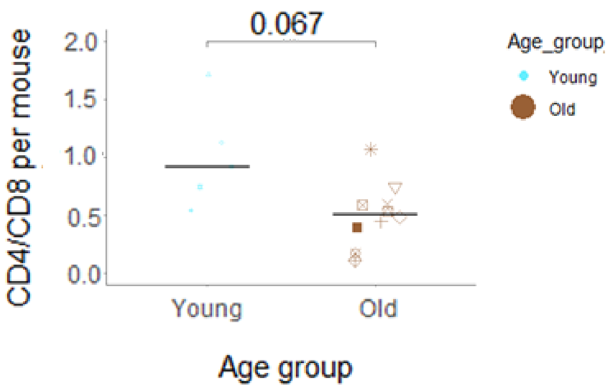


Figure 25: CD4/CD8 ratio per mouse. Each mouse is designated with a unique sign. Light blue signifies young mice, while brown indicates old mice. Black horizontal lines highlight the average ratios for each respective group.

The primary aim of our clustering process was to delineate $CD4^+$ T cell subsets accurately, as demonstrated in Fig. 26A. Our approach drew from methodologies established in previous studies [73,79,236]. We based the classification of these subsets on gene markers and their functional roles as identified in prior research [153,177,179,277,288]. Fig. 26B presents the expression of all $CD4^+$ T cell subsets across both the young (1-3 months) and old (18-30 months) age groups.

Our data showed parallels with aging human cohorts, where the naïve T cell population decreases with age. This was evident in our aging mice, where a significant reduction in the naïve subset with age was observed (Fig. 26C). Similar trends in aging populations, our data also reflected an increase in the TEM subset (Fig. 26G) and increases in the exhausted and aTreg (activated regulatory T cell) subsets (see Figs. 26H and 26F).

However, not all subsets aligned with these aging trends. The rTregs subset did not exhibit significant age-related shifts, corroborating with some prior research [92], as evident by a p-value of 0.86 in Fig. 26D. While the cytotoxic subset showed an upward trajectory, it was not statistically significant, a phenomenon observed under certain conditions, as shown in Fig. 26E. The naïve_Isg15 subset, having a limited cell count (Fig. 26A), was excluded from statistical evaluation.

Such variability in T-cell subset responses highlights the complexities of aging-related immune responses, a phenomenon detailed in previous studies on immunosenescence [306]. Notably, while most subsets exhibited slight variations, the cytotoxic and exhausted subsets, crucial for immune system function, displayed marked distribution of cell proportion in older mice presented in Fig. 26E and 26H respectively.

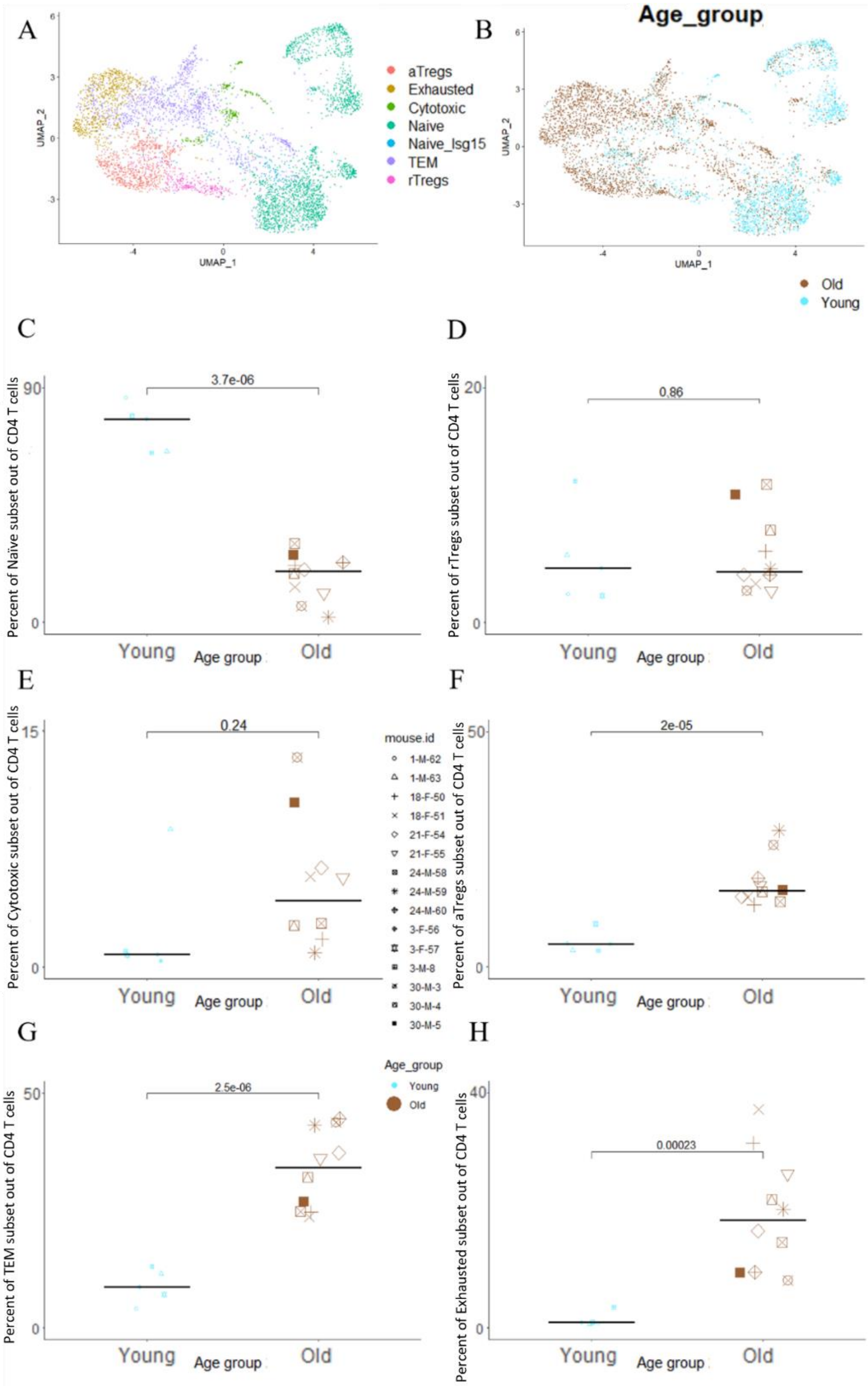


Figure 26: Analysis of CD4$^+$ T cell subsets within the TMS dataset. A) UMAP projection showcasing the seven identified CD4$^+$ T cell subsets: aTregs, rTregs, Exhausted, Cytotoxic, Naive,

Naive_Isg15, and TEM; B) The same UMAP projection as in (A), now color-differentiated by age groups— young (1-3 months, light blue) and old (18-30 months, brown); C-H) Comparative graphs elucidating the nuances among CD4⁺ T cell subsets. (C) Naive; (D) rTregs; (E) Cytotoxic; (F) aTreg; (G) TEM; (H) Exhausted. These visual representations convey a comprehensive comparison between the young and old age groups concerning the specified CD4⁺ T cell subsets.

In our initial hypothesis section, we posited a potential accumulation of aTregs in older individuals. T regulatory cells have two states, resting and activated when activated state presents a high level of activation markers, such as CD25 and CTLA-4 in addition to the regulatory markers. aTregs are responsible for the suppression of immune reactions. To validate this, we compared the ratio of aTregs to rTregs between different age groups. As illustrated in Fig. 27, older mice exhibited a significant increase in this ratio (p-value = 0.015) when compared to their younger counterparts. However, it is important to highlight that there is a considerable variability within the older mice group, with some mice showing a more pronounced ratio increase than others.

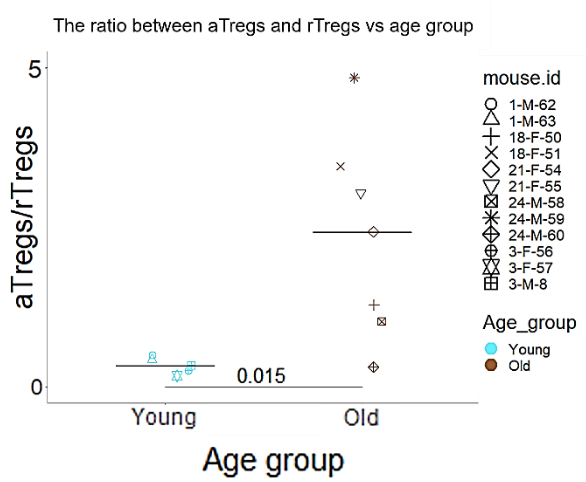


Figure 27: Comparative Analysis of aTregs/rTregs Ratios Across Age Groups in the TMS Dataset. Comparative analysis of the TMS dataset presenting the ratio between aTregs and rTregs across two distinct age groups - young and old. The data points from individual mice are color-coded: the young (light blue) age group and the old (brown) age group. Each mouse within the dataset is uniquely identified with a sign, facilitating a clear delineation of data points between the subjects. Horizontal lines are drawn across the scatter plot to indicate the average ratio of aTregs to rTregs for each age group, providing a visual cue for assessing the central tendency within the dataset concerning the aTregs/rTregs ratio across young and old age groups. Through this graphical representation, the analysis aims to elucidate any significant differences in the aTregs/rTregs ratio that might correlate with aging.

4.3.2 Comparison of the Differentially Expressed Genes in the TMS and Aging Mice CD4 Datasets' CD4 Subsets

Following the identification of the CD4$^+$ T cell subsets, the subsequent aim of our analysis was to establish the validity of these subsets by comparing the differentially expressed genes in the TMS dataset with those in the aging mice dataset. To confirm the reliability of the CD4$^+$ T cell subsets identified in the TMS dataset by assessing the similarities and differences between the two datasets. This validation was critical in providing a reliable foundation for subsequent correlation analyses exploring the role of CD4$^+$ T cell subsets in aging processes. We employed methodologies similar to those used by Maaten and Hinton for this validation [256]. Notably, the TMS dataset revealed a smaller number of cells per mouse compared to the more substantial cell counts in the aging mice dataset, echoing findings from studies on immune aging by Goronzy and Weyand [115,142,151].

In our exploration of the markers for the cytotoxic subset in these datasets, we identified 26 genes consistent with both our datasets and findings from Elyahu et al. [92]. Examples of these genes include Ccl5, Nkg7, Gzmk, Ccl4, Ctla2a, Eomes, and others. These genes could serve as potential marker genes (Fig. 28).

The variations in the gene set might be attributed to the different query gene sets used during sequencing, a challenge also highlighted by Wang et al. [307]. Additionally, the volcano plot in Fig. 29 comparing the cytotoxic to the TEM subsets, which displays a similar distribution of cytotoxic genes across both datasets, was constructed using methodologies discussed by Cui and Churchill [308].

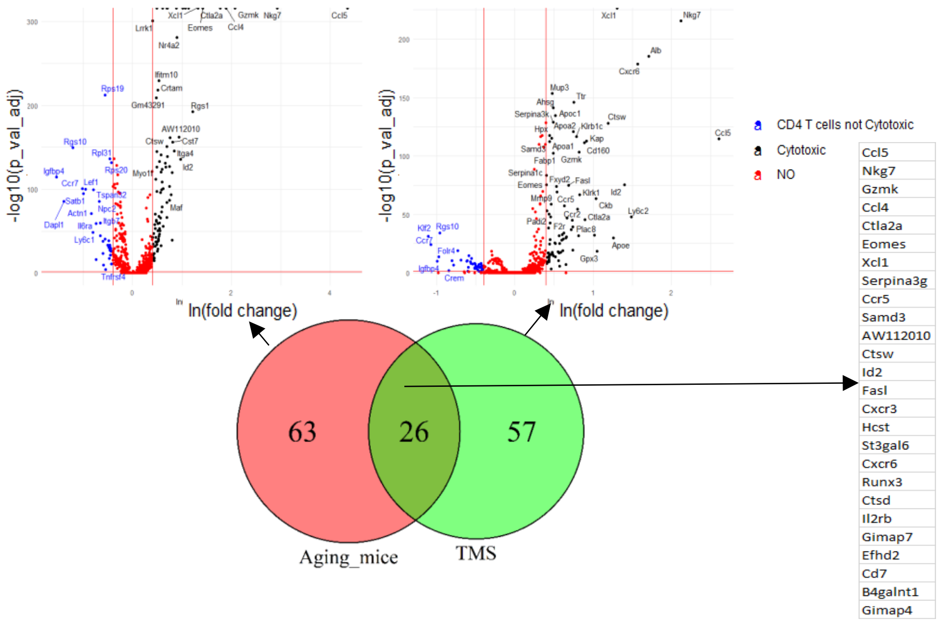


Figure 28: Differential Gene Expression in Cytotoxic Subsets: Comparing Aging Mice and TMS Datasets. Aging Mice Dataset Volcano Plot: This visual representation captures the differential gene expression within the aging mice dataset. Cytotoxic CD4$^+$ T cell subset - specific

differentially expressed genes are highlighted in black, "NO" is non-differentially expressed genes are denoted in red, and genes differentially expressed in other CD4⁺ T cell subsets are marked in blue. TMS Dataset Volcano Plot: The differential gene expression within the TMS dataset is displayed, with color coding mirroring the previous plot. The plot emphasizes genes that exhibit unique and shared expression patterns compared to the aging mice dataset. Venn Diagram of Cytotoxic CD4⁺ T Cell Subset Genes: This figure illustrates the overlap of differentially expressed genes between the aging mice (green) and TMS datasets (red). The intersection prominently features the 26 genes that are consistently differentially expressed in both datasets.

The comparison of the CD4⁺ subsets' differentially expressed genes in the TMS and aging mice datasets, we observed that about 30% of the upregulated genes in the cytotoxic subset were consistent. Importantly, among these, there is a notable overlap of 2,730 genes that characterize both the cytotoxic and TEM subsets across the datasets. This includes the set of 26 consistent genes we previously highlighted. A significant finding is that nearly 89% of the marker genes pinpointed for the cytotoxic and TEM subsets in the aging mice dataset matched those in the TMS dataset (Fig. 29). This high degree of overlap reinforces the precision with which we have identified the cytotoxic and TEM subsets in both datasets.

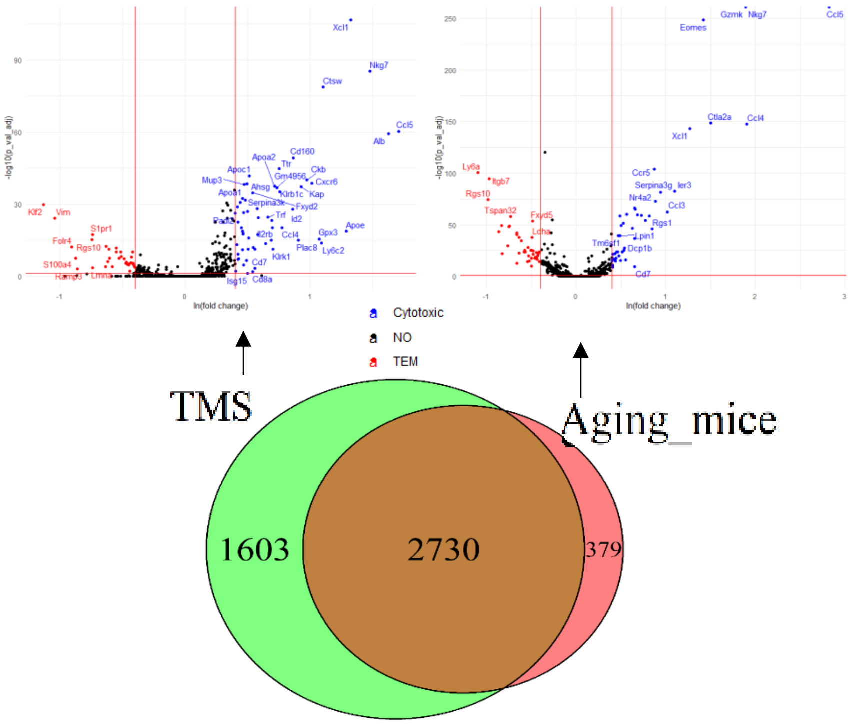


Figure 29: Differentially Expressed Genes between the Cytotoxic and the TEM subsets in the aging mice and the TMS datasets. In the upper plane, the aging mice and the TMS dataset are presented with volcano plots of DGE analysis between the Cytotoxic and the TEM subsets. In the lower plane, the Venn diagram represents the overlap of genes expressed in both the aging mice (red) and the TMS (green) datasets by the cytotoxic and the TEM subsets. Aging Mice

Dataset Volcano Plot: This visual representation captures the differential gene expression between the Cytotoxic CD4⁺ T cell subset (blue) and genes differentially expressed in the TEM CD4⁺ T cell subset (red), "NO" is non-differentially expressed genes are denoted in the black. TMS Dataset Volcano Plot: The differential gene expression within the TMS dataset is displayed, with color coding echoing the Aging Mice Dataset Volcano Plot. Venn Diagram of the Cytotoxic CD4⁺ T Cell Subset Genes and the TEM CD4⁺ T Cell Subset Genes: This figure illustrates the overlap of differentially expressed genes between the aging mice and the TMS datasets.

In our analysis of the TMS and aging mice datasets, we identified a significant overlap of 2,929 genes that characterize both the exhausted and TEM subsets. Approximately 88% of the marker genes identified for these subsets in the aging mice dataset aligned with those in the TMS dataset (Fig. 30). Additionally, we observed a pronounced expression of the gene Ccl5 in the cytotoxic subset, contrasting its subdued expression in the TEM subset. This differential expression suggests that Ccl5 may play a pivotal role in the proposed differentiation of cytotoxic CD4⁺ T cells from TEM cells.

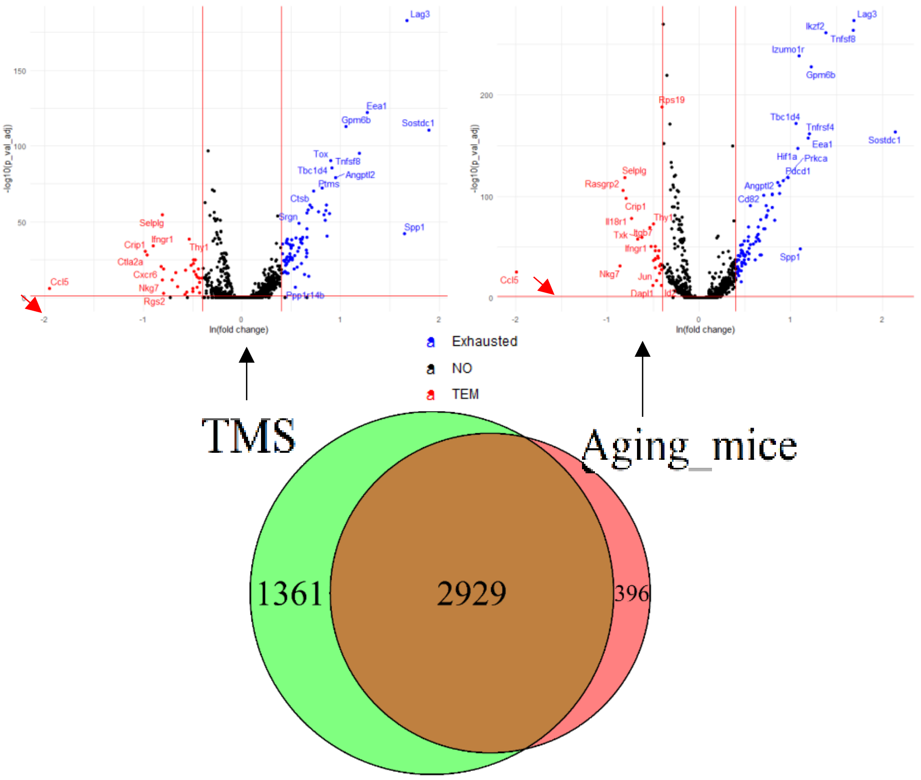


Figure 30: Differentially Expressed Genes between the Exhausted and the TEM subsets in the aging mice and the TMS datasets. In the upper panel, volcano plots of the DGE analysis are presented between the Exhausted and the TEM subsets for both the aging mice and the TMS dataset. In the lower plane, the Venn diagram illustrates the overlap of genes expressed in the Exhausted and the TEM subsets of the aging mice (red) and the TMS (green) datasets. Aging Mice Dataset Volcano Plot: This diagram illustrates the difference in gene expression between

the Exhausted CD4$^+$ T cell subset (blue) and the TEM CD4$^+$ T cell subset (red). "NO" is non-differentially expressed genes marked in black. TMS Dataset Volcano Plot: The differential gene expression within the TMS dataset is displayed, with color coding echoing the Aging Mice Dataset Volcano Plot. Venn Diagram of the Exhausted CD4$^+$ T Cell Subset Genes and the TEM CD4$^+$ T Cell Subset Genes: This figure illustrates the overlap of differentially expressed genes between the aging mice and the TMS datasets.

Both the TMS and aging mice CD4$^+$ subsets datasets revealed a notable overlap, which shared 2,876 genes characterizing the aTreg and rTreg subsets. Approximately 88% of the markers identified for these subsets in the aging mice dataset matched those in the TMS dataset, as depicted in Fig. 31. This consistency underscores the accuracy of the regulatory subset identifications, in line with the established importance of Tregs in maintaining immune homeostasis [290].

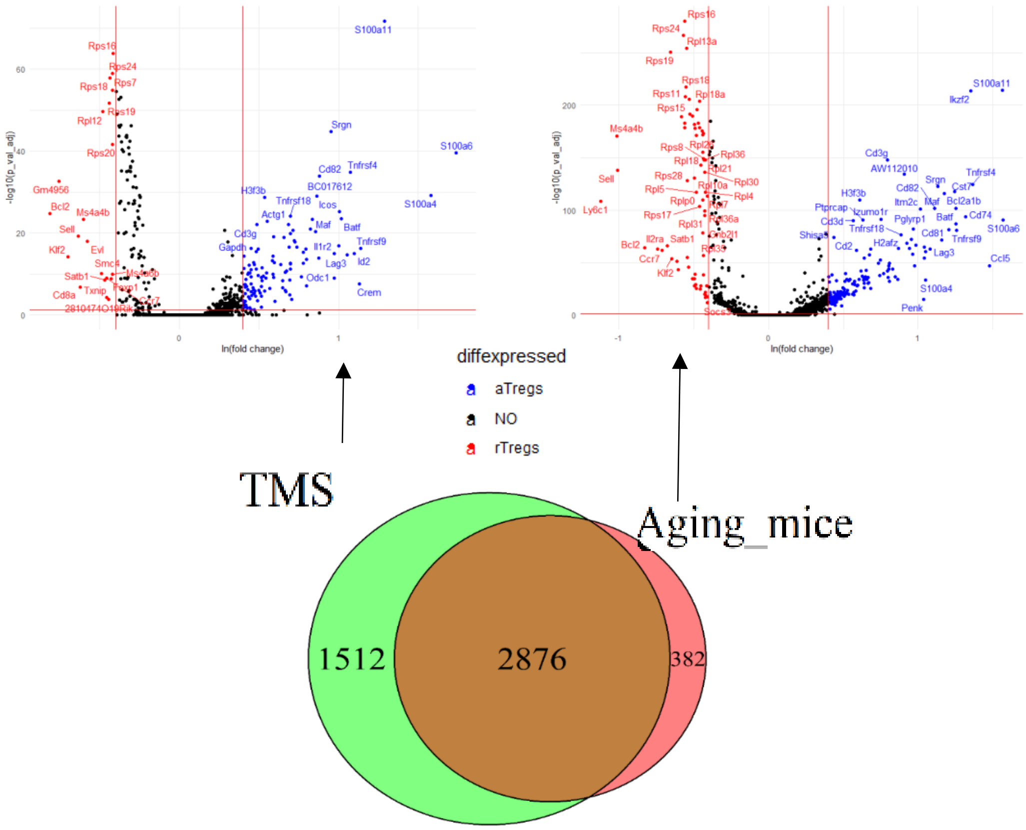


Figure 31: Differentially expressed genes between two Regulatory subsets, the activated (aTregs) and the resting (rTregs) subsets, in the aging mice and the TMS datasets. In the upper panel, the volcano plots of DGE analysis are between the aTregs and the rTregs subsets for both the aging mice and the TMS dataset. In the lower plane, the Venn diagram shows the shared genes between aTregs and rTregs subsets in aging mice (red) and TMS (green) datasets. Aging Mice Dataset Volcano Plot: This diagram compares gene expression between the aTregs (blue) and the rTregs (red) CD4$^+$ T cell subsets. "NO" is non-differentially expressed genes (black). TMS Dataset Volcano Plot: The TMS dataset displays DGE with color coding similar

to the Aging Mice Dataset Volcano Plot. Venn Diagram of the aTregs CD4$^+$ T Cell Subset Genes and the rTregs CD4$^+$ T Cell Subset Genes: This diagram demonstrates the overlap of DGE between the aging mice and the TMS datasets.

5 Computational Characterization of Key Cytokines Involved in Cellular Senescence

5.1.1 Senescent Cells Identified by p16 and p21

In our study, utilizing the TMS dataset, we identified senescent cells by the expression of both senescence markers CDKN2A/p16 and CDKN1A/p21 [309]. Based on the strategy of assessing the expression of these senescence markers, 1,707 cells were detected as illustrated in Figure 32, where the overlap between the two circles represents the identified senescent cells. Our study found that p16$^+$p21$^+$ cells are represented in all tissues, and their numbers increase with age, in alignment with previously documented patterns of senescence markers' upregulation [196]. The highest percentage of these cells is seen in 24-month-old mice, with a decline seen in 30-month-old mice. The data shows the accumulation of p16$^+$p21$^+$ senescent cells during aging, juxtaposed with a relatively small percentage found in the young group (1-3 months) [310]. This supports the observed trend for all p16$^+$ cells detected but not the observed trend for the p21$^+$ cells and supports the results obtained in the original article by the Tabula-Muris consortium et al. (Fig. 32A) [233].

Although senescent cells constitute a minuscule proportion of the total cellular pool (see Table S3), their impact on the overall tissue microenvironment, primarily through the SASP, can significantly influence the potential for healthy aging [108].

The prevailing hypothesis posits that the decline in p16$^+$p21$^+$ cells (i.e., the senescent subset) results from effective cellular clearance mechanisms [110]. A comprehensive analysis of the transcriptomes of senescent can shed light on the role of the SASP in facilitating or inhibiting cellular clearance. Such gene profiling endeavors promise to expose potential pathways for targeted interventions to bolster cellular health and longevity.

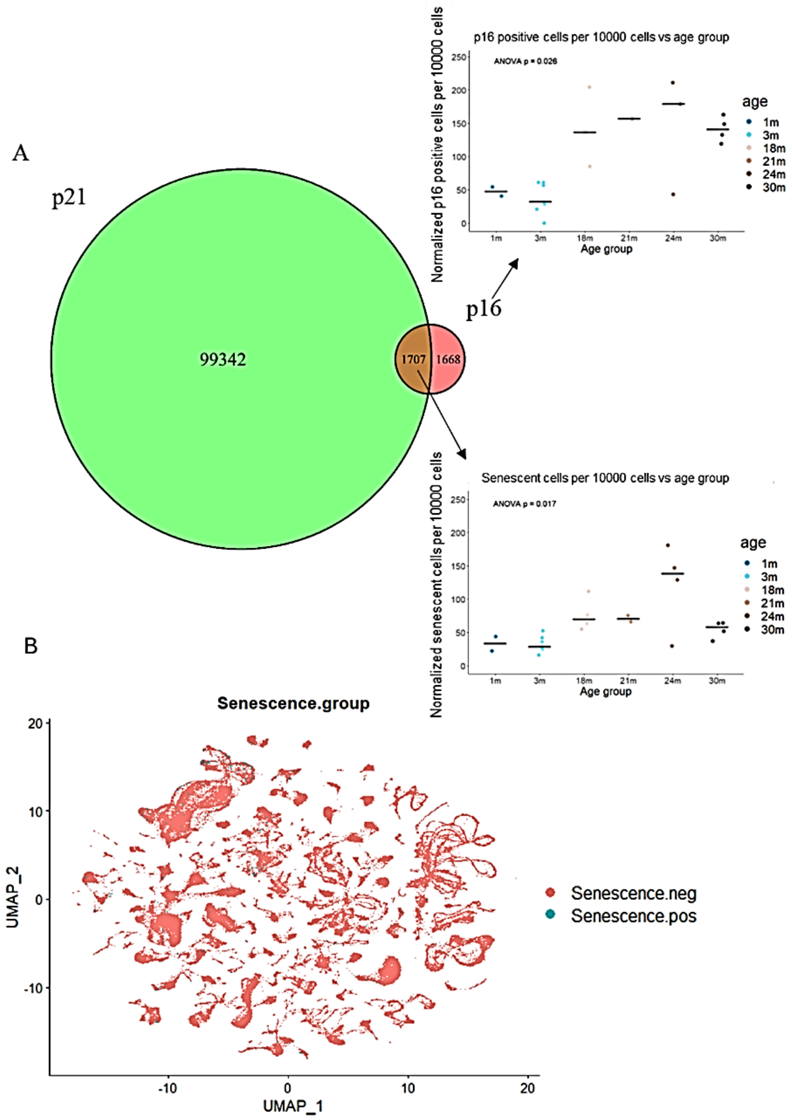


Figure 32: Visualization of senescent cells identification defined by p16^{+}p21^{+} in the TMS dataset. A) Venn diagram displays the overlap of cells expressing p21 (in green) and p16 (in red) in the left plane. p16-expressing cells are normalized to 10,000 cells for each mouse grouped by age, illustrating the accumulation of cells through age (top right). Senescent cells, represented by the overlap, are normalized to 10,000 cells for each mouse by age (bottom right), demonstrating the same accumulation of cells with age, a dispersion in cell amount in 24-month-old mice, and a drop in 30-month-old mice; B) UMAP projection of the TMS dataset is grouped into senescent named Senescence.pos (in light blue) and non-senescent named Senescence.neg (in red) groups, displaying the scattering of senescent cells throughout the tissues and age.

5.1.2 Senescent Cells' Exhibition of Age-Associated Transcriptome

Utilizing the TMS dataset, we conducted a DGE analysis using the *FindMarkers()* function within the Seurat package, contrasting senescent cells with non-senescent cells. This analysis highlighted an age-associated pro-inflammatory phenotype. Out of the 9,032 genes exhibiting differential expression between these two cellular states, 116 were expressed at significantly higher levels in the senescent cells, as illustrated in Fig. S3. Noteworthy genes from this list include Barx2, Emp2, Lypd3, Gm94, and Krt6a. Conversely, 36 genes, including Coro1a, Rac2, Ly6c2, and Ms4a1, were more prevalent in the non-senescent cells. We aimed to identify genes consistently associated with the senescence phenotype.

Building on our DGE analysis, we further explored the implicated gene sets using two reputable databases, GO and KEGG, to outline the interconnected pathways and processes relevant to cellular senescence. The main goal of the Gene Set Enrichment Analysis (GSEA) using the KEGG and GO databases was to graphically elucidate the relationships of processes, pathways, and diseases related to cellular senescence. This approach underscores the genetic foundations of senescence and paddles our understanding of potential therapeutic interventions for age-associated diseases.

In the upper portion of Fig. S3, a volcano plot is presented in which genes with higher expression in senescent cells appear in light blue, and genes predominant in non-senescent cells are highlighted in red. The bottom portion of Fig. 34 presents the top-30 genes for each senescent and non-senescent cells, providing a more precise snapshot of the most dominant genes in each category.

We explored genes defined by both random and ranked gene sets. Random gene sets are randomly assembled gene collections used as analysis controls. Ranked gene sets are genes organized based on specific criteria, such as DGE between senescent and non-senescent cells. We utilized two databases, GO and KEGG, for our analysis. Our investigation identified 431 enriched gene sets from GO and 42 from KEGG. Among these, pathways related to cellular senescence and aging were expectedly prominent, highlighting the senescent nature of the cells studied. Furthermore, there was notable enrichment in pathways associated with cancer, age-related diseases such as Alzheimer’s and Parkinson’s, and inflammation-related ailments.

The enrichment map, a visual network where nodes represent gene sets and edges indicate overlap between them, is sourced from the KEGG database unveils two main interconnected networks of processes. Five other processes were identified, yet they lacked a connection to the main networks; the estrogen signaling pathway is a prime example among these (Fig. S4 top). This pathway holds significance as it modulates cellular senescence, impacting notably bone health and age-related diseases [311]. Estrogen, the key player in this pathway, can prevent cellular senescence [312], mitigate bone loss [313], and inhibit stress-induced premature senescence in specific cells [314], although this beneficial effect may taper in older individuals. Hence, it emerges as an exhilarating finding in the GSEA of senescent cells. The GO analysis, focusing on biological processes, molecular functions, and cellular components, indicated high activation of cytokines, chemokines, and other typical senescence markers (Fig. S4, bottom). This analysis is visually represented in the GO enrichment map, which traces clusters of pathways and processes, encapsulating those involving the cytoskeleton, the extracellular matrix, and chemokines, thereby aiding in discerning biological themes.

In the KEGG analysis, cellular senescence genes were linked to various conditions, such as the coronavirus, Chagas disease, alcoholic liver disease, pertussis, Alzheimer's disease, Parkinson's disease, and multiple forms of cancer. Distinct signaling pathways, including the p53, estrogen, and age-related pathways, were also identified on the enrichment map (Fig. S4 top).

The gene-concept network plot (Fig. S5) represents the relationships between genes and biological concepts (presented in the enrichment map), assisting in understanding gene function and interactions within biological contexts. Through the visualization provided by the gene-concept network map, our objective was to identify critical cytokines and chemokines that might play pivotal roles in initiating and sustaining the clearance of senescent cells. It presents the involvement of chemokines Ccl4, Cxcl14, Cxcl2, Ccl2, Cxcl13, and Ccl8 in the gene set enrichment processes. Notably, the chemokines Ccl4 and Ccl2 also exhibit high expression levels in cytotoxic $CD4^+$ T cells. Moreover, research on Cxcl13 has identified its link to the Cxcr5 receptor, a marker for T follicular helper (Tfh) cells, a subset of $CD4^+$ T cells [315–319].

Exploring the genetic distinctions between supercentenarian mice and their older counterparts was crucial for validating our hypothesis, which suggests that the efficient clearance of senescent cells, mediated by shifts in $CD4^+$ T cell subsets, is a cornerstone for enhanced longevity. We conducted a DGE analysis on the senescent cells' subset extracted from the TMS dataset, comparing three age groups of mice: young (1-3 months), old (18-24 months), and supercentenarian (30 months). This analysis revealed a notable variation in the SASP of senescent cells among these age groups.

The comparison between old and supercentenarian mice identified 208 upregulated genes in the supercentenarian group and 41 upregulated genes in the old group, with only two shared genes between them. In contrast, we found 76 upregulated genes in the young group when comparing young and old mice. The comparison of supercentenarians versus old mice revealed 245 upregulated genes in the supercentenarian group with only three shared genes (Figs. S6B and S6E). This data underscores a significant difference in gene expression between the senescent cells of young mice and those of supercentenarians, hinting at a unique cytokine profile aiding in senescent cell clearance in supercentenarian mice, setting them apart from the other age groups.

As depicted in Figs. S6A-C and S6F, the old group revealed more inflammation marker genes, such as C1qb, within the transcriptome than the young group. Interestingly, when we re-organized the age groups (Age group 2) to compare young (1-3 months) and old (18-30 months) mice, we observed fewer upregulated genes in the united old group compared to the original old group (as shown in Fig. S6F). This indicates the possibility that some genes present in both young and supercentenarian groups may activate similar pathways in these age groups.

We conducted a GSEA on the senescent cells within the supercentenarian group, revealing the presence of general inflammatory pathways and markers of age-associated diseases as identified in the KEGG database analysis. This is illustrated in Figure S7. The GSEA disclosed 83 enriched gene sets in the KEGG database and 1,326 in the GO database, which are significant in understanding this group's molecular mechanisms of aging. Additionally, we identified upregulated genes that help differentiate between the age groups. In the restructured age classifications (Age Group 2), which comprises the young (1-3 months) and old (18-30 months) categories, with the old category combining both old (18-24 months) and supercentenarian (30 months) mice, we observed an enrichment of 25 gene sets in the KEGG database and 505 gene sets in the GO database, as depicted in Figure 40. The analysis of the GO database highlighted IL-6 production in the senescent cells of the supercentenarian group, alongside intracellular changes and molecular functions (MF) associated with age-related diseases. Notably, we observed amyloid-beta binding, associated with Alzheimer's disease, as a pertinent example of molecular function tied to age-related diseases (Fig. S7).

Building upon the GSEA delineated in Figure S7; we further explored the senescent cells within the supercentenarian group from the TMS dataset. This subsequent analysis unveiled a notable gene-set enrichment associated with immune system regulation processes. The cells exhibited a noticeable association with a spectrum of chemokines. Among the chemokines, Ccl4, Ccl8, and Cxcl13 were identified across all senescent cells analyzed. Conversely, Ccl24, Ccl12, Ccl6, Cxcl9, and Ccl3 were distinctly associated with this particular subset of senescent cells (Fig. S8). Employing a gene concept network analysis, we illuminated the molecular complexness intrinsic to this subset, thereby deepening our understanding of the chemokine associations and their potential implications in immune system regulation. This analysis serves as a refined exploration of the chemokine landscape within the supercentenarian senescent cells, complementing the initial findings of general inflammatory pathways and age-associated disease markers previously illustrated.

The GSEA conducted on the senescent cells' subset of the supercentenarian age group (Fig. S7-S8) was also applied to the senescent cells of the old group (18-24 months). In the old group, the cells are primarily characterized by inflammatory pathways, with a lesser emphasis on aging pathways, and an absence of p53 pathways. Conversely, in supercentenarians, the immune regulatory pathways are absent (Fig. S9). This scenario suggests a potential state of lower immune regulation coupled with a vast inflammatory reaction, which may exhaust the immune system, contributing to senescent cell accumulation and chronic inflammation. These findings align with the proposed hypothesis that aging induces significant changes in $CD4^+$ T cell composition. Furthermore, this analysis supports the broader hypothesis concerning age-related immunological shifts and their impact on cellular senescence, thus providing a foundational basis for further exploration into potential biomarkers and therapeutic strategies for healthy aging.

Following the previous results, the progression of our private GSEA facilitates the gene concept network, which emphasizes the significant roles of chemokines Cxcl13 and Ccl4 in cellular senescence (Fig. S10). We have identified pathways intrinsically linked to age-related diseases and metabolic syndromes comprehensively illustrated in Figures S9. This implies that various functional deviations may contribute to the aging process. Our findings thus provide a vital insight into the underlying mechanisms of age-related diseases and metabolic syndromes, which could pave the way for novel therapeutic interventions. This exploration resonates with the burgeoning perspective of perceiving aging as a pathological condition and augments our understanding of the molecular interplay at its core.

5.1.3 Correlation Between Senescent Cells and $CD4^+$ Subsets and between the Subsets

Age-associated $CD4^+$ T cell subsets show the same increasing trend in its proportion observed in previous studies [92]. The cytotoxic subset is suspected of having a significant reverse correlation with the senescent cell subset in the old age group (Fig. 34). Some reverse correlation between the cytotoxic and naïve subsets was seen, but not to a statistically significant extent in general and within the old group (Figs. 33 and 34); it also has a slight but not statistically significant positive correlation with the TEM, aTregs, and exhausted subsets in both age groups (Fig. 33) and a positive but not statistically significant correlation in the old group (Fig. 34). However, the cytotoxic subset has a slight reverse correlation with the exhausted subset in the old age group suggesting that the exhausted subset does not impact the cytotoxic subset (Fig. 34).

We found that the TEM subset has a high and statistically significant correlation with the aTregs subset at all ages and a statistically substantial reverse correlation with the naïve subsets (Fig. 33). The TEM subset also had a statistically significant reverse correlation with the rTregs subset in the old group (Fig. 34). The aTreg, cytotoxic, and TEM $CD4^+$ T cell subsets have great potential to serve as the regulator cells of the senescent cell accumulation, and further research is needed to explore their role in this.

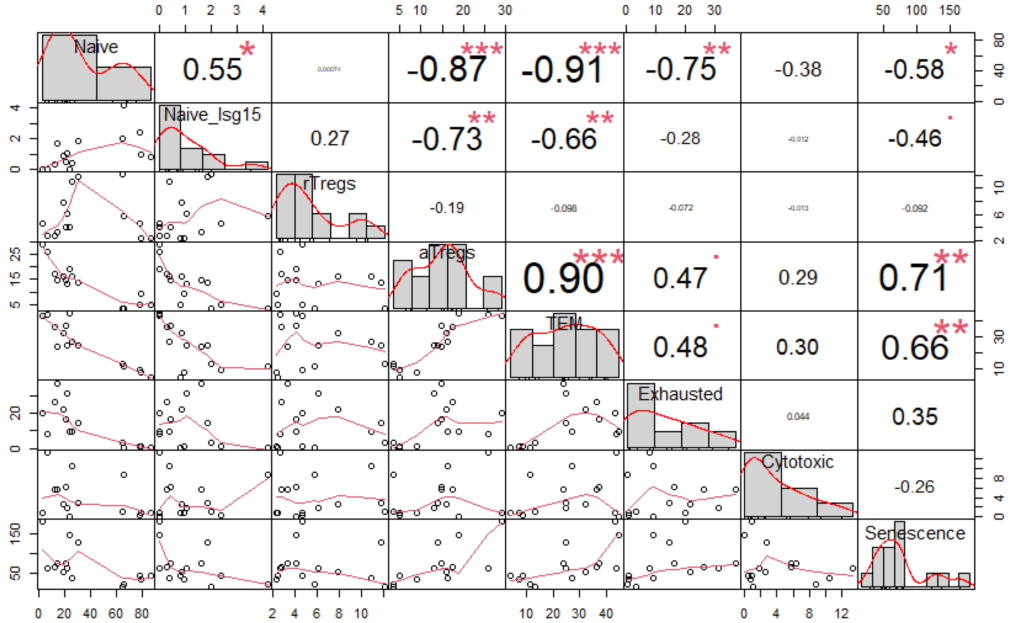


*Figure 33: Correlation matrix of $CD4^+$ T cell subsets (Naive, Naive_Isg15, rTregs, aTregs, TEM, Exhausted and Cytotoxic) and senescent cells contain both young (1-3 months) and old (18-30 months) mice groups. The scatter plots present the relationship between two $CD4^+$ T cell subsets or a $CD4^+$ T cell subset and senescent cells; each dot represents a mouse. The red line in scatter plots presents the relationship that best fits the trend line by the local polynomial regression. A histogram and density plot (red line) on the diagonal display the distribution of the subset. The Numerical values shown represent Pearson's correlation coefficient with symbols indicating the p-values (p-value: $0 < p \leq 0.001$ - "***", $0.001 < p \leq 0.01$ - "**", $0.05 < p \leq 0.1$ - "*", $0.1 < p \leq 1$ - "·").*

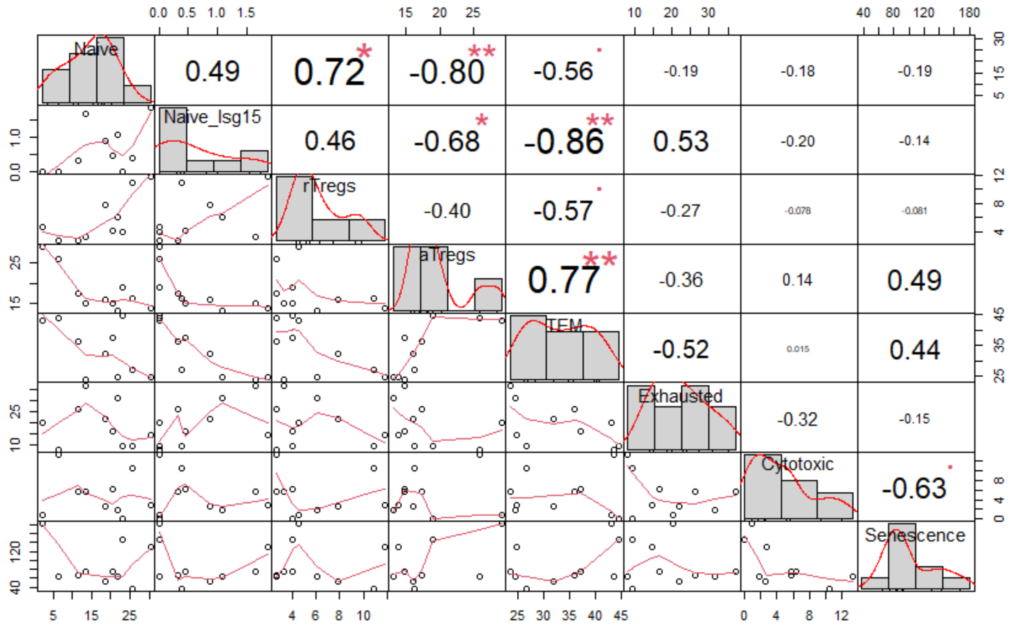


Figure 34: Correlation matrix of $CD4^+$ T cell subsets (Naive, Naive_Isg15, rTregs, aTregs, TEM, Exhausted and Cytotoxic) and senescent cells in the old (18-30 months) age group. In

*correspondence to Fig. 33, the scatter plots illustrate the association between two distinct subsets of CD4+ T cells or a CD4+ T cell subset and senescent cells. Each point on the plot signifies a mouse. The red line on scatter plots is the optimal trend line from local polynomial regression, representing data patterns. Diagonal plots display a histogram and density plot to represent the distribution of a subset. The Numerical values shown represent Pearson's correlation coefficient with symbols indicating the p-values (p-value: $0 < p \leq 0.001$ - "***", $0.001 < p \leq 0.01$ - "**", $0.05 < p \leq 0.1$ - "*", $0.1 < p \leq 1$ - "·").*

6 Discussion

In this study, we identified distinct features of senescent cells in different age groups in mice, particularly highlighting the unique features of these cells in older mice. In addition, we found significant representation of cytotoxic $CD4^+$ T cells in supercentenarians, and a novel categorization approach applied to the bladder cancer dataset identified distinct subsets of $CD4^+$ T cells. The comparative analysis across different species, age groups, and conditions (supercentenarian, aging mice, and bladder cancer datasets) highlighted distinct gene expression profiles. It revealed a set of variable genes, indicating species-specific variations in immune functions, elucidated species and age-specific characteristics of cytotoxic $CD4^+$ T cells. The consistent gene expression like GZMK, GZMB, and EOMES across species underpins evolutionarily conserved roles, while the distinct gene expression profiles highlight species-specific immune functions. Remarkably, the analysis of aging mice and supercentenarian datasets emphasized a pronounced representation of cytotoxic $CD4^+$ T cells, hinting at a pivotal role in immune responses related to aging and, possibly, cancer. Our research underscores the potential of cytotoxic $CD4^+$ T cells in senescent cell clearance, complementing the existing literature on immune aging. Our study showed a correlation between $CD4^+$ T cell subsets, especially the cytotoxic subset, and senescent cell clearance. The data suggests a potential therapeutic avenue in targeting cytotoxic $CD4^+$ T cells to counter senescent cell accumulation, thereby mitigating immune aging. The DGE analysis of the combined cytotoxic $CD4^+$ T cells revealed notable gene expression variations, such as the higher GZMH and GZMA expression levels in human cytotoxic $CD4^+$ T cells and age-associated alterations in T cell gene expression. The identified potential regulatory roles of aTreg, cytotoxic, and TEM subsets of $CD4^+$ T cells in regulating senescent cell accumulation underscore the necessity of further research to delineate their exact roles and implications in aging and age-associated diseases.

This study complements existing literature by not only corroborating known mechanisms but also by potentially extending understanding of the interplay between $CD4^+$ T cells, senescent cells, and aging. The correlation between $CD4^+$ T cell subsets and the clearance of senescent cells, primarily induced by the SASP in old mice, found in this study represents a valuable addition to the existing body of knowledge. The findings presented in this study are well aligned with the current literature on $CD4^+$ T cells, senescent cells, and aging. Our study highlights the role of cytotoxic $CD4^+$ T cells in managing senescent cell accumulation in aging mice, which is corroborated by recent research in which cytotoxic $CD4^+$ T cells were found to eliminate senescent cells in human skin, especially by targeting the human cytomegalovirus glycoprotein B (HCMV-gB) antigen [320]. Senescent T cells, known to have defects in proliferation and effector functions, accumulate during aging, chronic infections, and autoimmune disorders, all of which are utilized by tumors to induce T cell senescence [321]. This is aligned with our observation of senescent cell accumulation and its association with aging. Aging impacts $CD4^+$ T helper cell (Th) subset differentiation, especially during infections like influenza, as showcased in a study where senescence-induced changes were observed in $CD4^+$ T cell differentiation during influenza infection in aging mice [322]. Evidence suggests that alterations intrinsic to $CD4^+$ T cells contribute to chronic inflammation and could accelerate an organism-wide aging phenotype [290]. This is aligned with our observations on the changes in $CD4^+$ T cell composition and immune regulation pathways with aging and their potential association with chronic inflammation and senescent cell accumulation. In addition, we identified shared genes across datasets that supported the findings of prior research on the conservation of core genes in cytotoxic responses, emphasizing the universality of certain cytotoxic response genes across species and conditions. This supports our finding regarding the role of $CD4^+$ T cell subsets in aging mice. One study found that $CD8^+$ T cells might be more susceptible to aging than $CD4^+$ T cells, which supports our finding regarding the compensation of an increase in cytotoxic $CD4^+$ T cells for the reduction in $CD8^+$ T cells observed with aging [323].

Uncovering the role of $CD4^+$ T cells and senescent cell accumulation provides deeper insight into the biological mechanisms of aging. The correlations between $CD4^+$ T cell subsets and senescent cell clearance identified may inform the development of therapeutic interventions to mitigate aging-related conditions. The gene expression variations observed could aid in the identification of biomarkers for aging and age-related diseases. Comparative analysis across human and mouse datasets extends understanding of immune system alterations across species, which may be crucial for translational research. The similarity between human aging and bladder cancer data hints at potential common molecular pathways, opening avenues for exploring the interplay between aging and cancer. Our work lays a solid foundation for further research to delineate the complex interplay between genetics, aging, and disease and to explore the regulatory roles of different $CD4^+$ T cell subsets in senescent cell accumulation. In addition, our study suggests that targeting the cytotoxic $CD4^+$ T cell subset could be a viable strategy for countering senescent cell accumulation and potentially mitigate immune aging. The differentiation of $CD4^+$ T cells and their senolytic [324] (class of compounds or therapies aimed at clearing senescent cells from the body) action presents a possible therapeutic avenue for promoting longevity and combatting age-related diseases.

Utilizing advanced bioinformatic tools like Seurat for scRNA-seq data analysis, DGE analysis, and standard clustering algorithms. The identification of senescent cells through the expression of specific markers and GSEA-exposed pathways related to cellular senescence, age-related diseases, and inflammation-related ailments. We explored three other datasets to delve deeper into the cytotoxic $CD4^+$ T lymphocyte subset, particularly the cytotoxic variety. The supercentenarian and bladder cancer datasets were instrumental in identifying and subsetting human cytotoxic $CD4^+$ T lymphocytes, while the aging mice dataset was critical to the comparative analysis involving mice [92,93,234]. This integrated meta-analysis gave us a higher-resolution understanding of humans' and mice's cytotoxic $CD4^+$ T lymphocytes. The TMS dataset was used to characterize senescent cells and analyze their correlation with $CD4^+$ T lymphocyte subsets [233]. To identify $CD4^+$ T lymphocytes in the TMS dataset, the scRNA-seq of the aging mouse $CD4^+$ T lymphocyte subsets of the aging mice dataset of Elyahu et al. [92] were used. We replicated the analysis on the aging mouse dataset, and we had several compelling reasons for replicating this analysis. First, Elyahu et al. conducted their analysis using an earlier version of Seurat precisely because it was available then. Since then, a newer version has become available. This version update introduced enhancements and changes in the analysis pipeline, making it necessary to reanalyze the data to ensure compatibility and leverage the most up-to-date tools. Second, while Elyahu et al.'s analysis yielded valuable insights, it is essential to note that they did not directly compare the cytotoxic $CD4^+$ T cell subset between mice and humans. Their focus primarily revolved around characterizing this subset in the mouse dataset. Given the relevance of such a comparison to our study, we aimed to bridge this gap by conducting a comparative analysis involving mice and human datasets. We could not directly adopt their results due to the Seurat version update and differences in data preprocessing. Replicating their analysis allowed us to generate comparable results and maintain consistency in our study. Third, by conducting this analysis, we aimed to deepen our proficiency in the R programming language and better understand scRNA-seq data analysis. Lastly, our replication of the analysis served as a crucial step in validating the $CD4^+$ T cell subsetting in the TMS dataset. It enabled us to assess the robustness of the findings, confirm the presence of specific $CD4^+$ T cell subsets, and verify the biological insights gained from the data. This validation process was essential to ensure the reliability of our study's subsequent analyses and interpretations.

It is important to note the limitations of our work, some of which could be addressed in future work. The age ranges Examined in this study may not capture all variations in $CD4^+$ T cell behavior and senescent cell accumulation. Focusing mainly on mice might limit the translatability of our findings to humans. A lack of in vivo validation could restrict understanding of the actual physiological relevance of the findings. The absence of longitudinal data may hinder insights into the temporal dynamics of the observed phenomena. Moreover, the research faced limitations due to the composition of the TMS dataset, which lacked a significant number of $CD4^+$ T lymphocyte subsets. The Louvain clustering algorithm had difficulty recognizing rare subsets, and the high batch effect from tissue diversity in the TMS dataset posed challenges. The resolution of scRNA-seq and other analytical tools used may limit their ability to detect rare cell populations.

The study illuminates several avenues for future research, particularly in investigating the potential association between aging, immune system dynamics, and age-related diseases. It is imperative to extend this analysis to diverse datasets and conditions, fostering holistic understanding that could inform therapeutic strategies for a myriad of age-associated conditions.

Further research is recommended to validate the findings using scRNS-seq on a larger scale in old and “very old” mice, which we call supercentenarians. Additionally, an in vivo culture of TEM cells with suspected chemokines should be conducted to understand the differentiation mechanism of the cytotoxic CD4$^+$ T cell subset. These steps would contribute to more thorough understanding of the CD4$^+$ T cell-senescent cell interplay in aging. Delve deeper into the mechanisms through which CD4$^+$ T cells interact with and clear senescent cells, especially the cytotoxic subset. Conduct longitudinal studies to track the dynamics of CD4$^+$ T cell subsets and senescent cell accumulation over time in various age groups. Explore potential therapeutic interventions targeting cytotoxic CD4$^+$ T cells to mitigate senescent cell accumulation and age-related immune alterations. Extend the cross-species analysis to other animal models to validate and expand understanding of the CD4$^+$ T cell-senescent cell interplay. Investigate gene-targeting approaches to modulate the activity of CD4$^+$ T cells, exploring the impact on senescent cell clearance and aging. Explore the link between immune senescence, chronic diseases, and the role of CD4$^+$ T cell subsets in disease progression and management. Investigate the molecular pathways involved in the interactions between CD4$^+$ T cells and senescent cells and how these pathways are altered in aging and disease states. Leverage newer technologies and bioinformatic tools to enhance the resolution and accuracy of future analyses, further examining the complex interplay between CD4$^+$ T cells, senescent cells, and aging. Our study calls for more rigorous research to validate and expand upon the findings, exploring the therapeutic potential of modulating cytotoxic CD4$^+$ T cells to counter senescent cell accumulation and immune aging. The field may benefit from a concerted effort to understand the intricate interactions between the immune system and cellular senescence in aging and disease contexts. Validate and expand the current findings through larger-scale studies and across different species. Foster collaborations between immunologists, gerontologists, and bioinformaticians to delve deeper into the molecular mechanisms underpinning the observed phenomena. Initiate investigations into potential therapeutic interventions targeting cytotoxic CD4$^+$ T cells to combat cellular senescence and age-related diseases. Seek funding and support for advanced research and technology to continue exploring the interplay between immune system aging and cellular senescence.

The intricate interplay between CD4$^+$ T cells, senescent cells, and aging elucidates pivotal insights into the mechanisms underpinning immune system alterations over time. The distinct senescent cell identities across age groups contribute to nuanced understanding of cellular senescence, whereas the comprehensive analysis of $CD4^+$ T lymphocytes across aging mice and supercentenarians reveals substantial variations in their subsets, notably in cytotoxic $CD4^+$ T cells. The higher resolution attained through meta-analysis and the employment of updated analytical tools like Seurat and clustifyr significantly bolstered the robustness and validation of $CD4^+$ T cell subsetting, instrumental for deeper understanding of immune system dynamics. The findings contribute to broader understanding of immune system changes with aging. Specifically, they highlight a potential mechanism through which the immune system could be harnessed or modulated to tackle aging-related cellular senescence, which is crucial in the broader context of aging research and geriatric medicine. It also contributes to the field of immunology and aging biology. By exploring the roles of $CD4^+$ T cells and senescent cells, our work ties into more extensive discussions on immune system aging, longevity, and age-associated diseases. It intersects with ongoing efforts to unravel the molecular and cellular underpinnings of aging, potentially informing the development of interventions to promote healthy aging. Your study also complements a growing body of literature investigating immune-senescent cell interactions, shedding light on possible therapeutic targets to mitigate aging-related immune alterations and chronic diseases.

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Supplementary Material

Our analysis distinctly differentiates between human and mouse species, as demonstrated by the orthogonal trend of principal components (PCs) observed in Fig. S1A-B. This showcases the unique genetic architecture of each species and its impact on aging patterns, consistent with Treaster's findings [34]. However, it is essential to approach our next observation with caution. While the human aging data were encompassed within the PCs of the bladder cancer data, as shown in Fig. S1A. This is further illustrated in the UMAP projection (Fig. S1B), where a partial overlap between human aging and bladder cancer data suggests potential genetic expression similarities between these conditions. But it is crucial to understand that many genes participate in multiple biological processes, and an overlap in expression does not necessarily imply a direct relationship between aging and bladder cancer. Further studies are essential to draw any definitive conclusions. Additionally, the distinct patterns seen in the volcano plots emphasize the differences in upregulated genes between humans and mice, but also show a notable similarity between human aging and bladder cancer data (Fig. S1C).

Age group	Mouse ID	Number of CD4 cells
Young	1	4097
Young	2	3927
Young	3	2614
Young	4	2548
Old	5	3068
Old	6	3332
Old	7	2044
Old	8	2377

Table S 1: Number of $CD4^+$ T cells per mouse in the aging mice dataset, categorized by age group and corresponding mouse ID.

	Treatment	Number of CD4 cells	Number of cytotoxic $CD4^+$ cells	Precent of cytotoxic subset
1	Anti-PD-L1	1984	518	26.11%
2	Anti-PD-L1	2459	787	32%
3	Anti-PD-L1	3562	610	17.125%
4	Anti-PD-L1	3546	641	18.08%
5	Chemotherapy	1350	344	25.48%
6	Untreated	3031	496	16.36%
7	Untreated	961	143	14.88%

Table S 2: Number of $CD4^+$ and cytotoxic $CD4^+$ T cells per individual in the bladder cancer dataset, categorized by treatment type.

	$p21^+p16^-$	$p21^-p16^+$	$p21^+p16^+$
Number of cells	99,342	1,668	1,707
Percentage of total number of cells	40.48%	0.68%	0.695%

Table S 3: Percentage of cells expressing p21 and p16 in the TMS dataset. The table is a direct extension of the Venn diagram in Figure 32A. It illustrates the intersection of three sets, namely $p21^+p16^-$ (green), $p21^-p16^+$ (red), and $p21^+p16^+$ (overlap).

Tissue	Age (months)	Number of cells
Bladder	1, 3, 18, 24	8945
Fat	18, 21, 30	6777
Heart and Aorta	1, 3, 18, 21, 24, 30	8613
Kidney	1, 3, 18, 21, 24, 30	21647
Large Intestine	30	1887
Limb Muscle	1, 3, 18, 21, 24, 30	28867
Liver	1, 3, 18, 21, 24, 30	7294
Lung	1, 3, 18, 21, 30	24540
Mammary Gland	3, 18, 21	12295
Marrow	1, 3, 18, 21, 24, 30	40220
Pancreas	18, 21, 24, 30	6201
Skin	18, 21	4454

Spleen	1, 3, 18, 21, 24, 30	35718
Thymus	3, 18, 21, 24	9275
Tongue	1, 3, 18, 24	20680
Trachea	3	7976

Table S 4: TMS dataset cells distribution in tissues by age. Tissues highlighted with yellow include samples from all six ages.

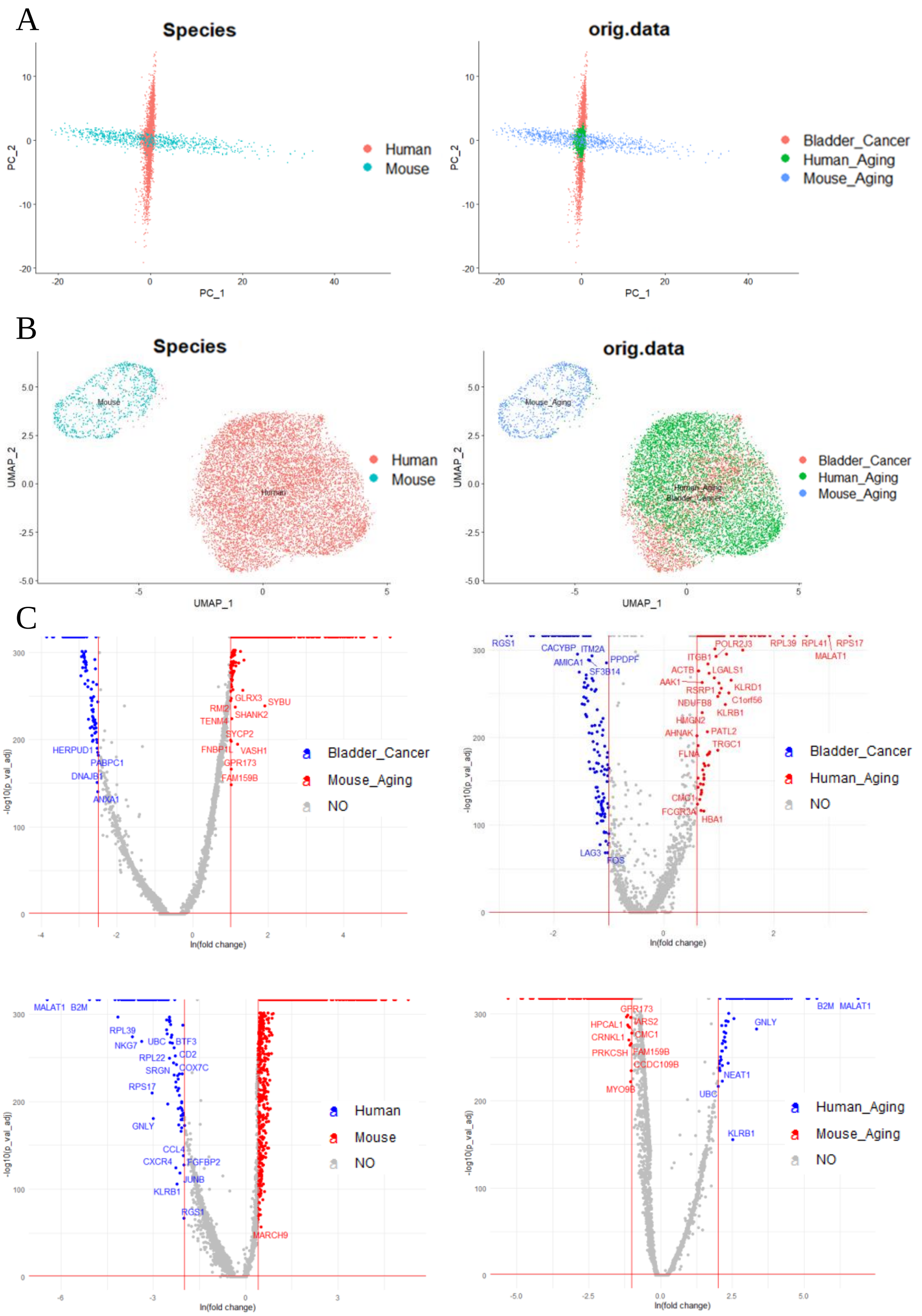

Figure S 1: Comparison of cytotoxic CD4$^+$ T cells in three datasets: human aging, bladder cancer, and aging mouse datasets. A) The left plot displays the PCA projection, which depicts principal components (PC) grouping by species. On the right, the original dataset is grouped *accordingly; B) The UMAP projection corresponds to the PCA plots demonstrated in A; C) The volcano plots showcase the DGE (differentially expressed genes) analysis comparing the bladder cancer dataset (blue) with the aging mouse dataset (red) in the upper left plane, the bladder cancer dataset (blue) with the human aging dataset (red) in the upper right plane, human datasets (blue) with the Mouse (red) in the lower left plane, and the Human aging dataset (blue) with the aging mouse dataset (red) in the upper left plane. The non-differentially expressed genes are marked as "NO" and shown in grey. For both A and B plots, the following colors represent each species: Human (red), Mouse (light blue), Bladder cancer (red), Human aging (green), and Aging mouse (blue).*

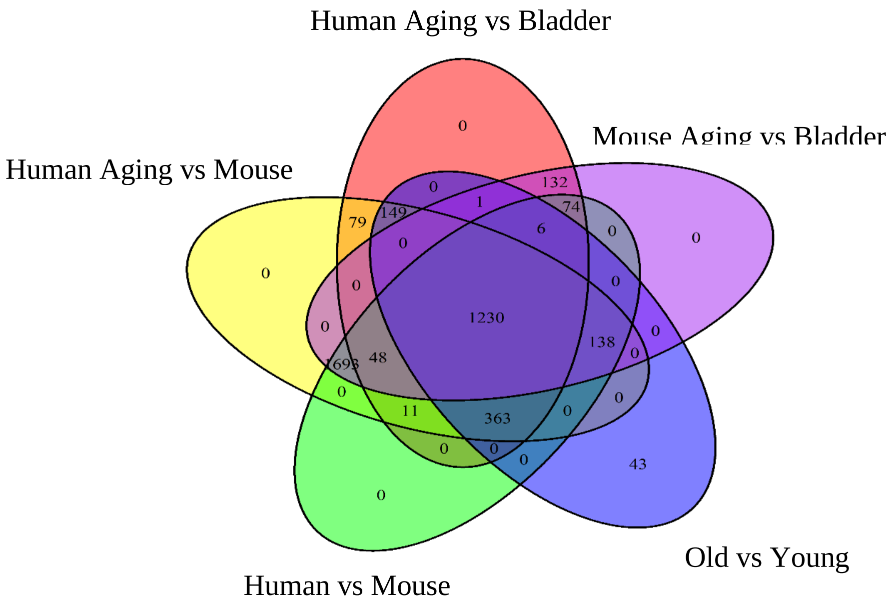

Figure S 2: Venn diagram shows gene expression in cytotoxic subsets from three datasets: supercentenarians, aging mice, and bladder cancer. It compares human aging to mouse aging (yellow), human aging to bladder cancer (red), mouse aging to bladder cancer (purple), and old to young age (blue). It also compares human data to mouse data (green).

Age group reorganized	Age group	Age (months)	Mouse ID	Number of all cells
Young	Young	1	1-M-62	9809
Young	Young	1	1-M-63	16171
Young	Young	3	3-M-5/6	4929
Young	Young	3	3-M-7/8	5296
Young	Young	3	3-M-8	6265
Young	Young	3	3-M-8/9	460
Young	Young	3	3-M-9	4959
Young	Young	3	3-F-56	15218
Young	Young	3	3-F-57	8475
Old	Old	18	18-F-50	12062

Old	Old	18	18-F-51	9779
Old	Old	18	18-M-52	17857
Old	Old	18	18-M-53	4947
Old	Old	21	21-F-54	16844
Old	Old	21	21-F-55	18984
Old	Old	24	24-M-58	12166
Old	Old	24	24-M-59	12107
Old	Old	24	24-M-60	9656
Old	Old	24	24-M-61	3731
Old	Supercentenar-ian	30	30-M-2	25405
Old	Supercentenar-ian	30	30-M-3	9757
Old	Supercentenar-ian	30	30-M-4	7863
Old	Supercentenar-ian	30	30-M-5	12649

Table S 5: Total Cell Count per Mouse in the TMS Dataset by Age Group. The table categorizes mice from the TMS dataset into reorganized age groups: Young (1-3 months) and Old (18-30 months), with a further distinction in the Old group between Old (18-24 months) and Super-centenarian (30 months). Each mouse is uniquely identified by a 'Mouse ID' composed of its age in months, sex, and a unique number. The 'Number of All Cells' column displays the total cell count for each mouse.

Age group re-organized	**Age group**	**Age (months)**	**Mouse ID**	**Number of T cells**	**Percent of T cells per mouse**
Young	Young	1	1-M-62	343	3.5%
Young	Young	1	1-M-63	1635	10.11%
Young	Young	3	3-M-5/6	123	2.495%
Young	Young	3	3-M-7/8	292	5.51%
Young	Young	3	3-M-8	531	8.475%
Young	Young	3	3-M-8/9	0	0%
Young	Young	3	3-M-9	11	0.22%
Young	Young	3	3-F-56	3876	25.5%
Young	Young	3	3-F-57	1614	19.04%
Old	Old	18	18-F-50	2171	18%
Old	Old	18	18-F-51	1309	13.39%
Old	Old	18	18-M-52	1131	6.33%
Old	Old	18	18-M-53	255	5.15%
Old	Old	21	21-F-54	3125	18.55%
Old	Old	21	21-F-55	3036	15.99%
Old	Old	24	24-M-58	1164	9.57%
Old	Old	24	24-M-59	836	6.905%

Old	Old	24	24-M-60	1108	11.47%
Old	Old	24	24-M-61	149	3.99%
Old	Supercente-narians	30	30-M-2	183	0.72%
Old	Supercente-narians	30	30-M-3	1052	10.78%
Old	Supercente-narians	30	30-M-4	1188	15.11%
Old	Supercente-narians	30	30-M-5	1194	9.44%

Table S 6: T cell distribution in the TMS dataset.

	Age group 2	**Mouse ID**	**Number of $CD8^+$ T cells**	**Number of $CD4^+$ T cells**	**CD4/CD8**
1	Old	18-F-50	622	280	0.450160772
2	Old	18-F-51	607	363	0.598023064
3	Old	21-F-54	913	443	0.485213582
4	Old	21-F-55	842	629	0.747030879
5	Old	24-M-58	549	323	0.588342441
6	Old	24-M-59	306	327	1.068627451
7	Old	24-M-60	660	74	0.112121212
8	Old	30-M-3	652	112	0.171779141
9	Old	30-M-4	641	343	0.535101404
10	Old	30-M-5	650	257	0.395384615
11	Young	1-M-62	109	123	1.128440367
12	Young	1-M-63	459	786	1.712418301
13	Young	3-F-56	1383	753	0.544468547
14	Young	3-F-57	702	524	0.746438746
15	Young	3-M-8	216	199	0.921296296

Table S 7: CD4/CD8 ratio and $CD4^+$ and $CD8^+$ T cell distribution in the TMS dataset.

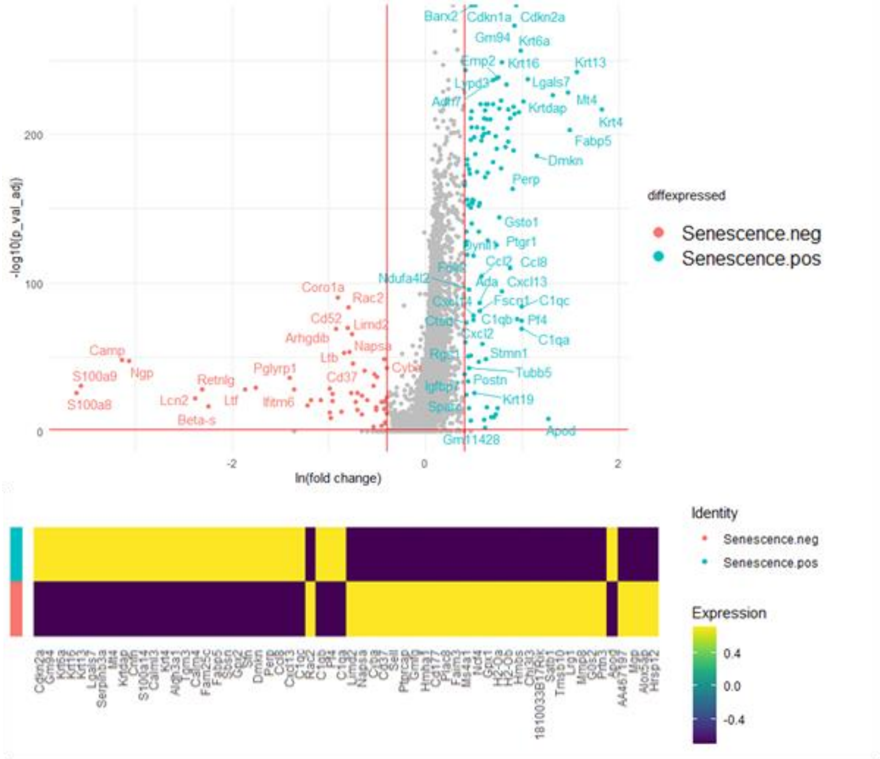

Figure S 3: DGE of senescent cells vs non-senescent cells in the TMS dataset. The volcano plot in the top plane shows the genes that are differentially expressed in senescent cells (light blue) named Senescence.pos, and non-senescent cells (red) named Senescence.neg. The lower plane displays the average expression of 30 highly expressed genes and 30 down-expressed genes for the same cell groups as in the upper plane. The expression scale ranges from yellow (highly expressed) to purple (down-expressed).

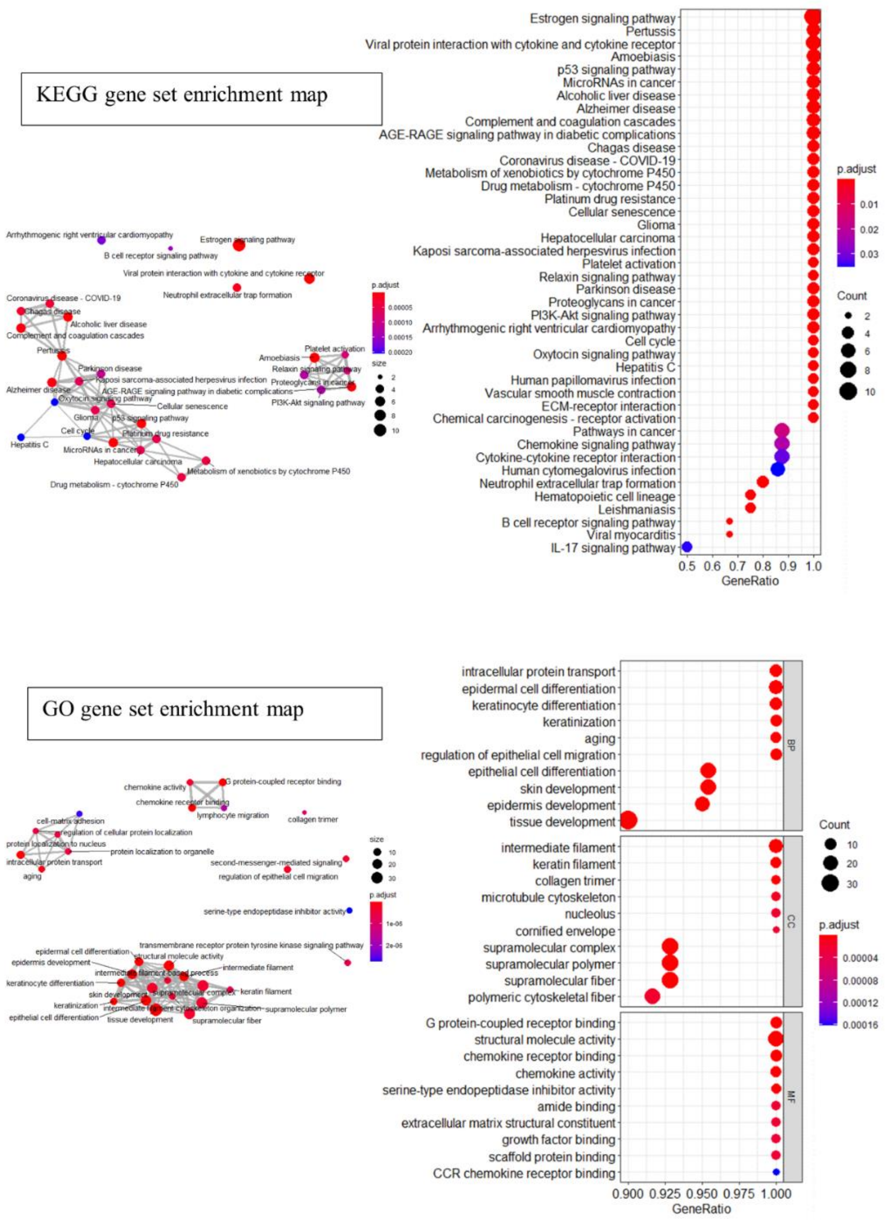

Figure S 4: Gene-set enrichment maps of mice senescent cells of the TMS dataset. The maps use dot plots and hypergeometric tests to gain insights into aging and senescent cells. The upper plot displays an enrichment map of senescent cells enriched according to the KEGG database, highlighting pathways involved in inflammation, p53 signaling, estrogen signaling, and chemokine signaling pathways. The lower plot showcases the expression of the same cell data enrichment, according to the GO database, revealing insights into Biological Processes (BP), such as the developmental process, Molecular Functions (MF), such as receptor bindings, and Cellular Components (CC), such as filaments and nucleus. Both plots present relationships between the biological pathways, facilitating a better understanding of the functions of senescent cells and their interactions with supercentenarians.

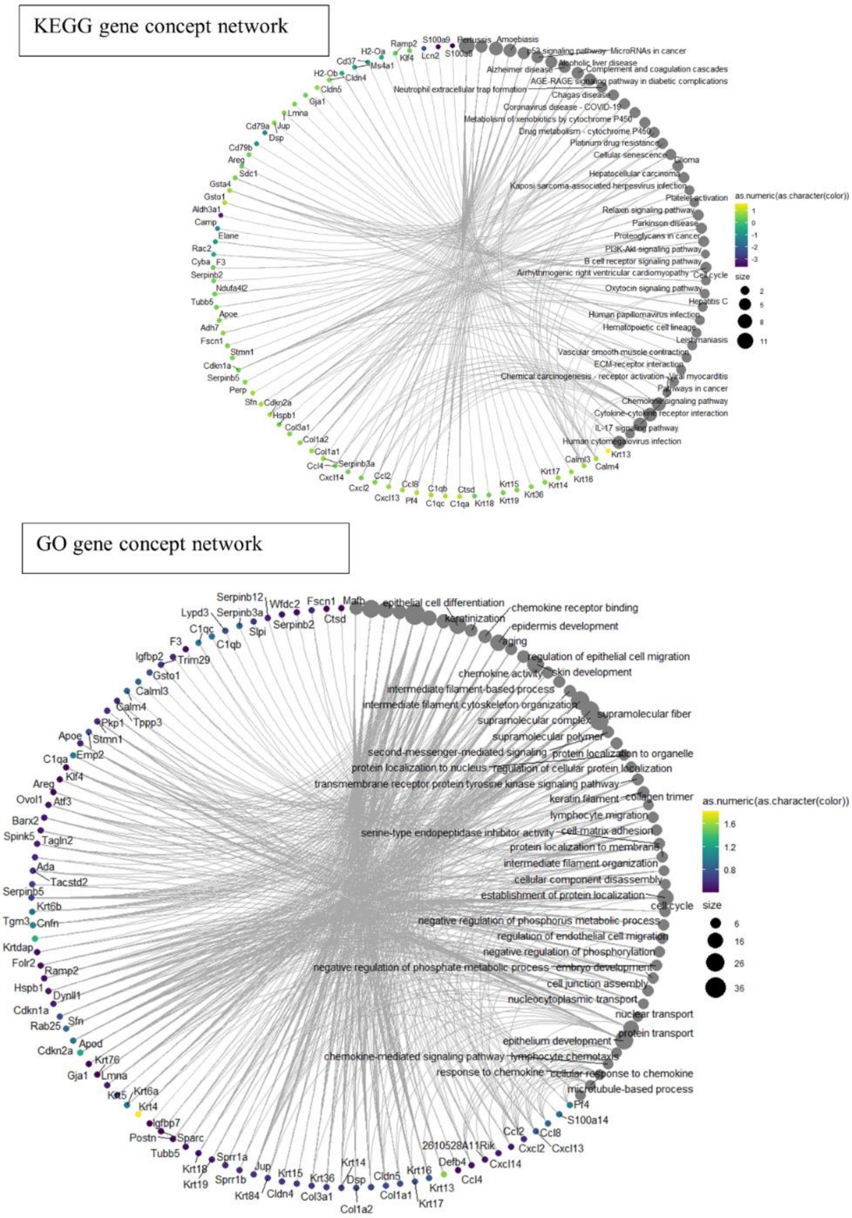

Figure S 5: Gene-concept network of mice senescent cells of the TMS dataset. To determine which genes are involved in the pathways presented in the enrichment map shown in Fig. S4, we need to take into account the fact that a gene may belong to multiple annotation categories, which adds a level of biological complexity. To help us understand this complexity, we created a concept network. The upper plot of the network shows the concept network for senescent cells enriched based on the KEGG database, while the lower plot shows the same cell data concept network enriched according to the GO database.

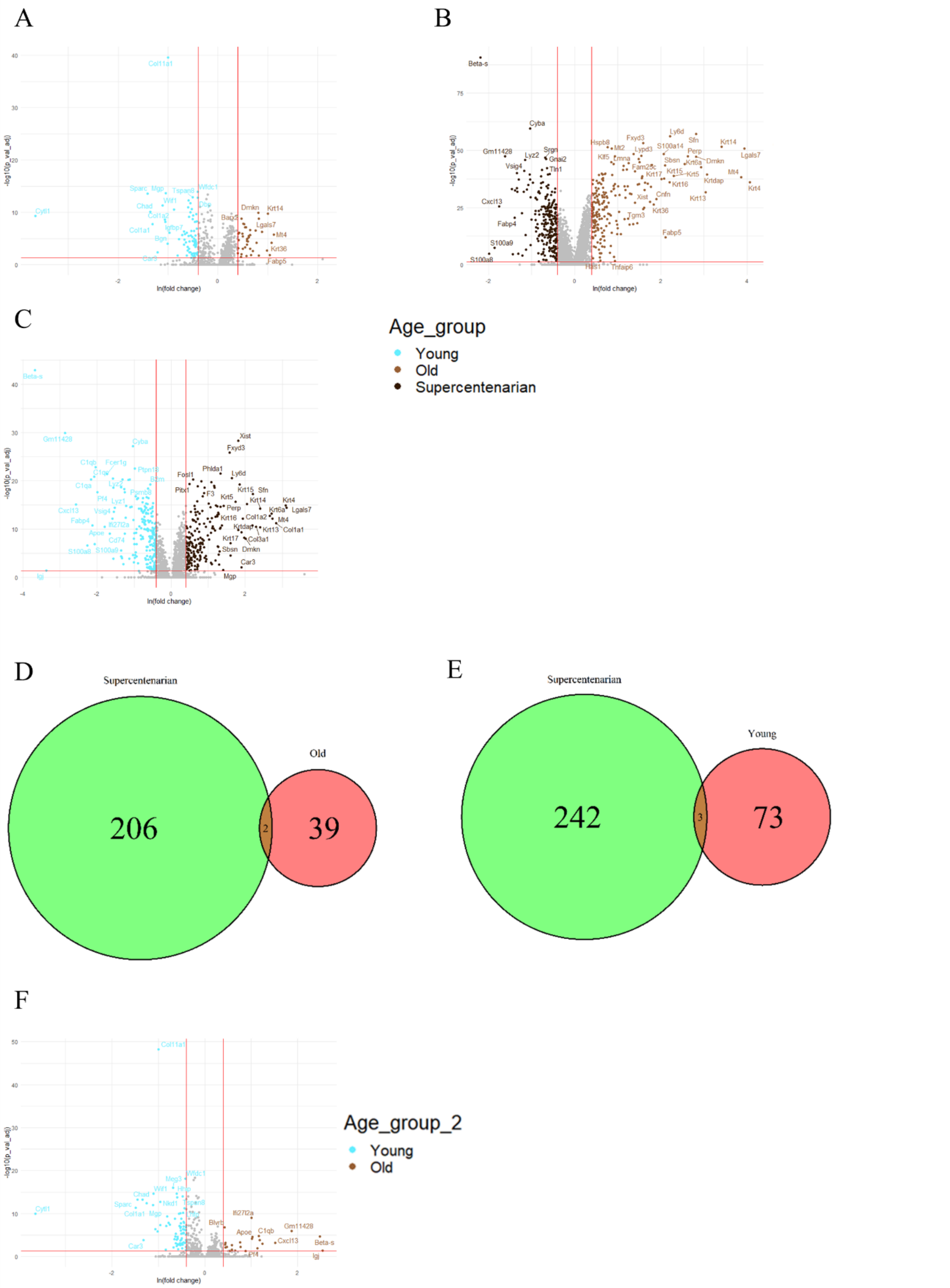

Figure S 6: Comparative DGE analysis and Intersectional Genomics in Young, Old, and Supercentenarian Mice on senescent cells, using the TMS dataset.H

A) The volcano plot showcases the DGE analysis, with the young group (1-3 months) represented in light blue and the old group (18-24 months) depicted in brown; B) In another volcano plot, the supercentenarians group (1-3 months) is shown in dark brown, while the old group

(18-24 months) is still represented in brown; C) Lastly, the volcano plots display the DGE analysis comparing the young group (1-3 months) in light blue and the supercentenarian group (30 months) in dark brown; D) A Venn diagram is presented, showing the intersection between the supercentenarians group (green) and the old group (red); E) Venn diagrams are presented, showing the intersection between the supercentenarians group (green) and the young group (red) senescent cells (D), as well as the intersection between the young and supercentenarians senescent cells (C); F) Volcano plot comparing gene expression of senescent cells in old (18-30 months, brown) and young (1-3 months, light blue) groups. "NO" is a non-differentially expressed gene (grey).

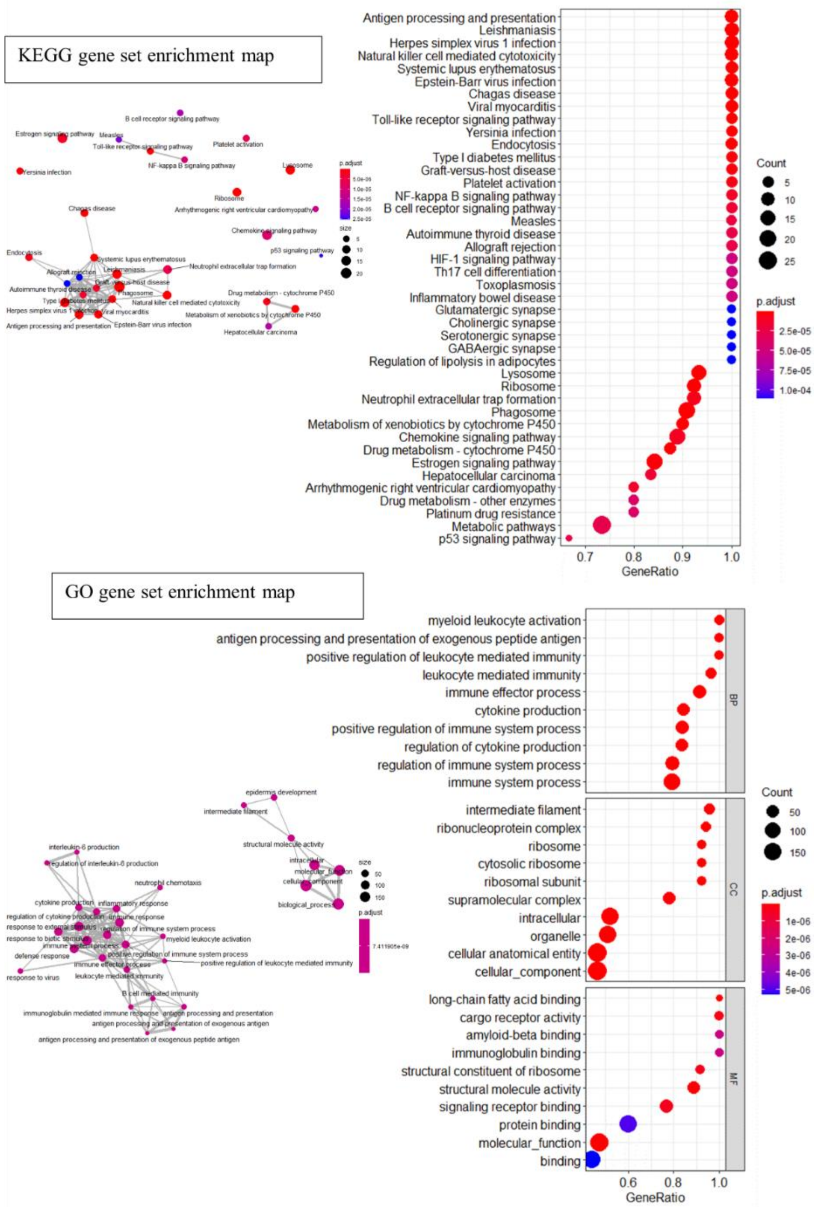

Figure S 7: Gene-set maps of senescent cells from supercentenarian mice (30 months old) reveal the molecular mechanisms that underlie aging and senescence. The maps use dot plots and hypergeometric tests to gain insights into the aging process in supercentenarians and assist in *developing new therapeutic strategies. The upper field displays an enrichment map of senescent cells from the old mice group, enriched according to the KEGG database, highlighting pathways involved in inflammation, p53 signaling, estrogen signaling, and chemokine signaling pathways. There are no aging disease and aging degradation pathways. The lower field displays the expression of the same cell data enrichment, according to the GO database, revealing insights into Biological Processes (BP), such as the immune effector process, Molecular Functions (MF), such as long-chain fatty acid binding, and Cellular Components (CC), such as a ribosome. Both plots present relationships between the biological pathways, making it easier to understand the functions of senescent cells in interactions with supercentenarians.*

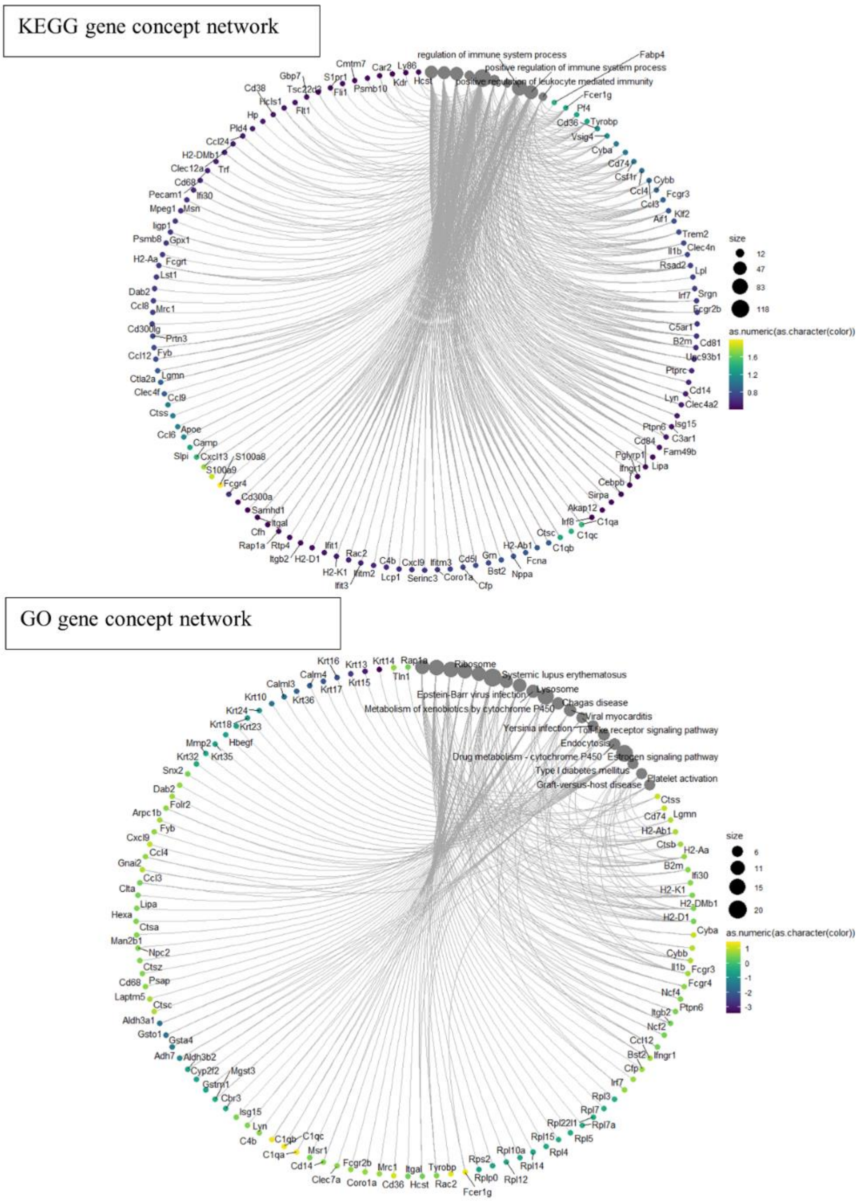

Figure S 8: Gene-concept network of senescent cells from supercentenarian mice (30 months old) in the TMS dataset. In order to identify the genes involved in the pathways depicted on the enrichment map featured in Fig. S7, it is crucial to consider the possibility that a gene could be associated with multiple annotation categories, thus introducing a level of biological intricacy. To gain a better understanding of this complexity, we generated a concept network.

The top diagram of the network illustrates the concept network for senescent cells enriched using the KEGG database. In contrast, the bottom diagram displays the same cell data concept network enriched with reference to the GO database.

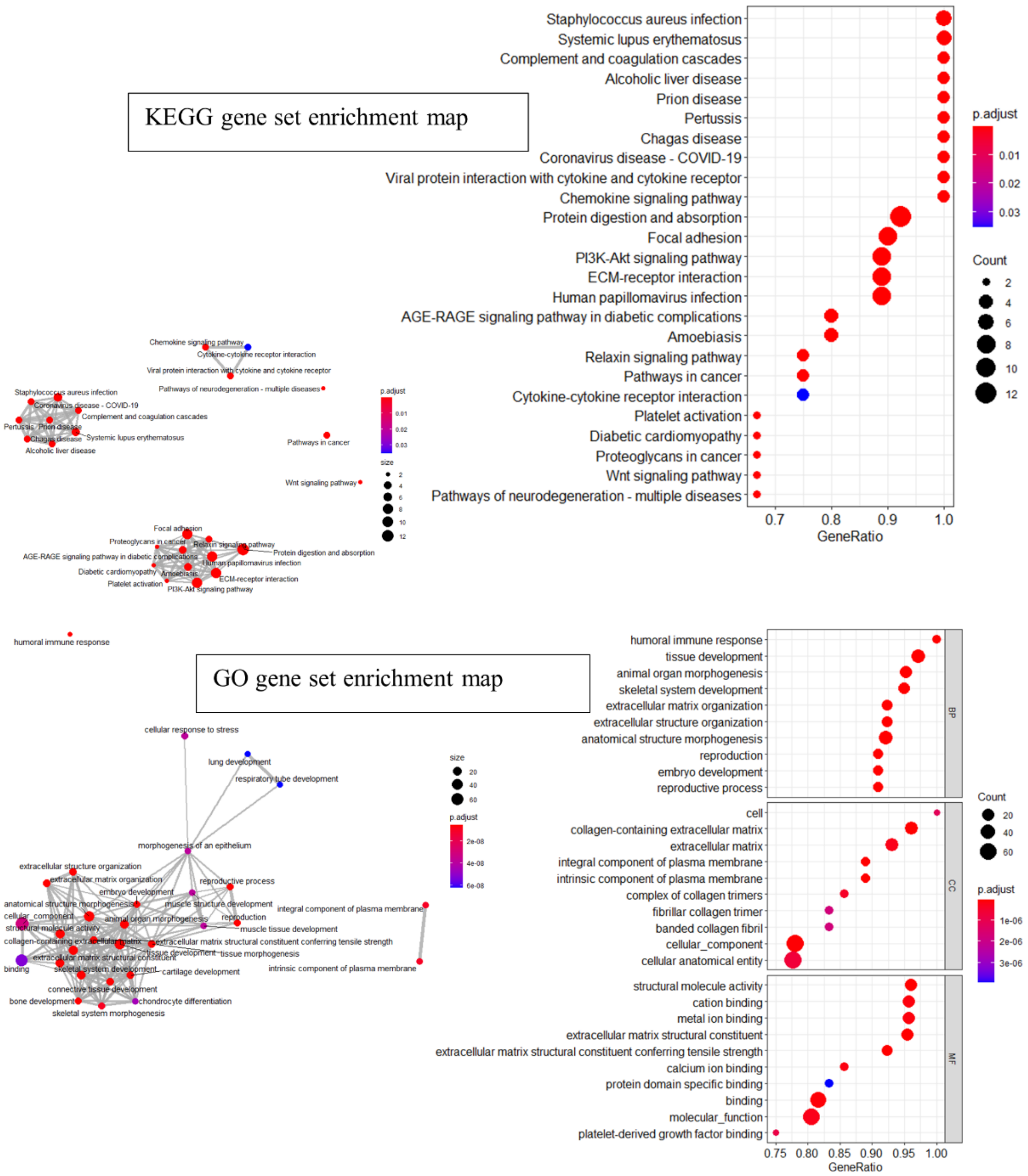

Figure S 9: Dot plots and hypergeometric tests present gene-set enrichment maps of senescent cells from old mice (18-30 months) in the TMS dataset. The findings offer significant insights into the molecular mechanisms that play a role in the aging process and senescence. The KEGG database indicates that the top panel displays the enrichment map of senescent cells from the old mice group, highlighting pathways contributing to inflammation, aging, and age-related diseases. The lower panel demonstrates the expression of the same cell data enrichment, according to the GO database, and reveals insights into various biological processes (BP), molecular functions (MF), and cellular components (CC), such as humoral immune response, structural molecular activity, and extracellular matrix, respectively. Both graphs demonstrate *the relationships between the biological pathways, thereby facilitating a comprehensive understanding of senescent cell functions and their interactions with aging.*

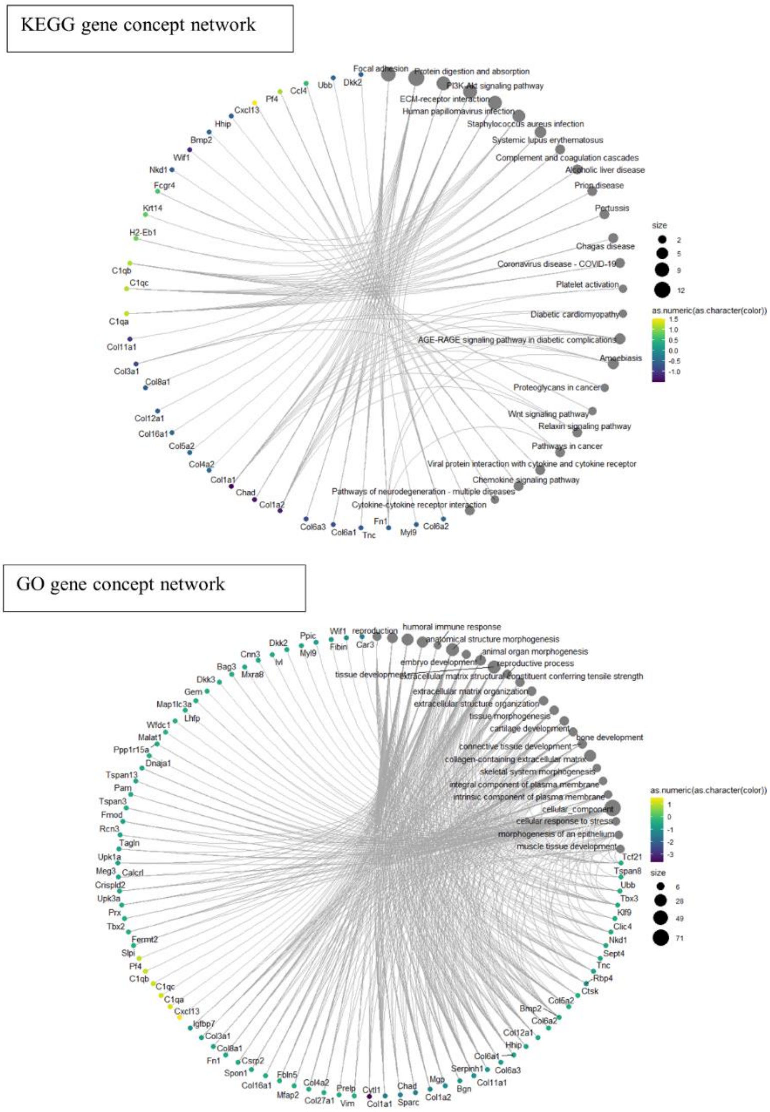

Figure S 10: Gene-concept network of senescent cells from old mice (18-30 months old) in the TMS dataset. In order to identify the genes involved in the pathways depicted in the enrichment map showcased in Figure S9, we must consider the possibility that a particular gene may be linked to multiple annotation categories, which adds a layer of biological complexity to the analysis. To gain a deeper understanding of this intricate network, we created a concept network. The upper diagram of the network depicts the concept network for senescent cells that were enriched using the KEGG database, while the lower diagram displays the same cell data concept network enriched with reference to the GO database. By comparing these two diagrams, we can gain valuable insights into the biological functions associated with the genes and their interconnections.

תקציר

על פי נתוני ארגון הבריאות העולמי (WHO) והמחלקה של האו"ם לענייני כלכלה וחברה, החטיבה לאוכלוסייה, אוכלוסייה העולם מתבגרת באופן שוטף. מצב זה דורש שימור בריאות ויכולות קוגניטיביות לאורך זמן ממושך יותר, כלומר "הזדקנות מוצלחת" עם פחות מחלות הקשורות בגיל. קיימת קבוצה מיוחדת של מבוגרים שהגיעו לגילאי 110 ומעלההמייצגת הזדקנות מוצלחת, חקר קבוצה זו יכול לתרום להשגת תובנות ומסקנות כיצד להישמר מפני מחלות הזדקנות למשך זמן רב יותר ולתת איכות חיים טובה יותר עבור הזקנים.

הזדקנות תאית, עצירת צמיחת תאים בלתי הפיכה, היא מאפיין מרכזי בתהליך ההזדקנות. תאים מזדקנים מבטאים את הפעילות המטבולית שלהם על ידי שחרור פנוטיפ ההפרשה הקשורים לסנס(SASP) תערובת של ציטוקינים וכימוקינים. מעניין שבעוד שה-SASP מושך תאי מערכת חיסון הוא גם מקדם דלקת מקומית ומקל על הצטברות של תאים מזדקנים. באמצעות ה-SASP תאים מזדקנים משפיעים על תאים סמוכים, ובכך גורמים להם לעבור סנס.

ההזדקנות הסלולרית מלווה בהזדקנות חיסונית (המכונה גם זיקנה חיסונית), המתייחסת להזדקנות מערכת החיסון. הזדקנות זו משפיעה על התגובות האימוניות וככל הנראה תורמת להצטברות של תאים מזדקנים עקב פינוי לא יעיל של תאים אלה. שינויים הקשורים לגיל ברפרטואר תאי ה-$CD4^+$ T הינם חלק חיוני ממערכת החיסון האדפטיבית, נצפו הן בבני אדם והן בעכברים, כאשר אנשים מבוגרים מציגים עלייה משמעותית בתאי זיכרון T אפקטוריים (TEM), תת-קבוצות רגולטוריות מופעלת (aTregs), מותשות וציטוטוקסית בהשוואה לאנשים צעירים יותר .

במחקר שלנו, יישמנו כלי ניתוח ביואינפורמטיים על מערכי נתונים של רצף RNA חד-תא(scRNS-seq) כדי לבחון את ההצטברות המשותפת של לימפוציטים הקשורים לגיל ותאים מזדקנים בהזדקנות. המטרה העיקרית שלנו הייתה לזהות SASP הקשור לגיל ולחקור את פינוי תאים מזדקנים על ידי לימפוציטים $CD4^+$ T ציטוטוקסיים. החקירה שלנו גילתה הבדלים ב-SASP בקרב אנשים צעירים, מבוגרים ומבוגרים מאוד, והצביעה על מעורבותם של כימו-קינים ספציפיים בפינוי תאים מזדקנים. לסיכום, תוצאות הניסוי תומכות בהשערה שלנו ששינויים בהרכב הלימפוציטים, כולל עלייה בתאי T מווסתים משופעלים וירידה בתאי $CD4^+$ T ציטוטוקסיים ו-SASP הקשור לגיל ספציפי תורמים להצטברות של תאים מזדקנים עם הגיל. תובנות אלו מהוות חשיבות מכרעת בהבנת מנגנוני ההזדקנות וסוללות את הדרך להתערבויות המקדמות הזדקנות מוצלחת ומפחיתות שכיחות מחלות הקשורות לגיל.

השפעות דיסרגולציה של לימפוציטים על הזדקנות תאית

חיבור לשם קבלת התואר "מוסמך במדעי הרפואה" (M.Med.Sc.)

בפקולטה למדעי הבריאות

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שם המנחה: פרופסור אלון מונסונגו

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אוניברסיטת בן-גוריון בנגב

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מאת אנה דרנקו

תאריך עברי ב' באדר א' תשפ"ד תאריך לועזי 11/02/2024